3rd-year Bachelor's degree in Software Engineering - FSM 2025-2026

K-MEANS ALGORITHM

Elaborated by Med Yassine Khlif & Sedki Lajmi

A powerful approach to unsupervised data clustering...

1. Introduction

In the world of data science and machine learning, clustering is an unsupervised learning technique used to group similar data points. Unlike supervised learning, clustering does not require labeled data. One of the most popular clustering algorithms is K-Means, which organizes data into K distinct, non-overlapping clusters based on the similarity of features.

2. Definition

K-Means is an iterative algorithm that partitions a dataset into K clusters, where each data point belongs to the cluster with the nearest mean (centroid).

→ “K” number of clusters chosen by the user.

→ “Means” refers to the average (mean) of data points in a cluster, representing its center.

Objective:

Maximizing similarity within groups: It sorts data into clusters so that the items within each cluster are much more alike than the items in other clusters.

Examples:

K-Means can be used to detect anomalies in data, group similar songs by tempo and rhythm, segment customers based on buying behavior, or classify images and documents by content.

K-Means clustering is widely used when we want to group similar items without knowing the categories beforehand. For example, in music recommendation systems (like Spotify, YouTube Music, Apple Music…), millions of songs are analyzed based on numerical features such as:

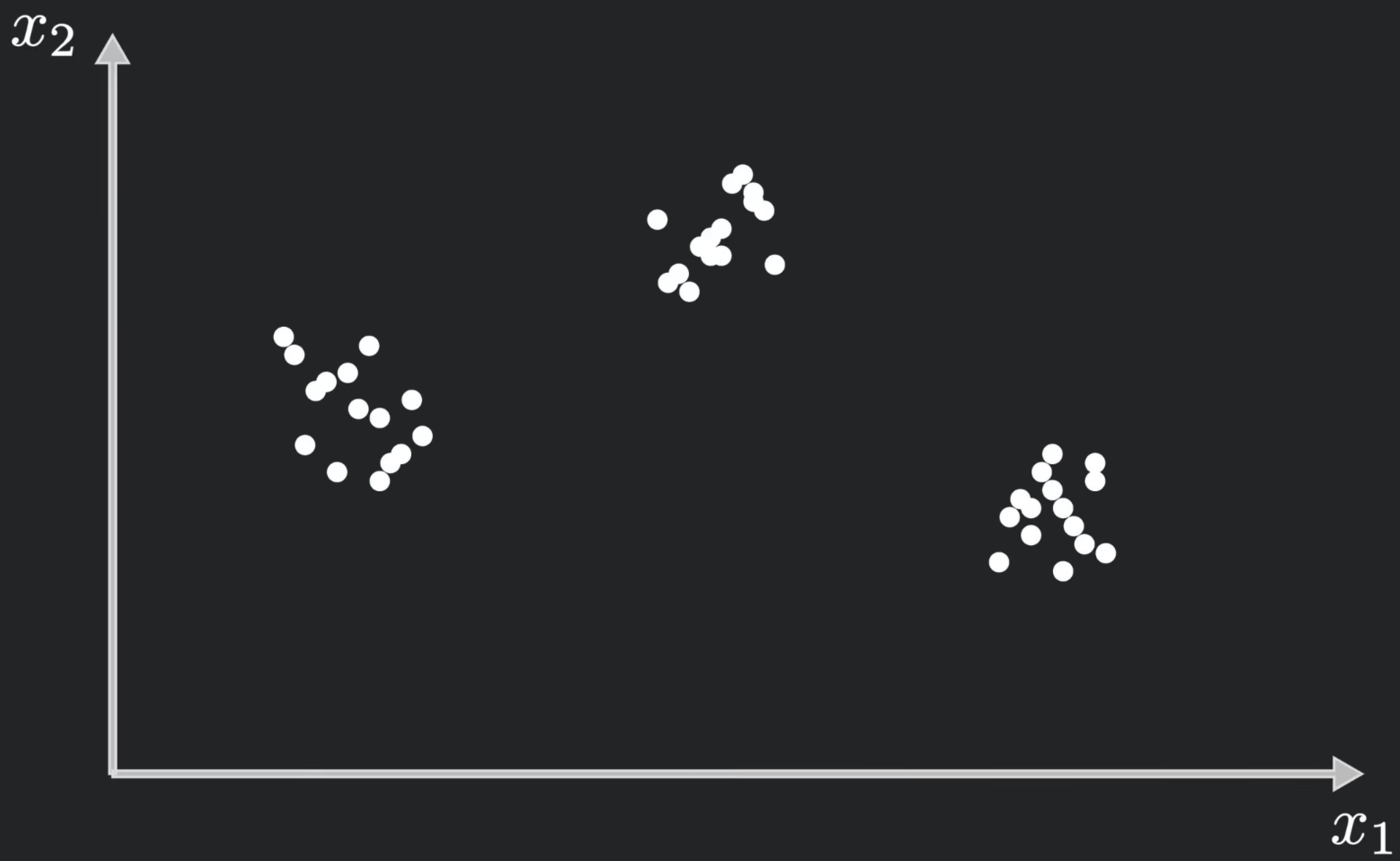
- **Tempo (how fast the song is)**
- **Energy level (calm vs. intense)**
- **Danceability**
- **Instrumentation (acoustic vs. electronic)**
- **Loudness**

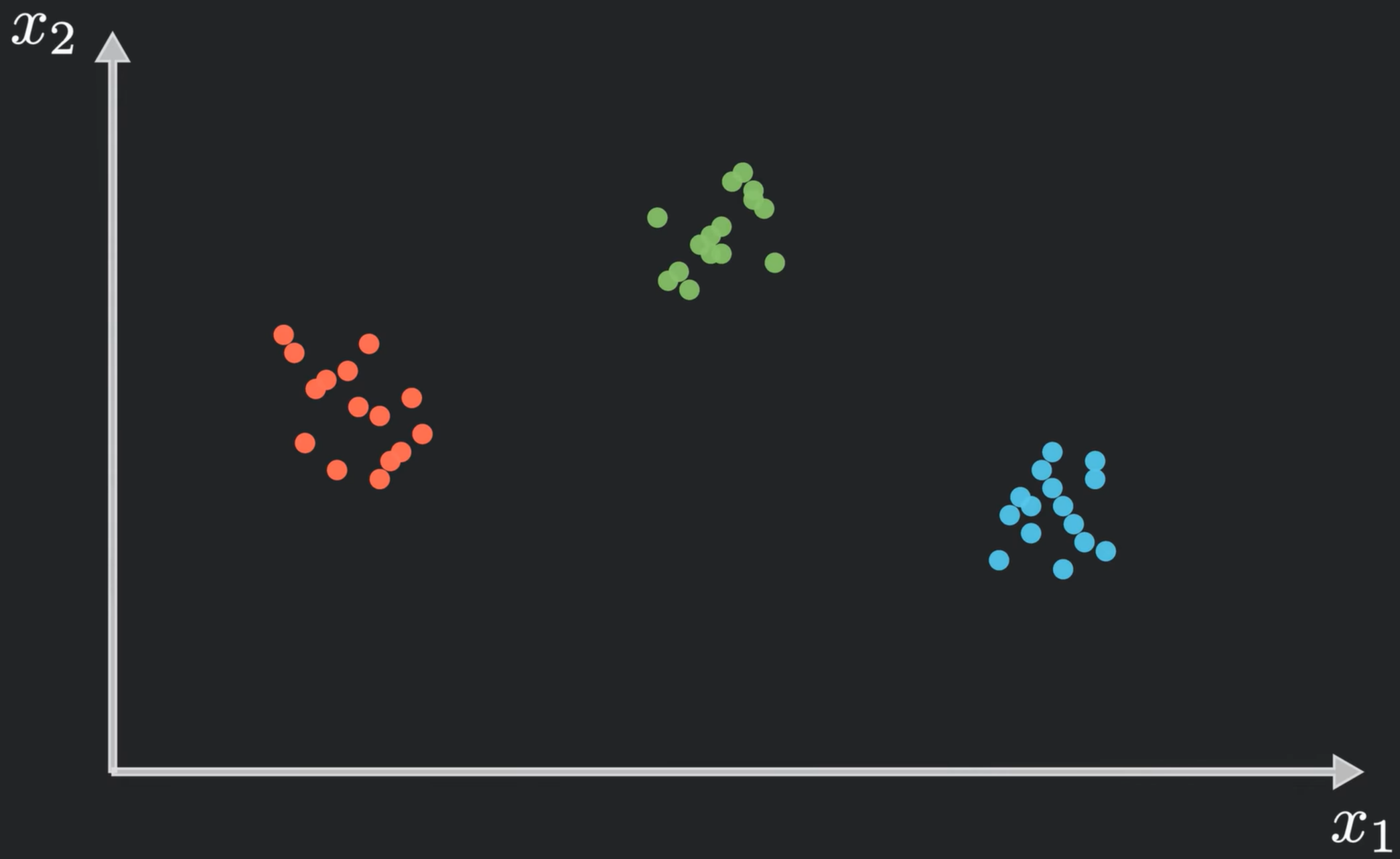
Each song becomes a data point with those attributes (tempo, energy, etc.).

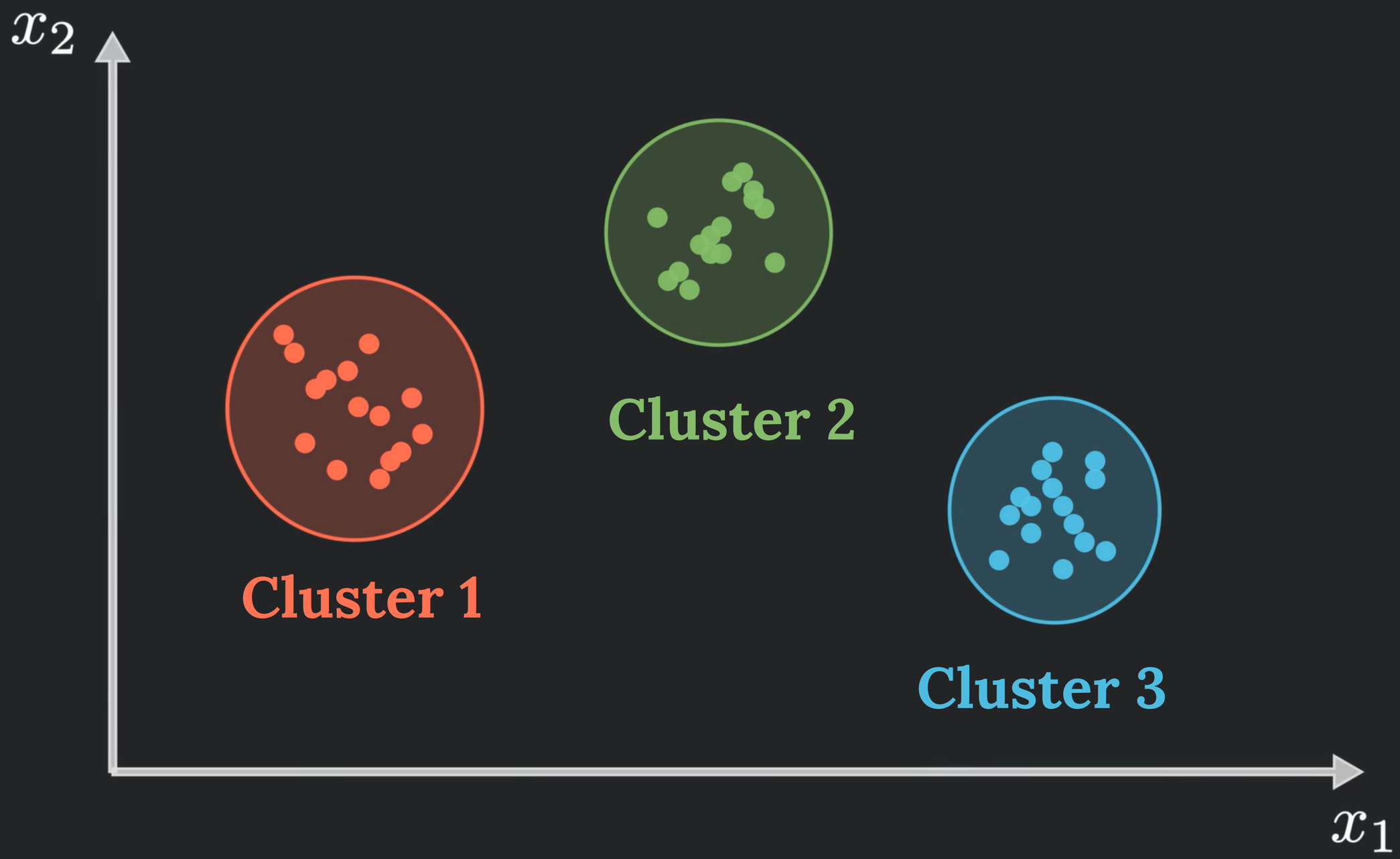
K-Means groups them into clusters based on similarity:

- **Cluster 1: Calm, slow, acoustic songs (study/relax playlist)**
- **Cluster 2: High-energy, electronic songs (gym playlist)**
- **Cluster 3: Medium-tempo pop songs (party playlist)**

The algorithm doesn't know "genre" or "mood." It just groups songs mathematically based on the distance between features.


x_2
x_1


x_2
x_1

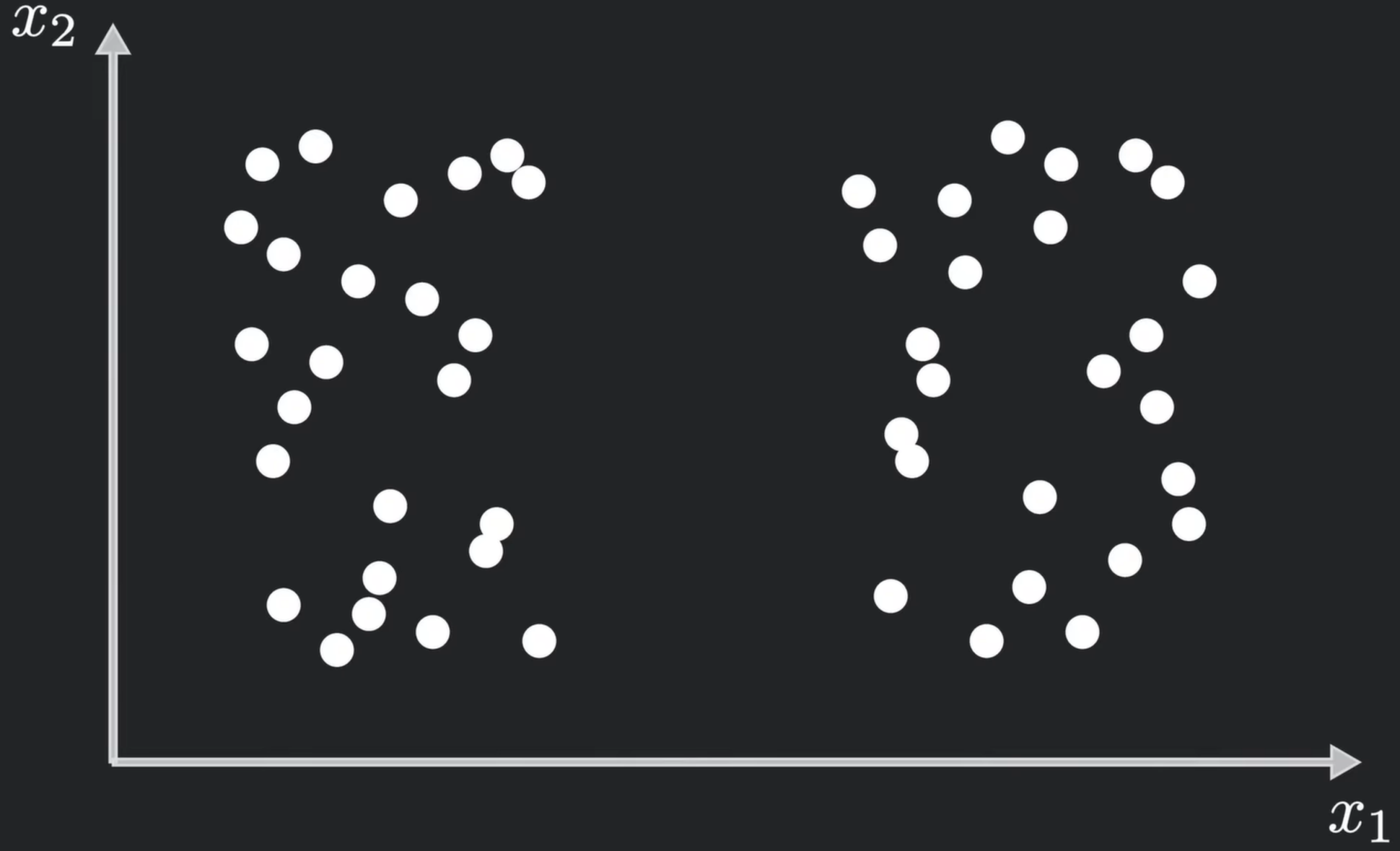

x_2
Cluster 1
Cluster 2
Cluster 3
x_1

Let's dive deep into how the K-means algorithm works!

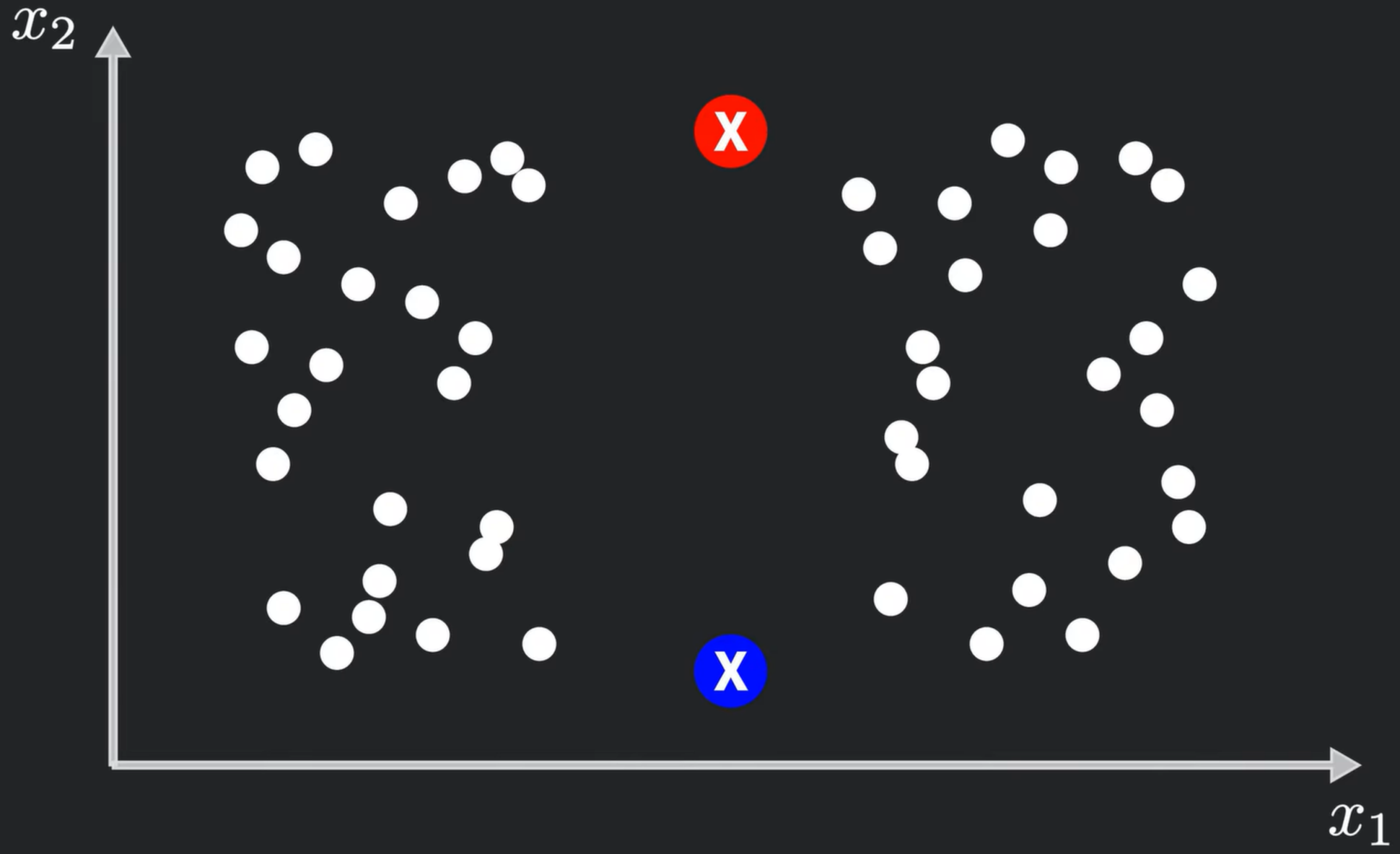
3. Steps of the K-Means Algorithm

1. Choose the number of clusters (K). Example: K = 3 for three clusters.
2. Initialize centroids randomly.
3. Randomly select K points from the dataset as initial cluster centers.
4. Assign each data point to the nearest centroid.
5. Compute distance (usually Euclidean) and assign to the closest centroid.
6. Update the centroids.
7. Recalculate each centroid as the mean of the points in its cluster.
8. Repeat steps 3–4 until centroids do not change significantly or a maximum number of iterations is reached.

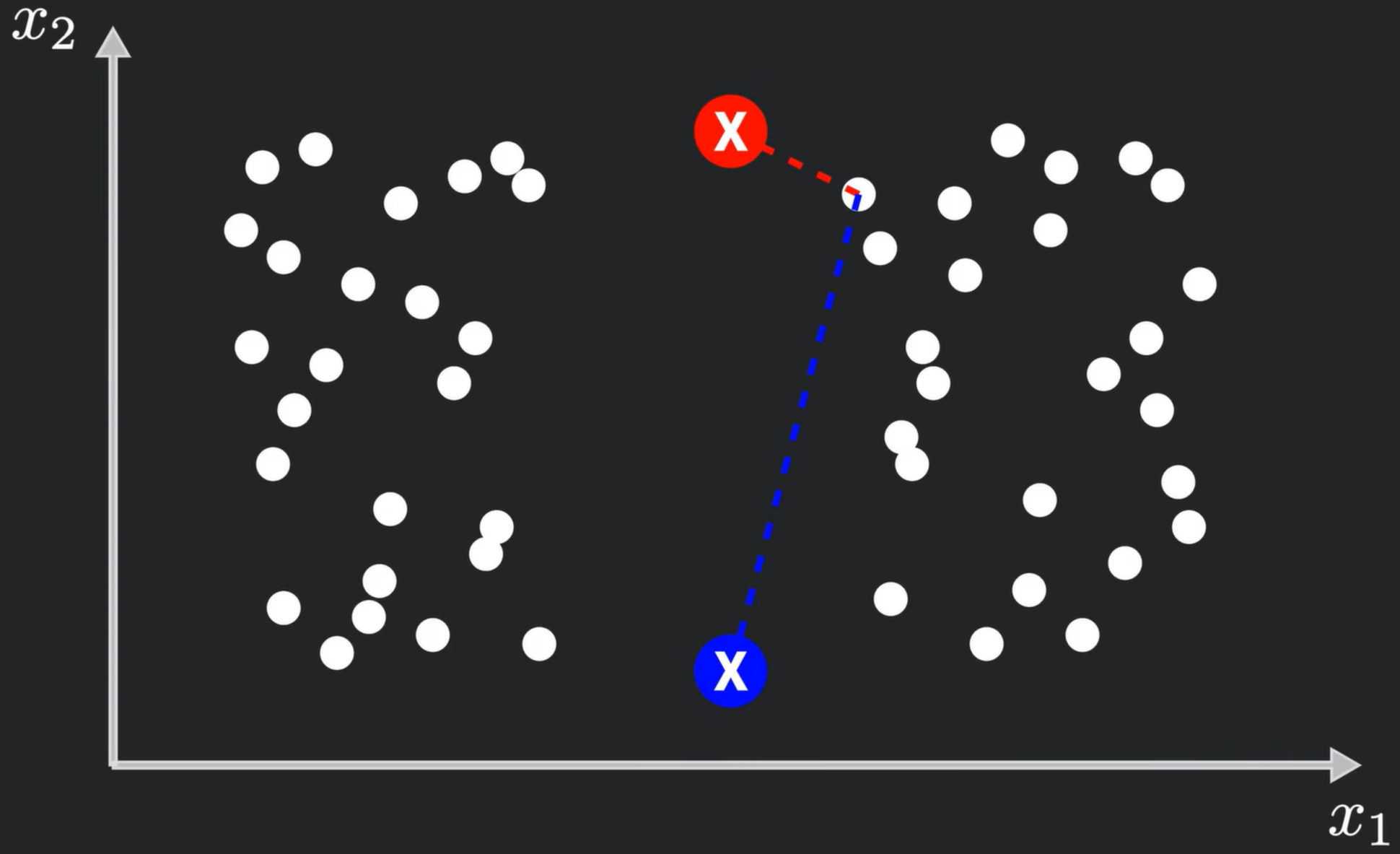
Choose the number of clusters (K = 2 for two clusters)

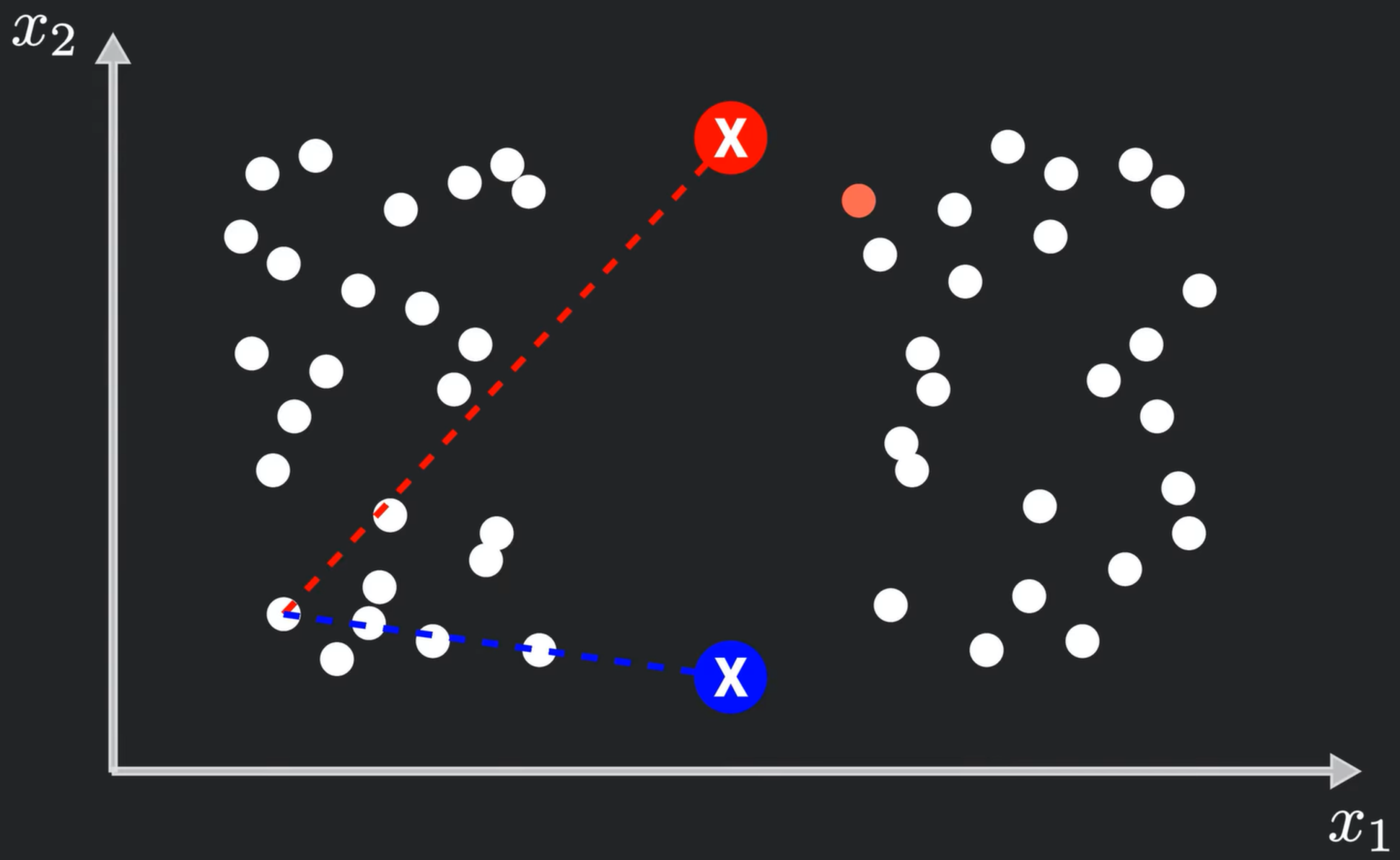


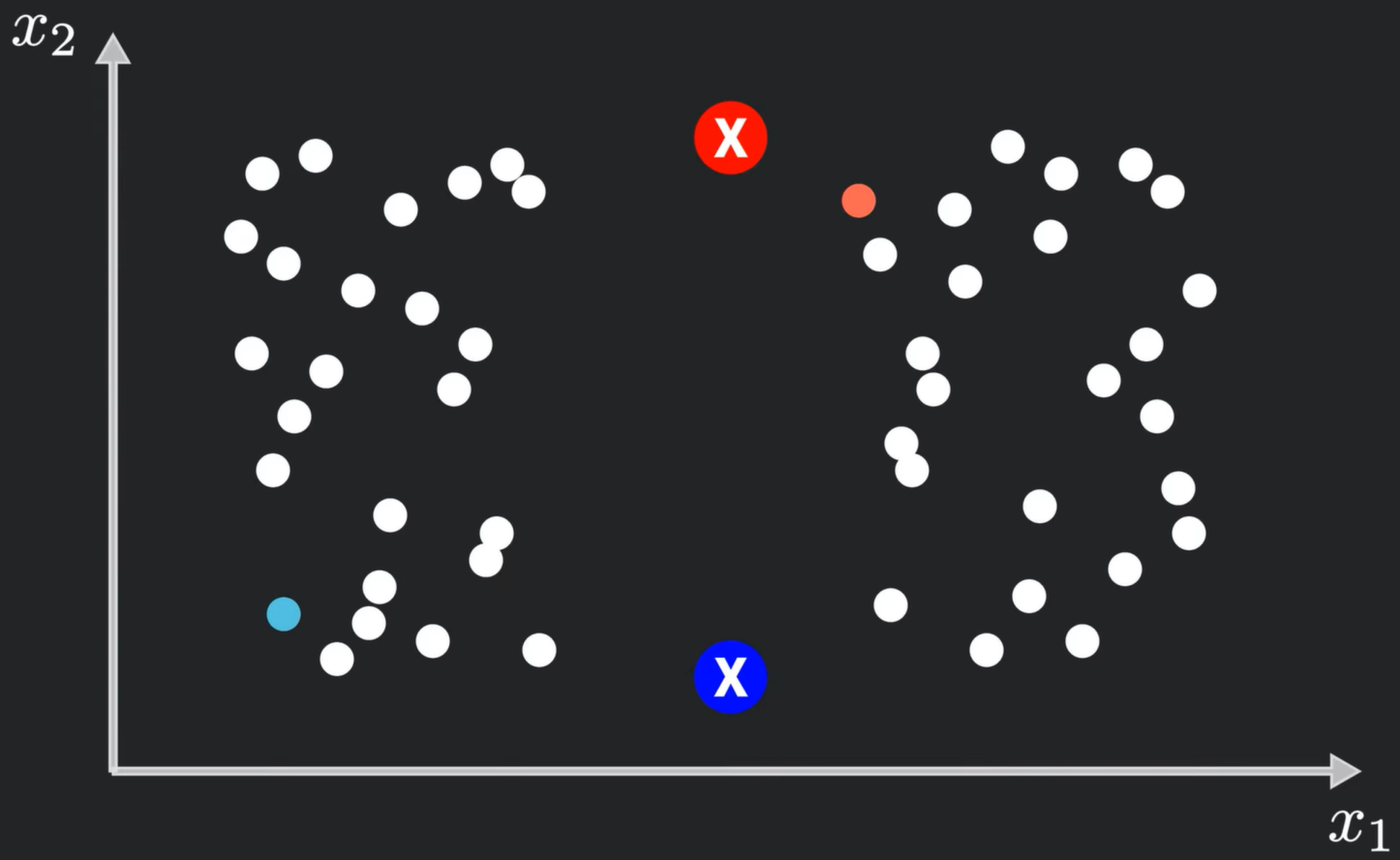
Initialize centroids randomly

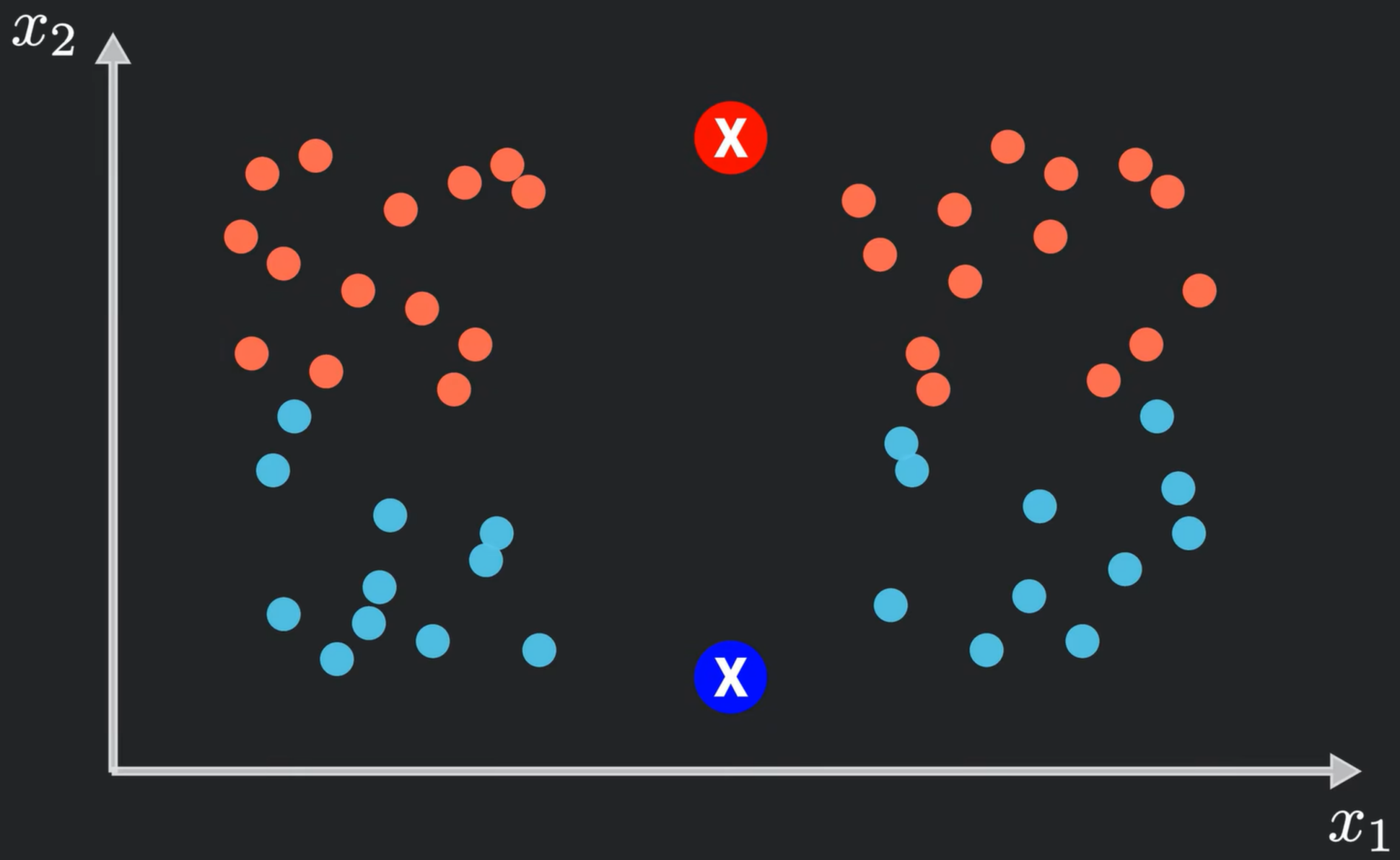


Assign each data point to the nearest centroid.

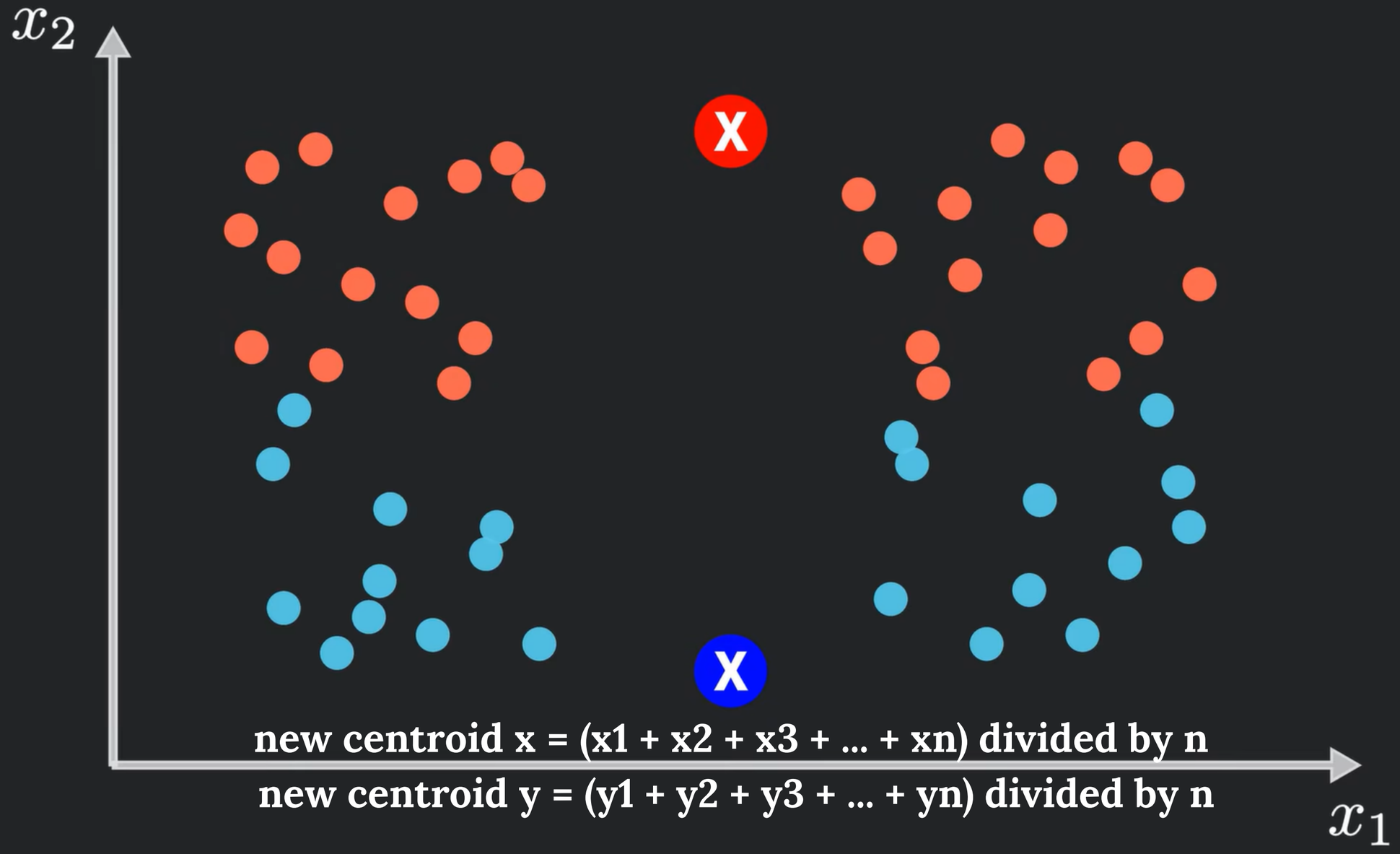



x_2
X
X
x_1


x_2
X
X
x_1

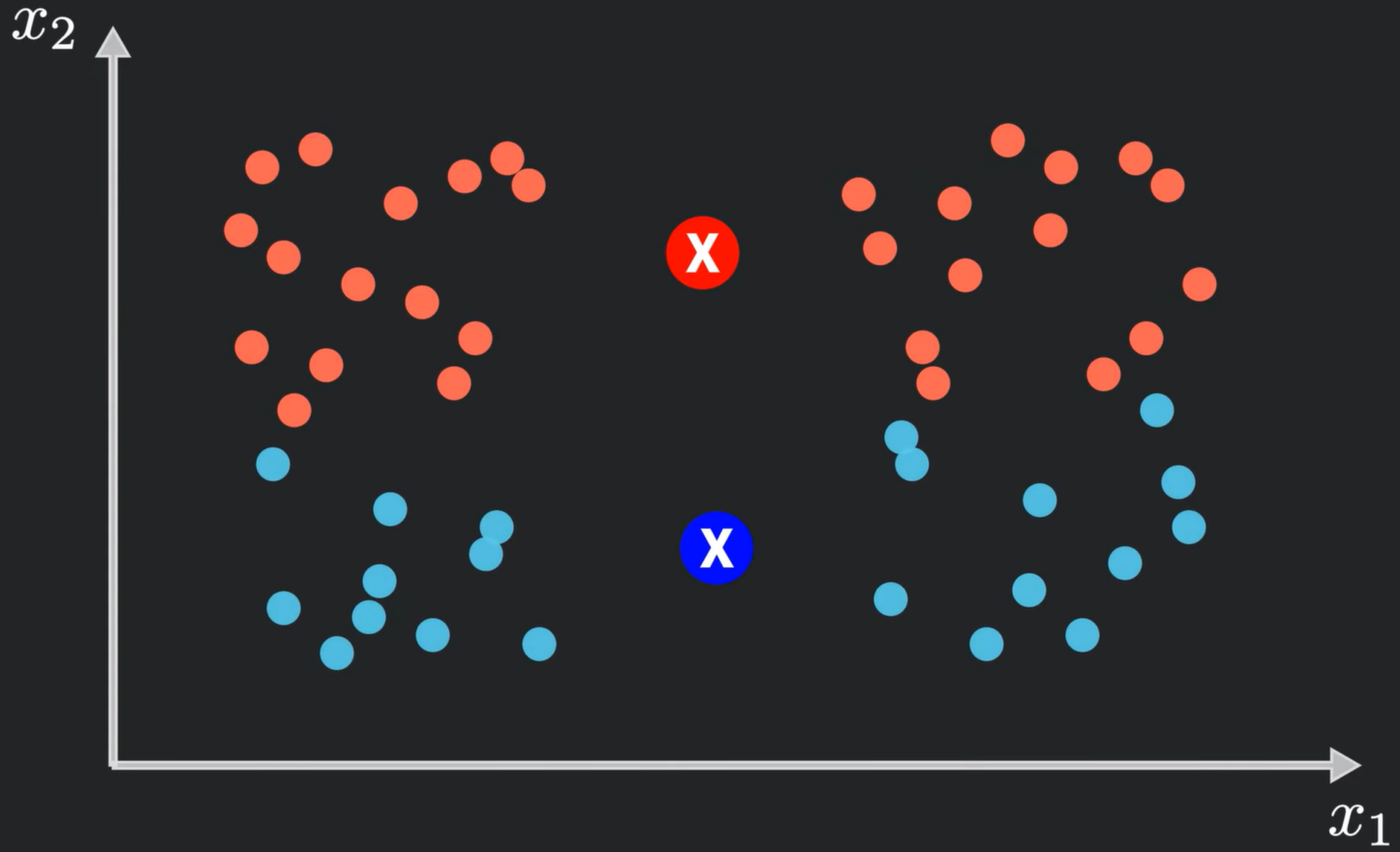

x_2
X
X
x_1

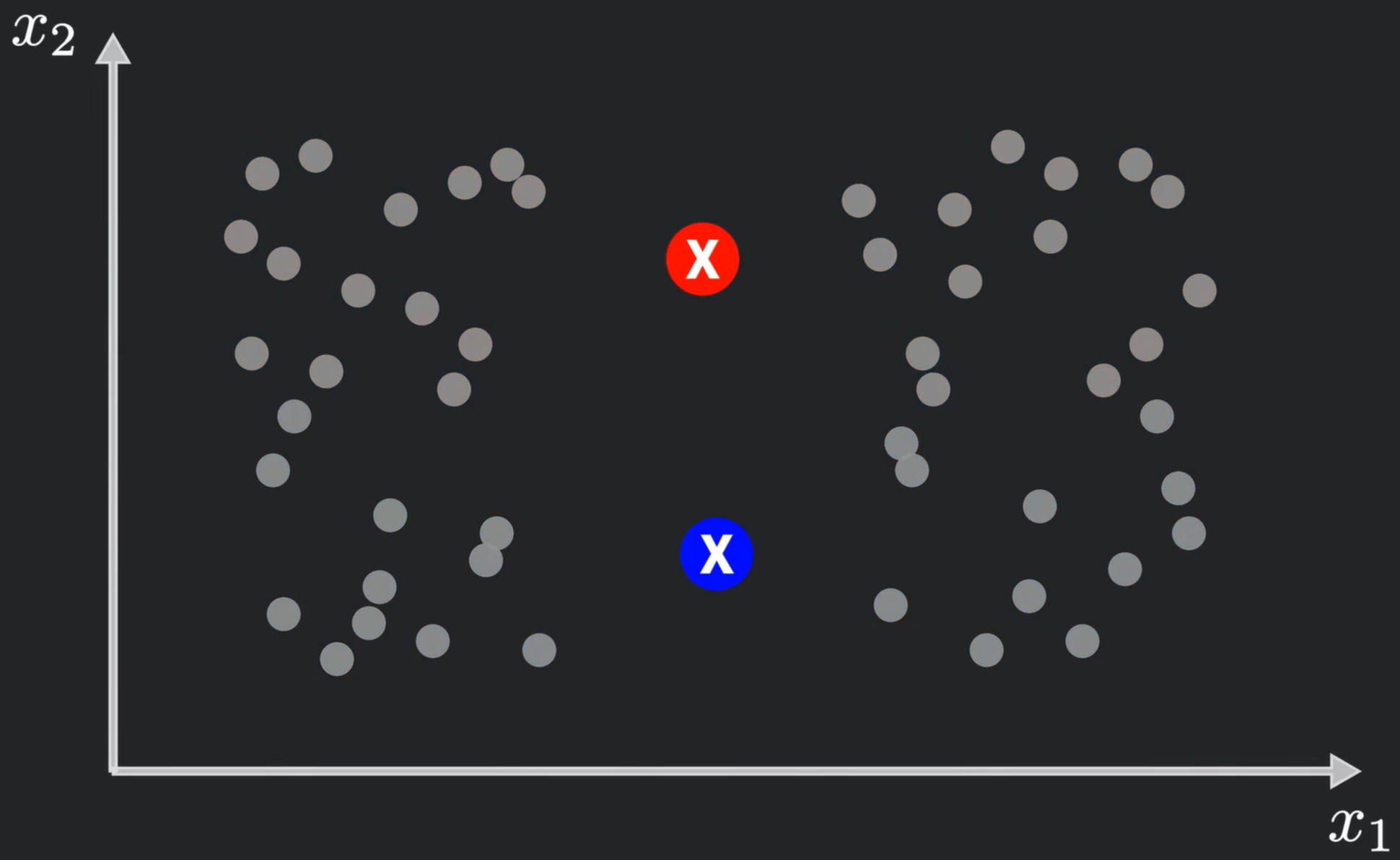
Measure how close each point is to the centroids and assign it to the nearest one



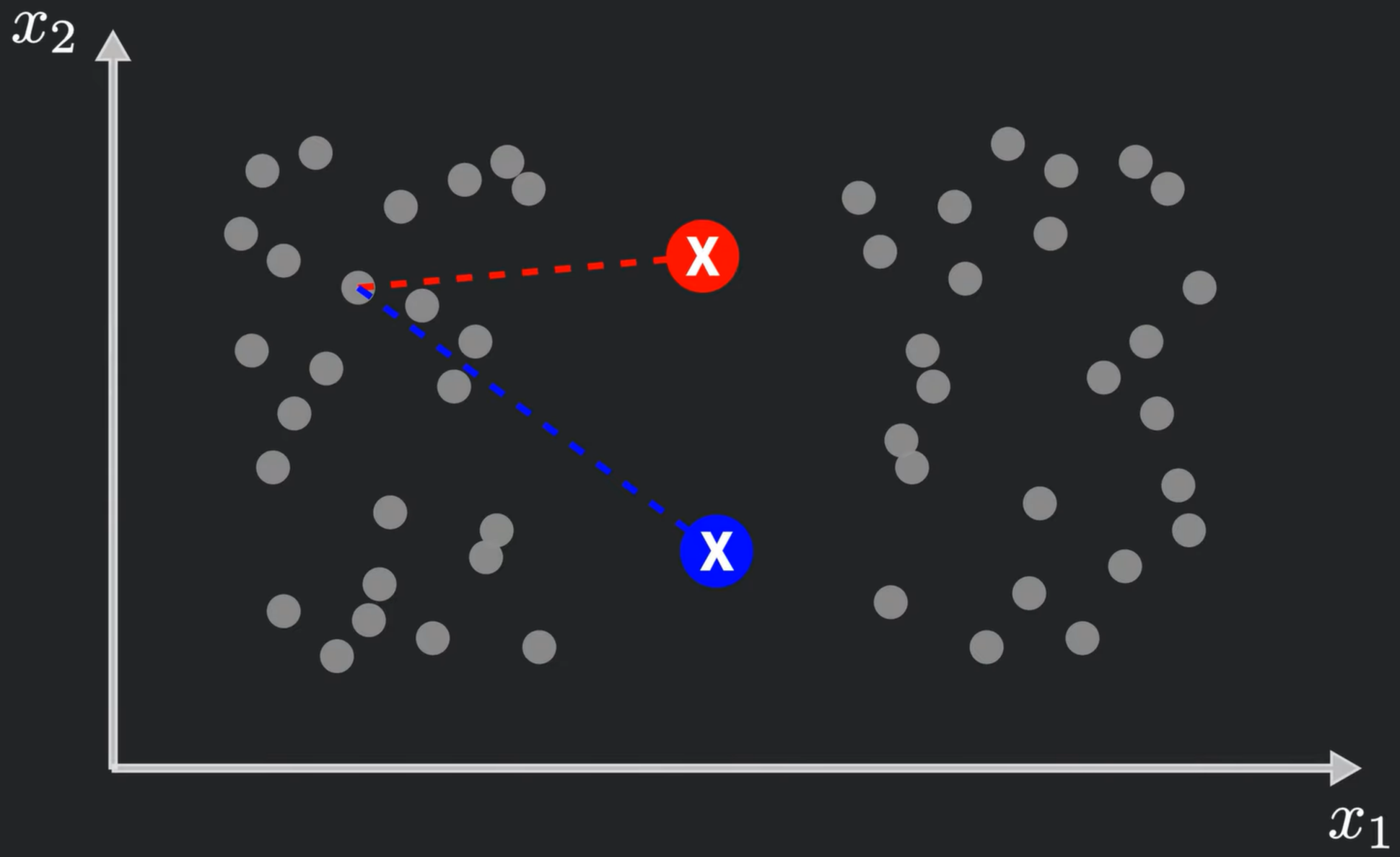
Where x1, x2, ..., xn are the X coordinates, y1, y2, ..., yn are the Y coordinates, and n is the number of points in each cluster.

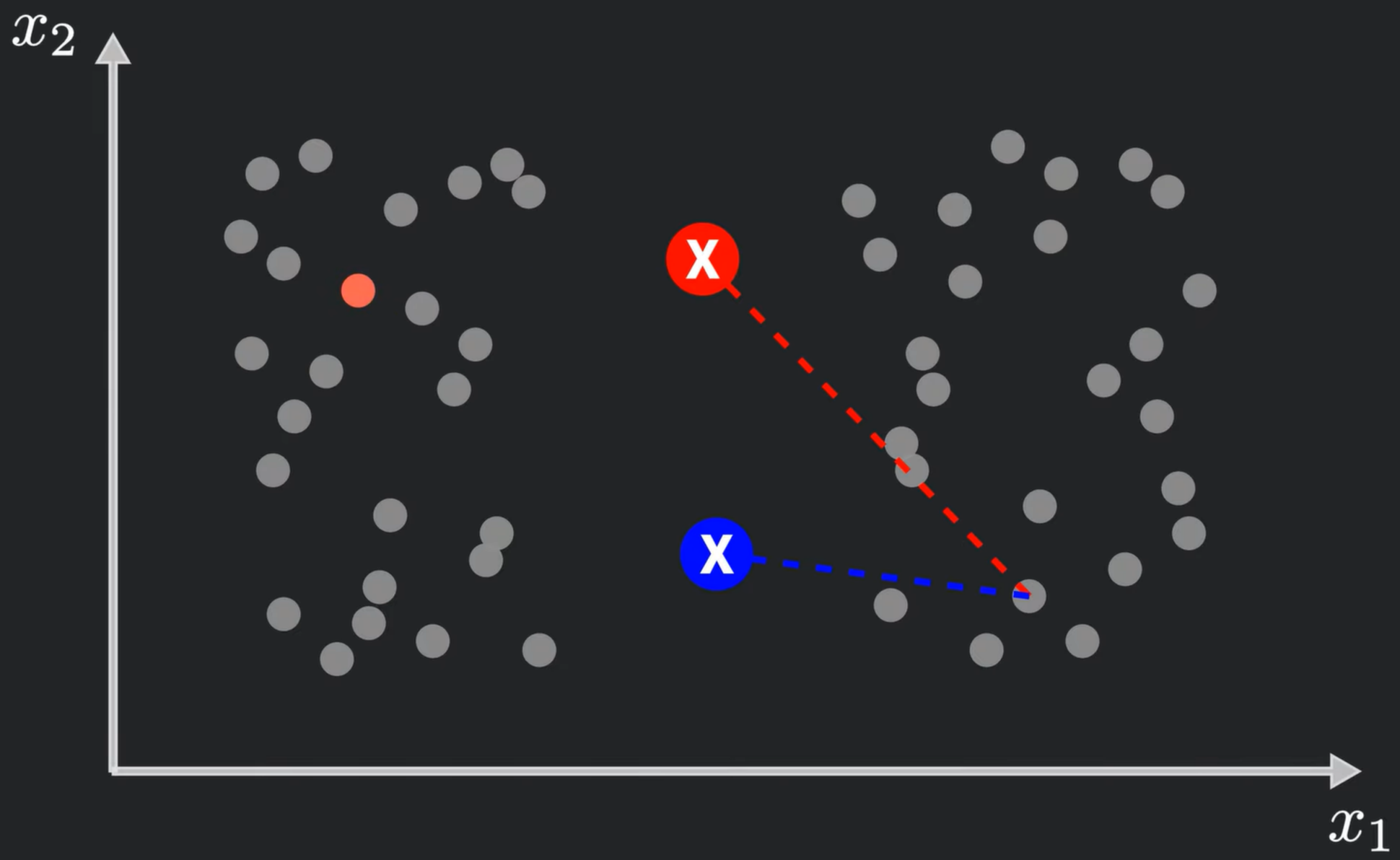
Update the centroids

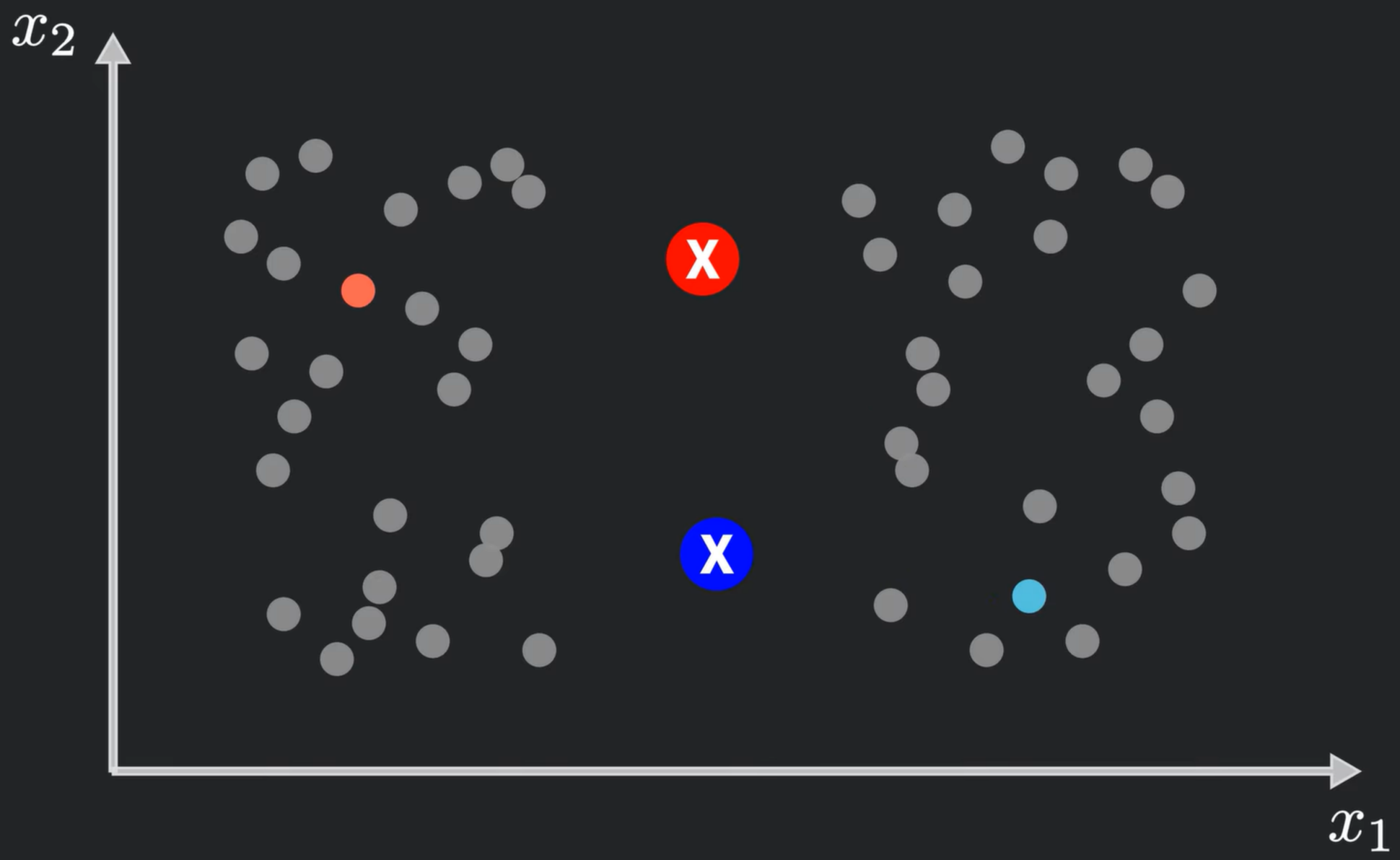


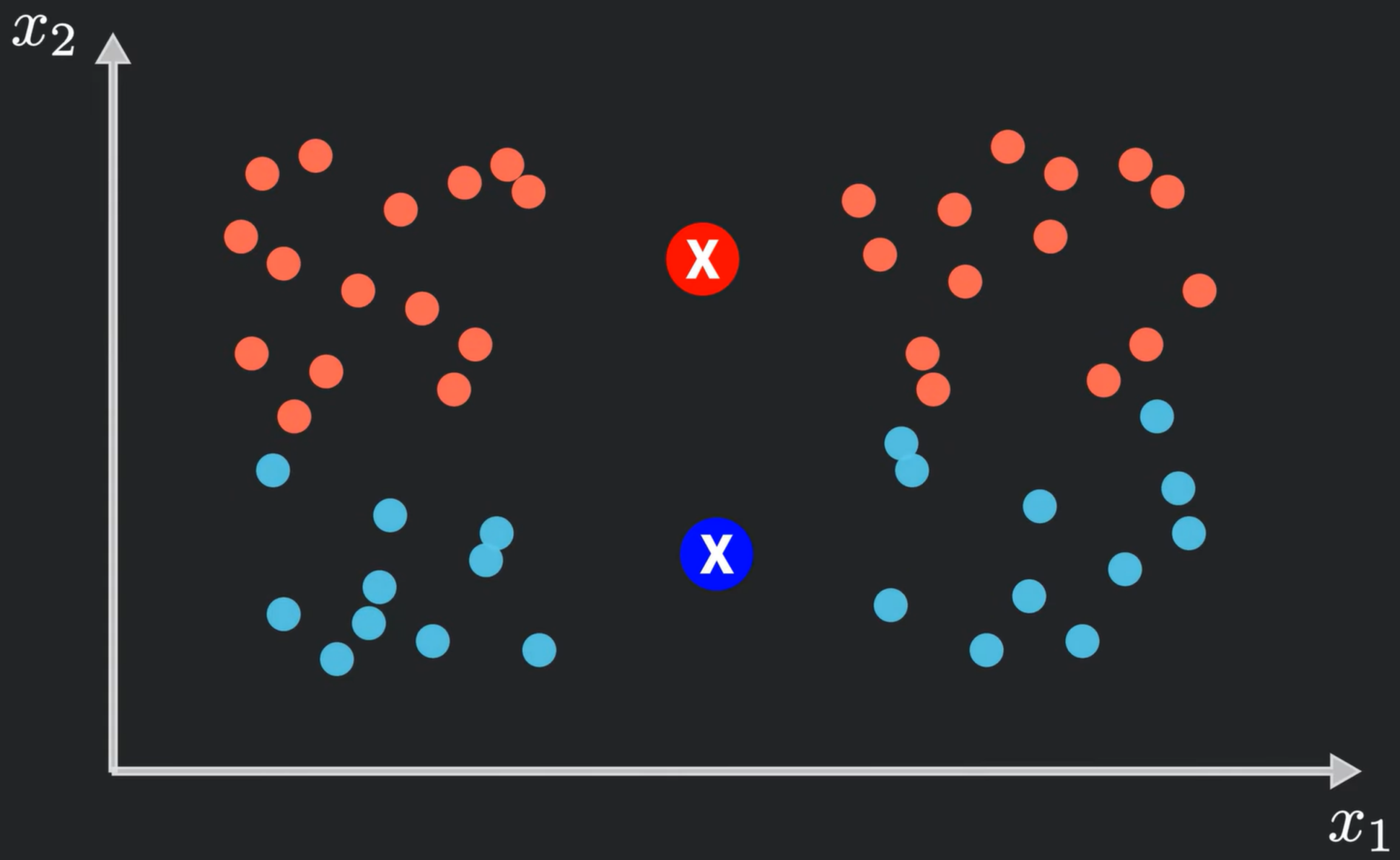

x_2
X
X
x_1

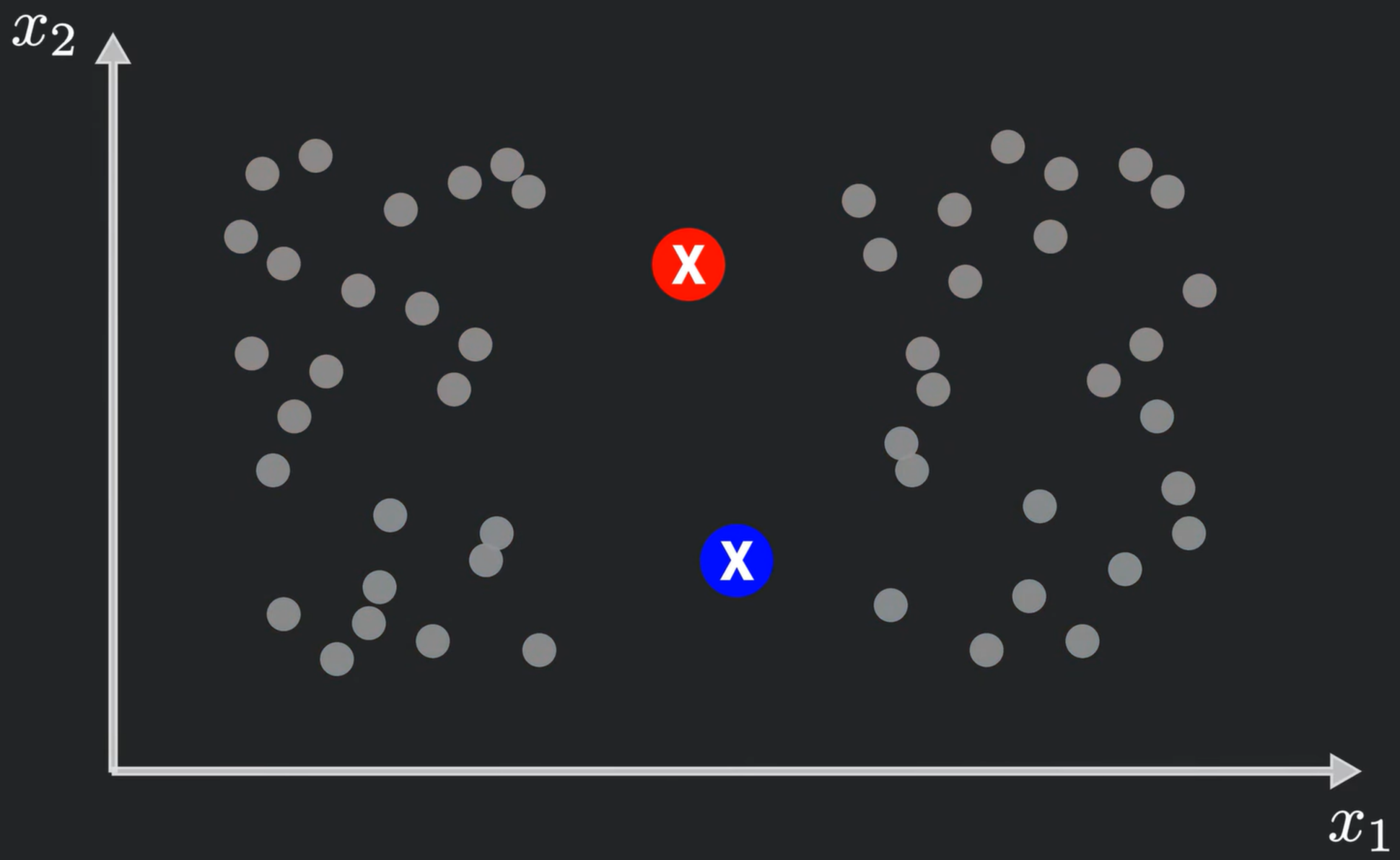
Assign each data point to the nearest centroid again, and keep repeating the algorithm

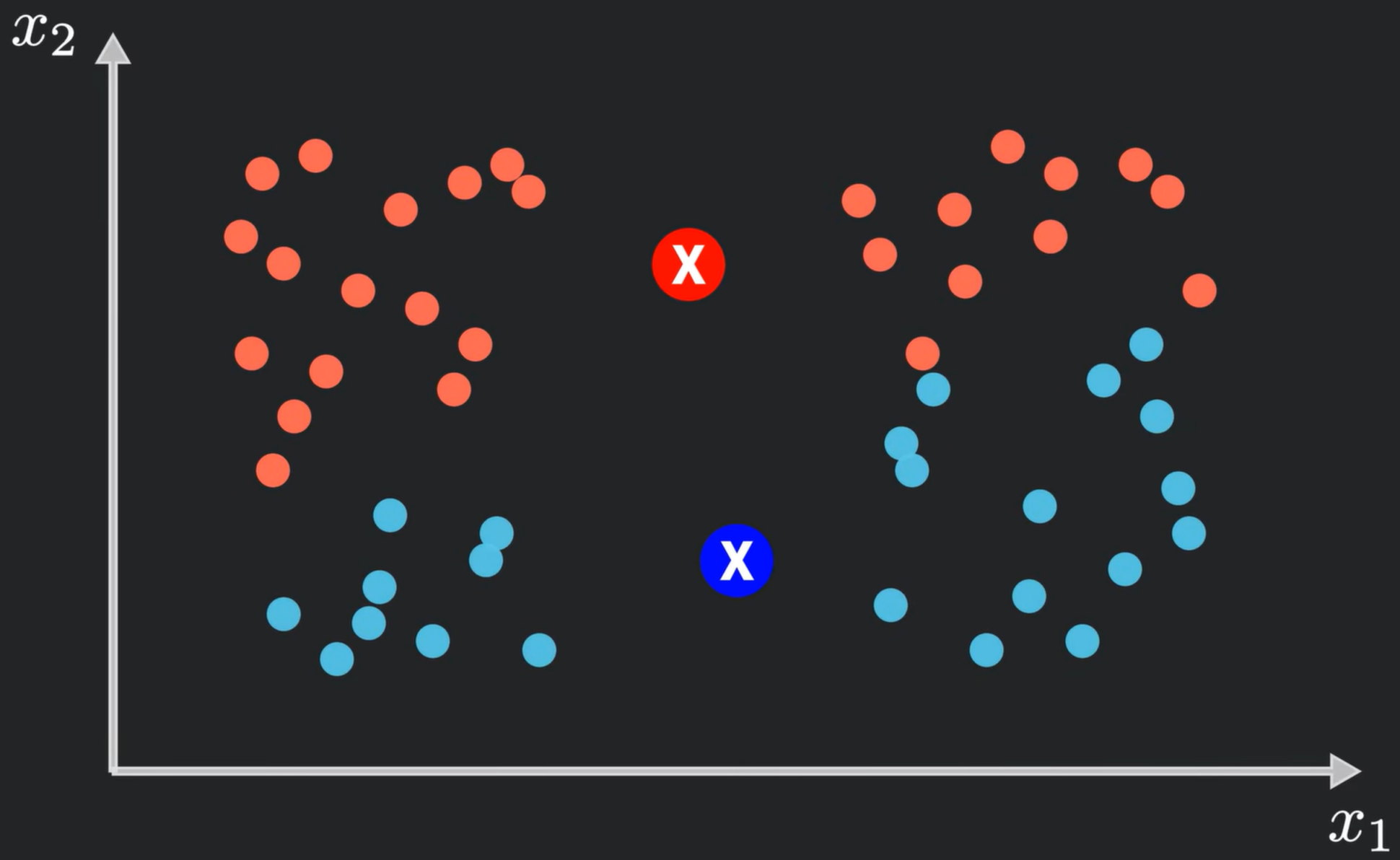


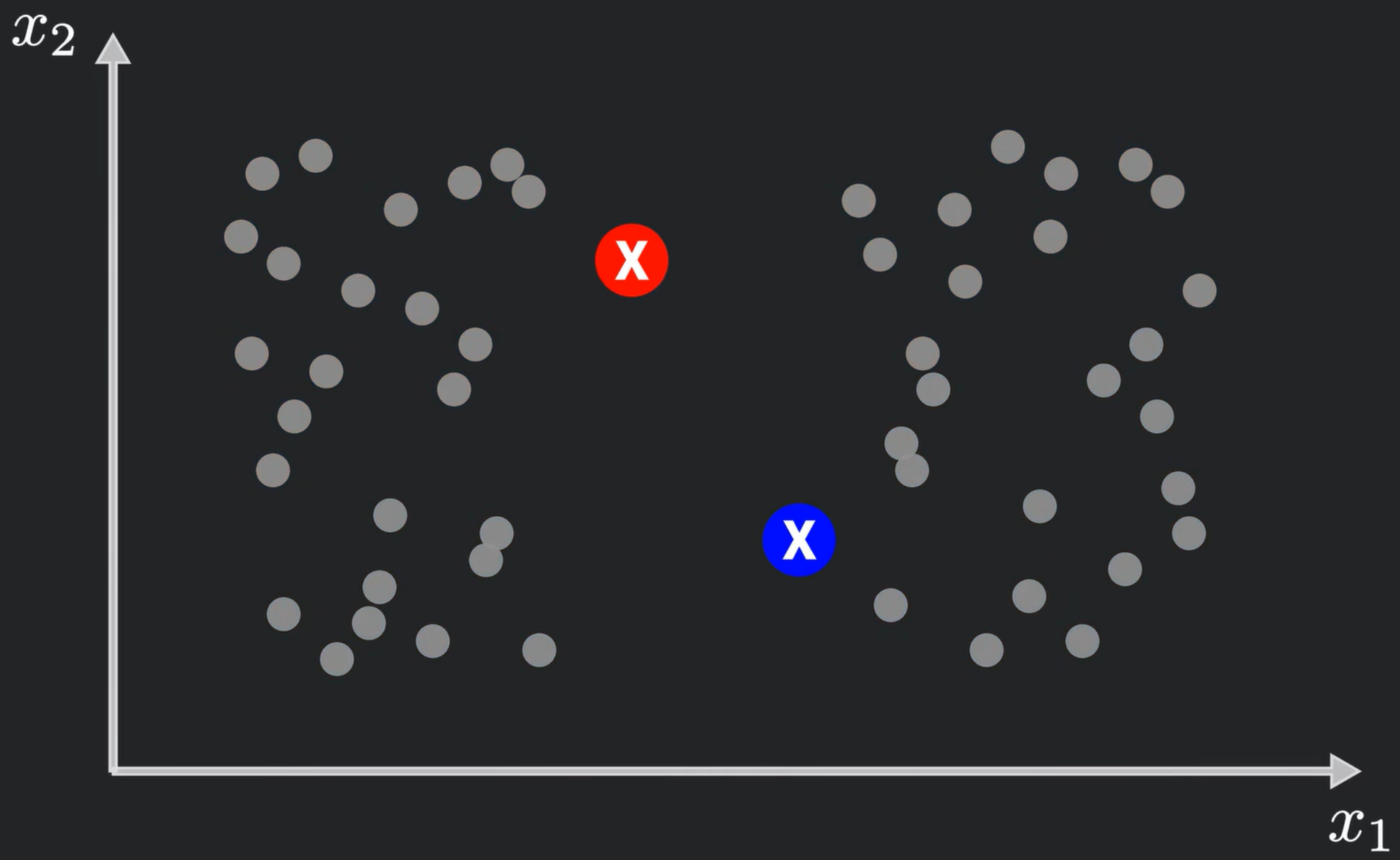

x_2
X
X
x_1

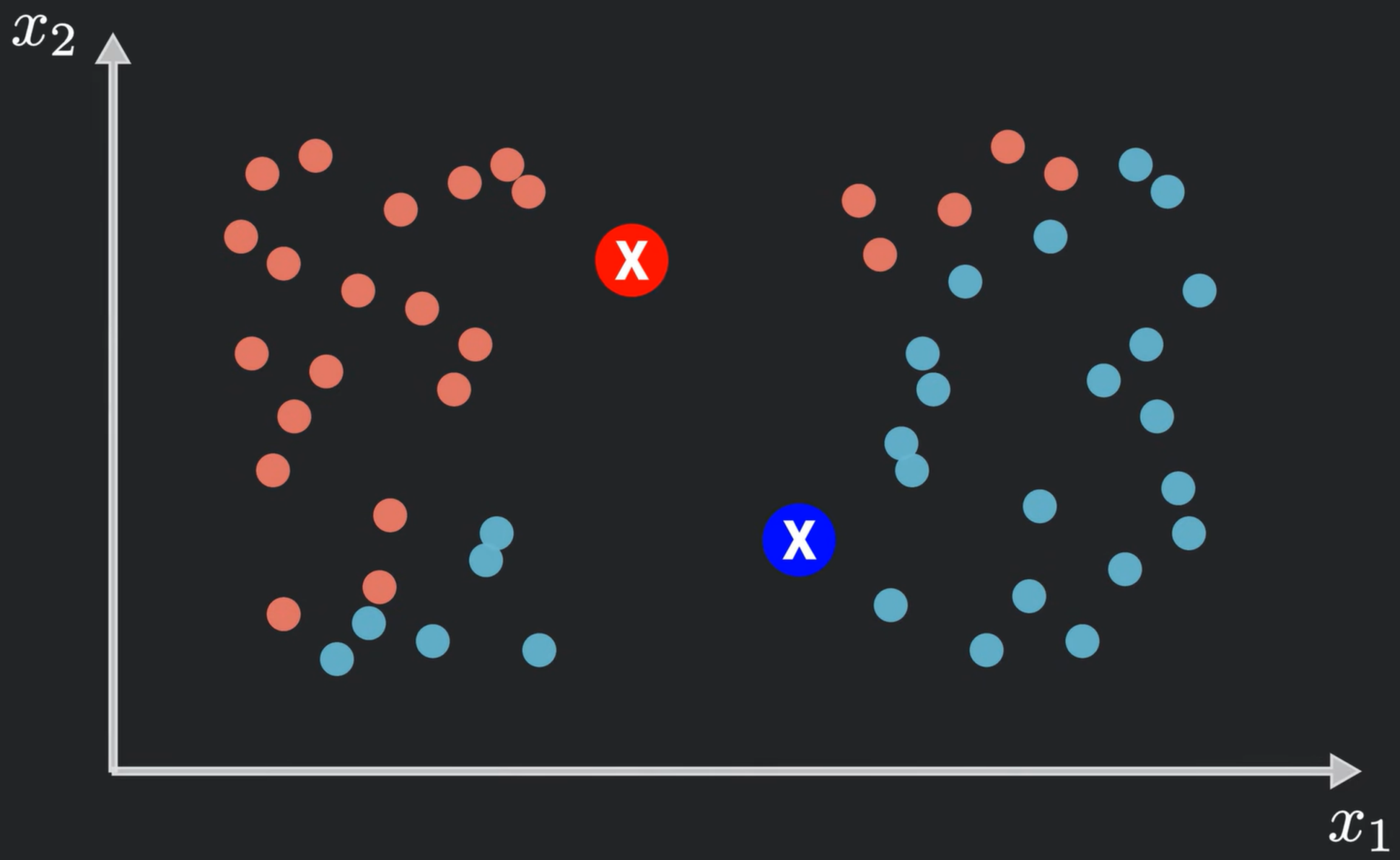

x_2
X
X
x_1

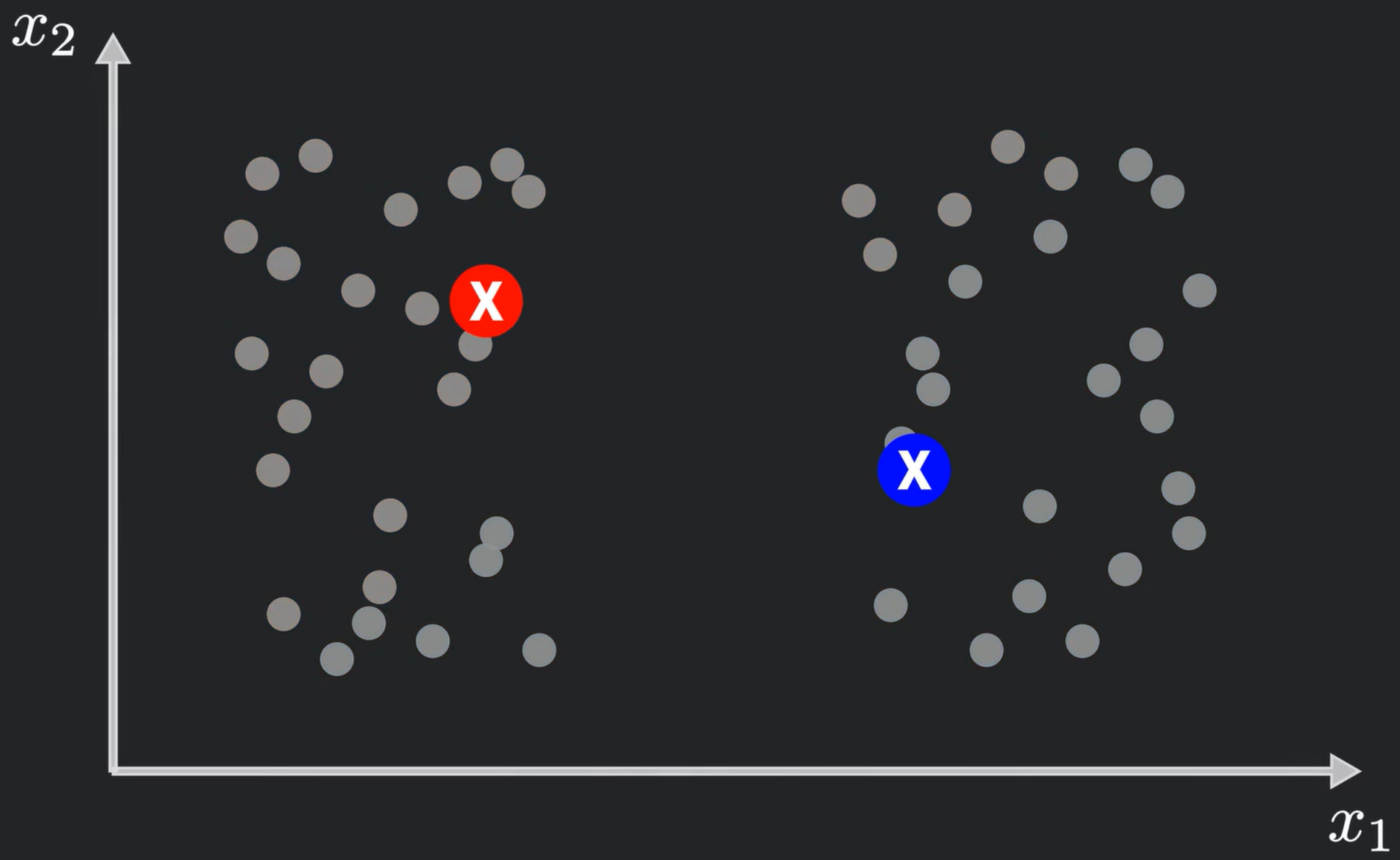

x_2
X
X
x_1

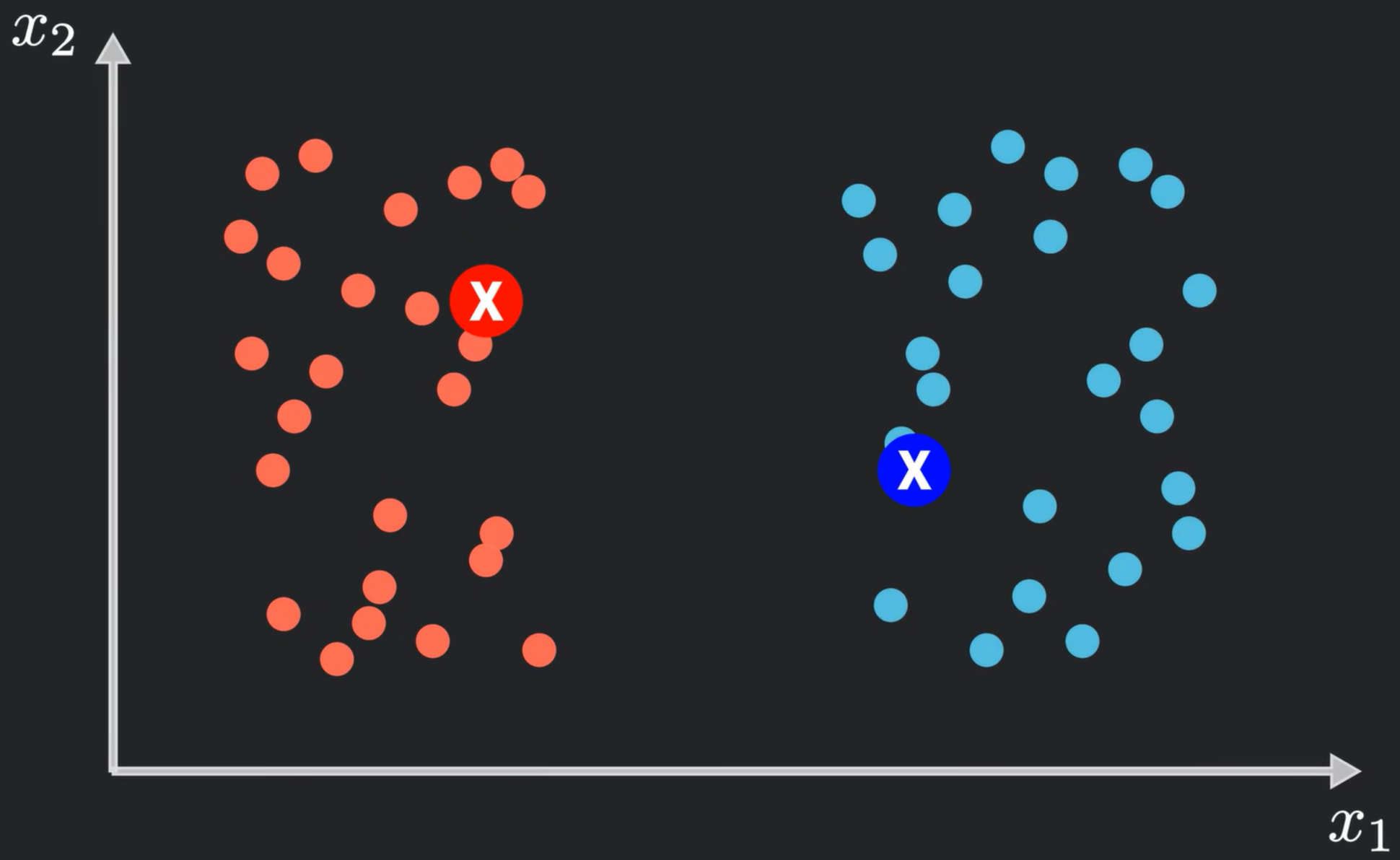

x_2
X
X
x_1

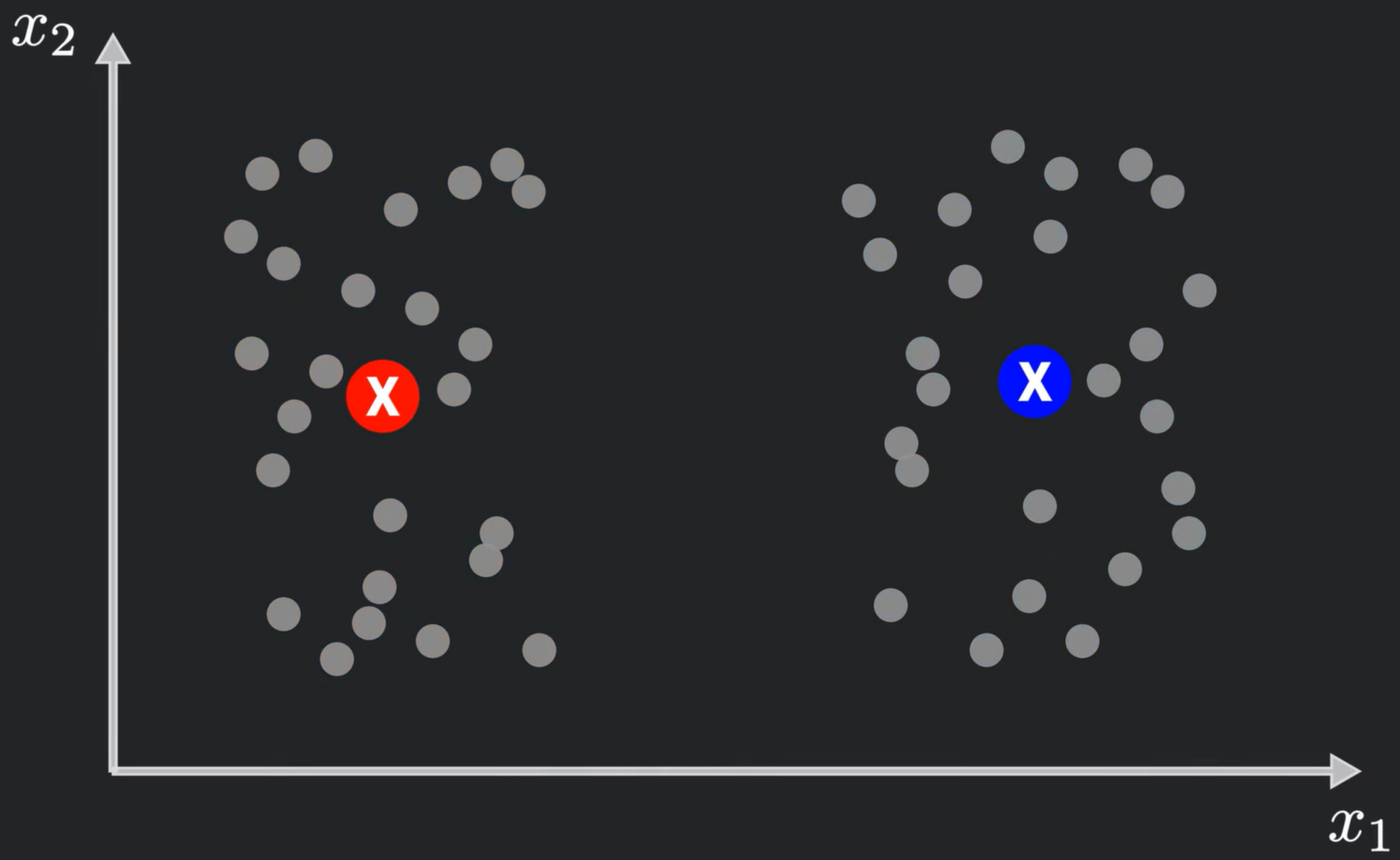

x_2
X
X
x_1


x_2
X
X
x_1

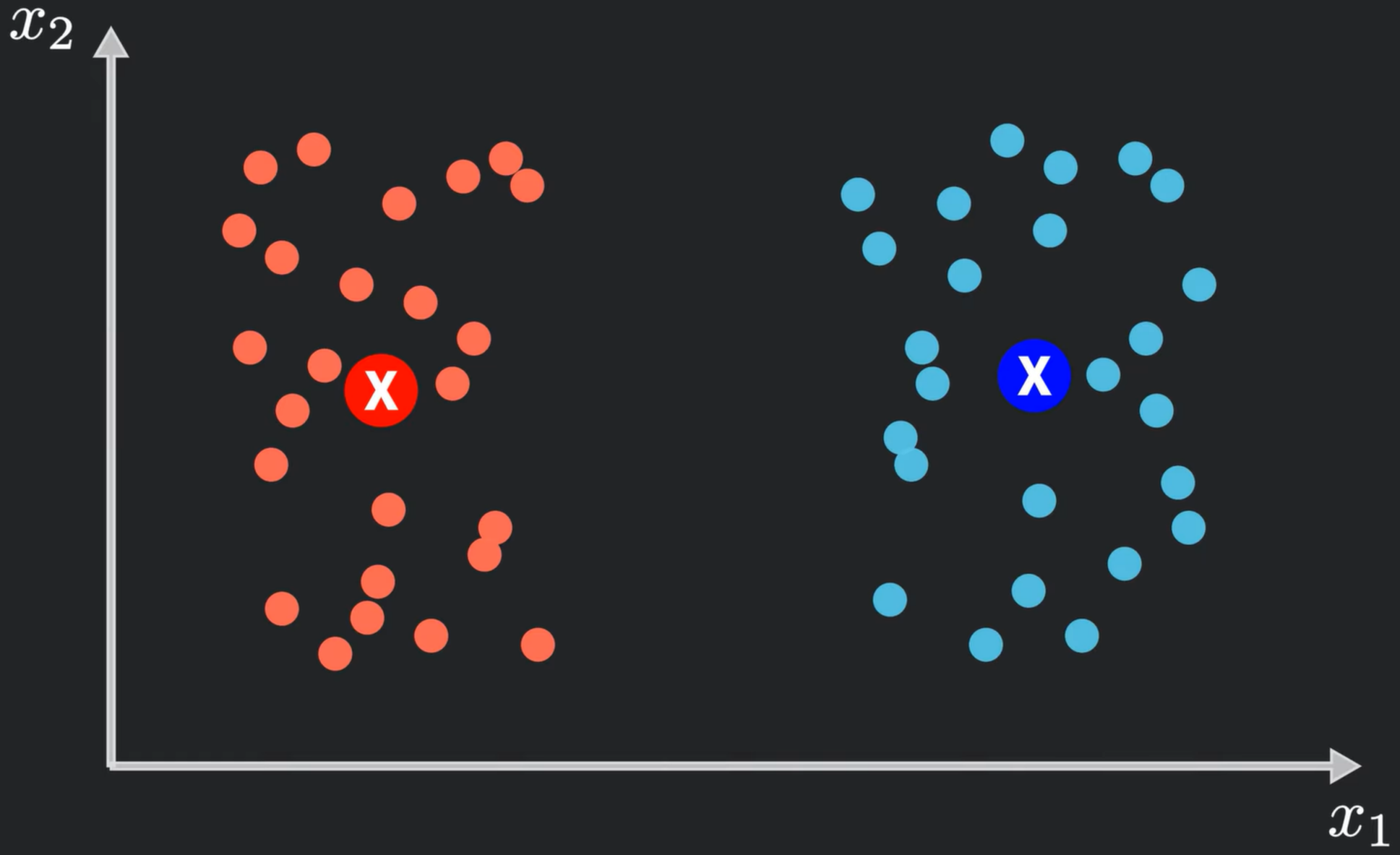

x_2
X
X
x_1

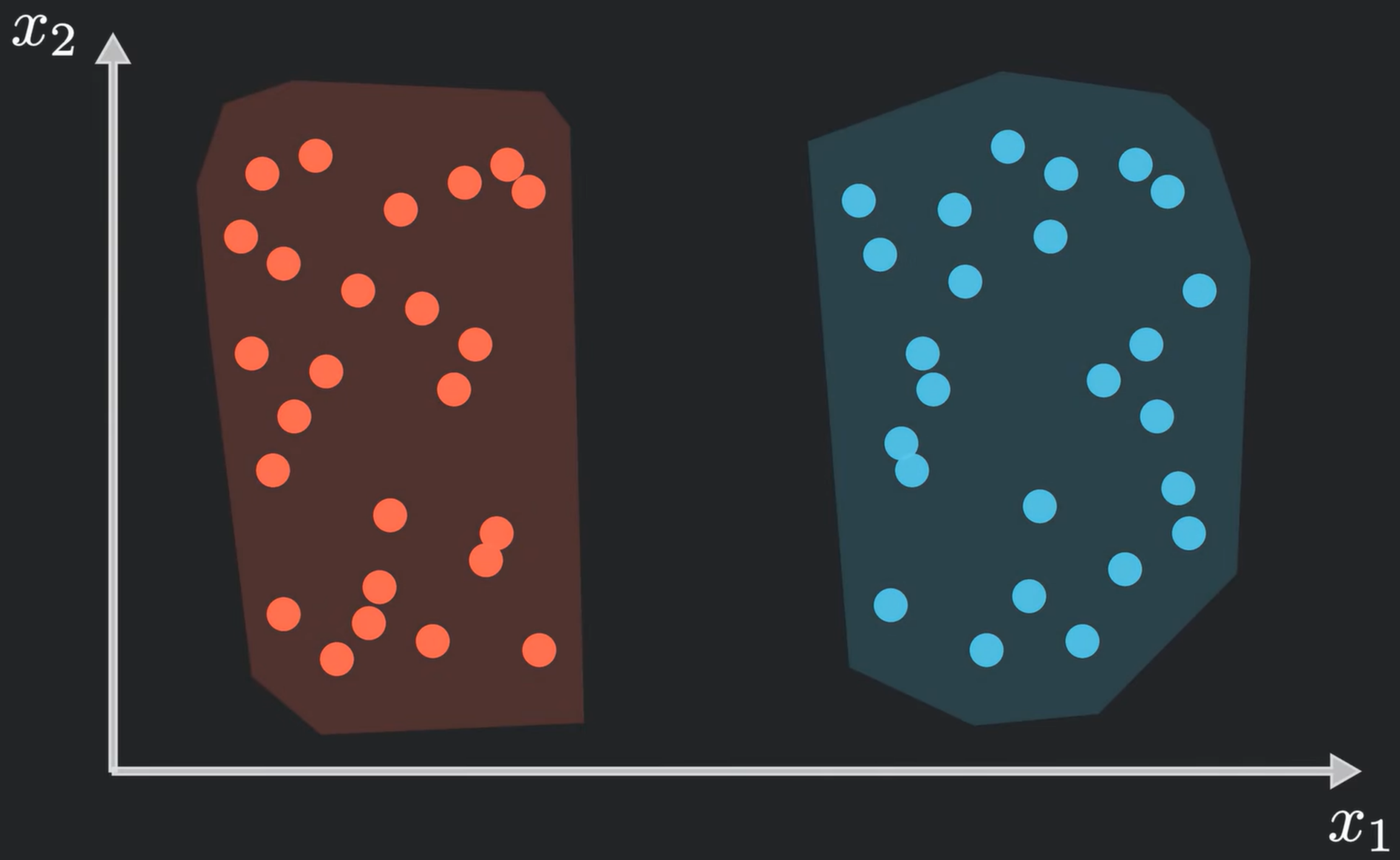

x_2
X
X
x_1


x_2
X
X
x_1


x_2
x_1

As we can see, the centroids are not moving anymore, they have converged, and we have finally two clusters




x_2
x_1

Mathematics and the cost function

$$X = \{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(n)}\},\ x^{(i)} \in \mathbb{R}^d$$

This represents the entire dataset.
Each x(i) is one data point (a sample) in the dataset, and there are n data points total in a D-general dimensional space.

Think of both the data points and the centroids as vectors.

$$X = \{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(n)}\},\ x^{(i)} \in \mathbb{R}^d$$

$$\mu_1,\ \mu_2,\ \mu_3,\ \ldots,\ \mu_K$$

$$X = \{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(n)}\},\ x^{(i)} \in \mathbb{R}^d$$

$$\mu_1,\ \mu_2,\ \mu_3,\ \cdots,\ \mu_K$$

These are the centroids of the clusters.
There are K clusters, so there are K centroids.

$$X = \{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(n)}\},\ x^{(i)} \in \mathbb{R}^d$$

$$\mu_1,\ \mu_2,\ \mu_3,\ \ldots,\ \mu_K$$

$$c_i = \arg \min_{k \in \{1,\ldots,K\}} \underbrace{\left\| x^{(i)} - \mu_k \right\|^2}$$

$$X = \{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(n)}\},\ x^{(i)} \in \mathbb{R}^d$$

$$\mu_1,\ \mu_2,\ \mu_3,\ \ldots,\ \mu_K$$

$$c_i = \arg \min_{k \in \{1,\ldots,K\}} \underbrace{\left\| x^{(i)} - \mu_k \right\|^2}$$

Instead of using the raw distance, we square the distance. We do this mainly for mathematical convenience. Squaring the distance does not change which centroid is the closest, but it makes the calculations easier because it removes square roots from the formula, which simplifies the math and the optimization process.

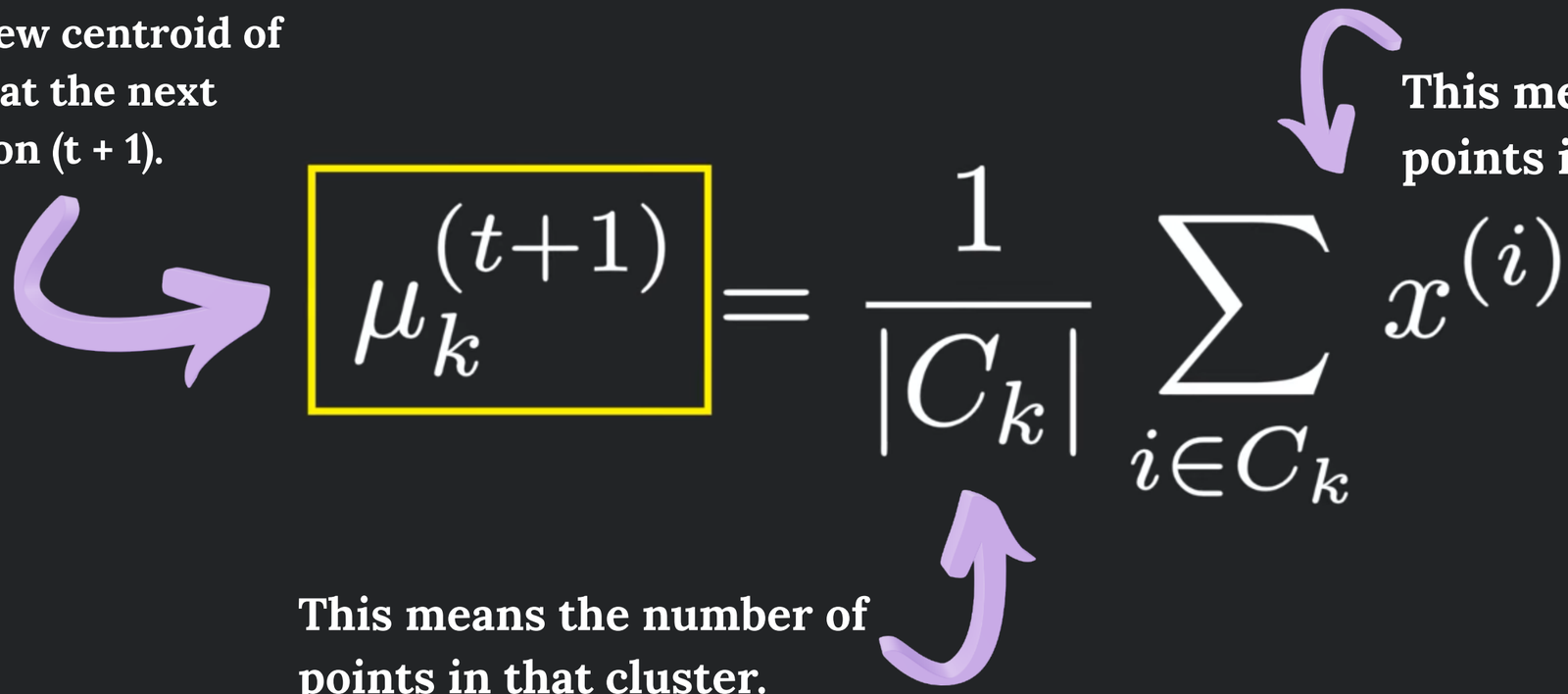
After computing the squared distances to all K centroids, we choose the smallest one and assign the data point to that cluster.

$$\boxed{\mu_k^{(t+1)}} = \frac{1}{|C_k|} \sum_{i \in C_k} x^{(i)}$$

Once the points are assigned to a cluster, the next step is to update the centroids. For each cluster, we calculate the mean position of all data points assigned to it. That average becomes the new centroid for the next iteration.

This is the new centroid of cluster k at the next iteration (t + 1).

This means sum all the points in that cluster.



This means the number of points in that cluster.

Once the points are assigned to a cluster, the next step is to update the centroids. For each cluster, we calculate the mean position of all data points assigned to it. That average becomes the new centroid for the next iteration.

This is the cost function, also called the Within-Cluster Sum of Squares.

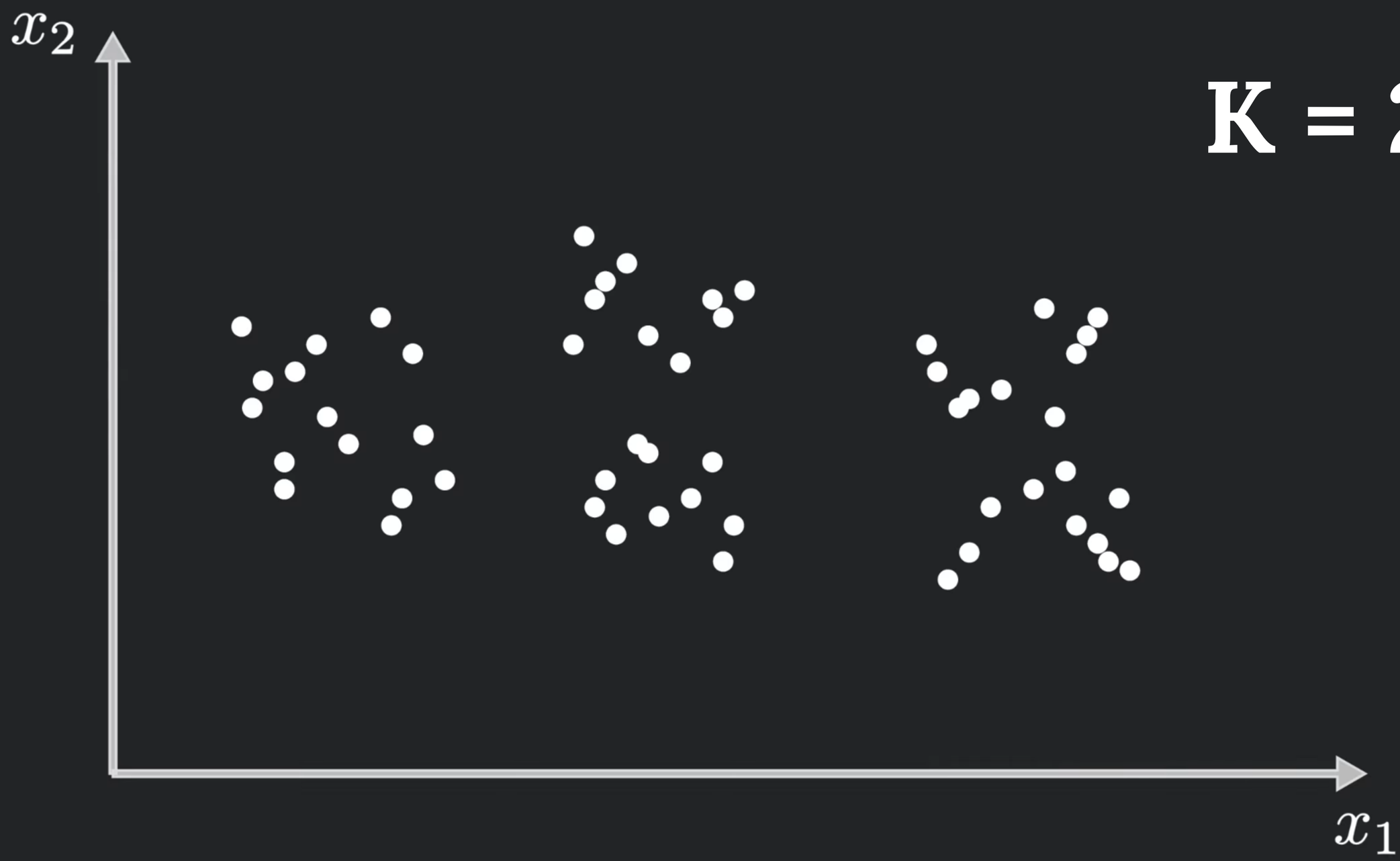
$$J(c, \mu) = \sum_{j=1}^{k} \sum_{x^{(i)} \in C_j} \|x^{(i)} - \mu_j\|^2$$

This is the cost function, also called the Within-Cluster Sum of Squares.

$$J(c, \mu) = \sum_{j=1}^{k} \sum_{x^{(i)} \in C_j} \|x^{(i)} - \mu_j\|^2$$

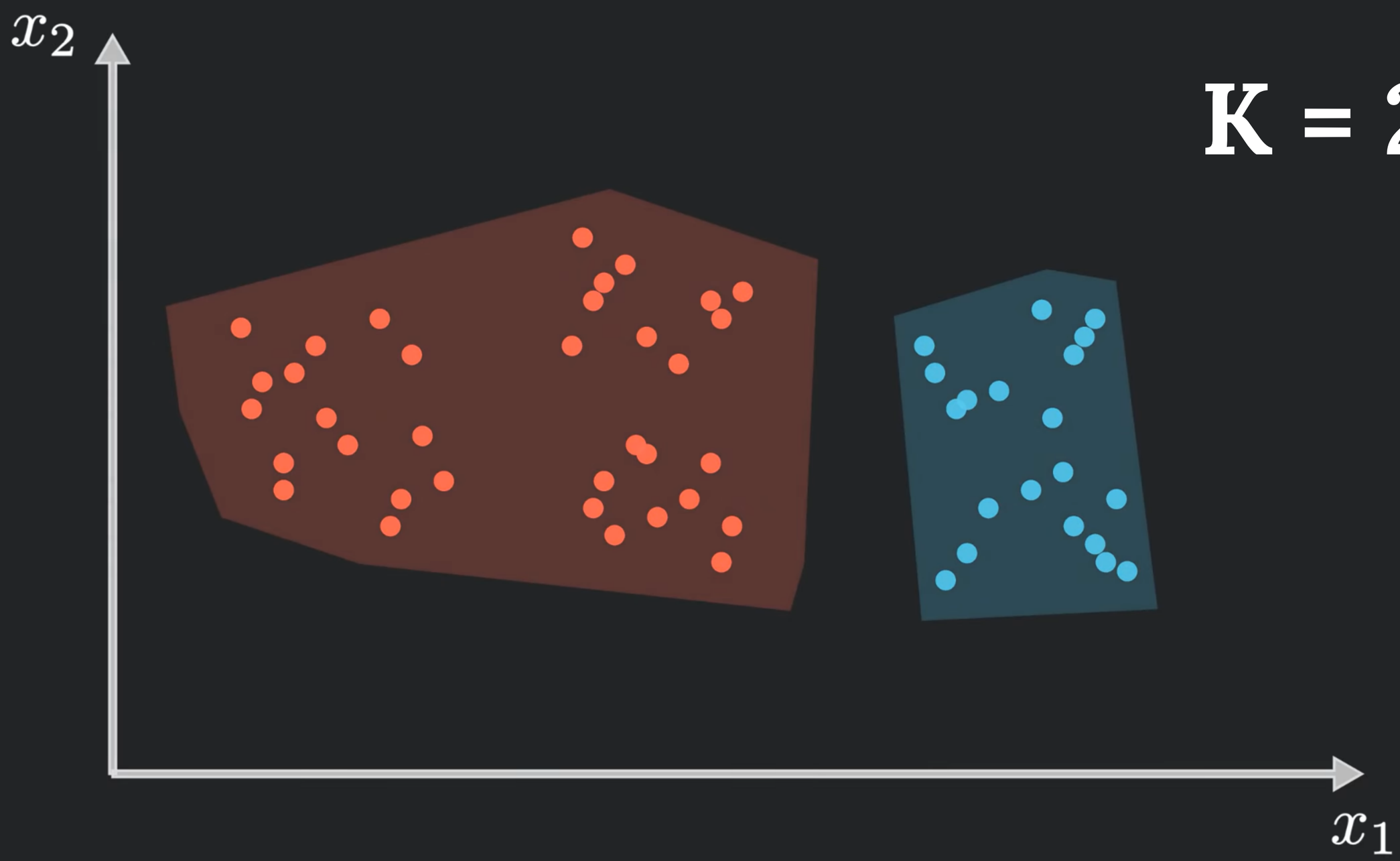
What we are actually doing in K-Means is minimizing a cost function, usually denoted as J. This function measures how far each data point is from its assigned centroid. Mathematically, it is written with a double summation: the inner summation represents the distances of the data points within one cluster, and the outer summation repeats this for all K clusters.

Now the question is: how do we decide how many clusters we actually need? For example, if we set K = 2, the data splits into two clusters. If we set K = 3, it splits further. As we increase K, the data continues to be divided into smaller and smaller groups.



K = 2


x_2
x_1
K = 2

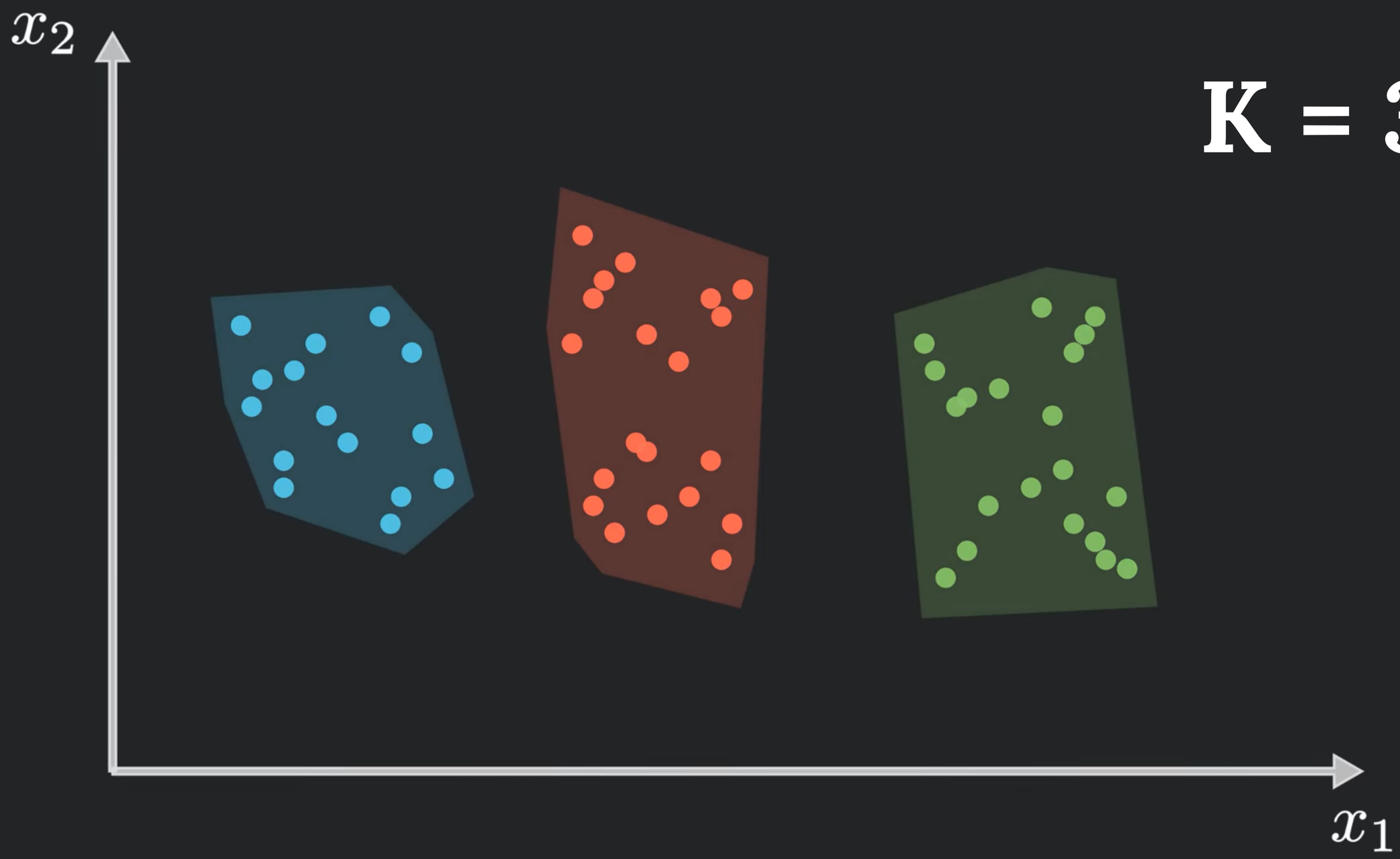


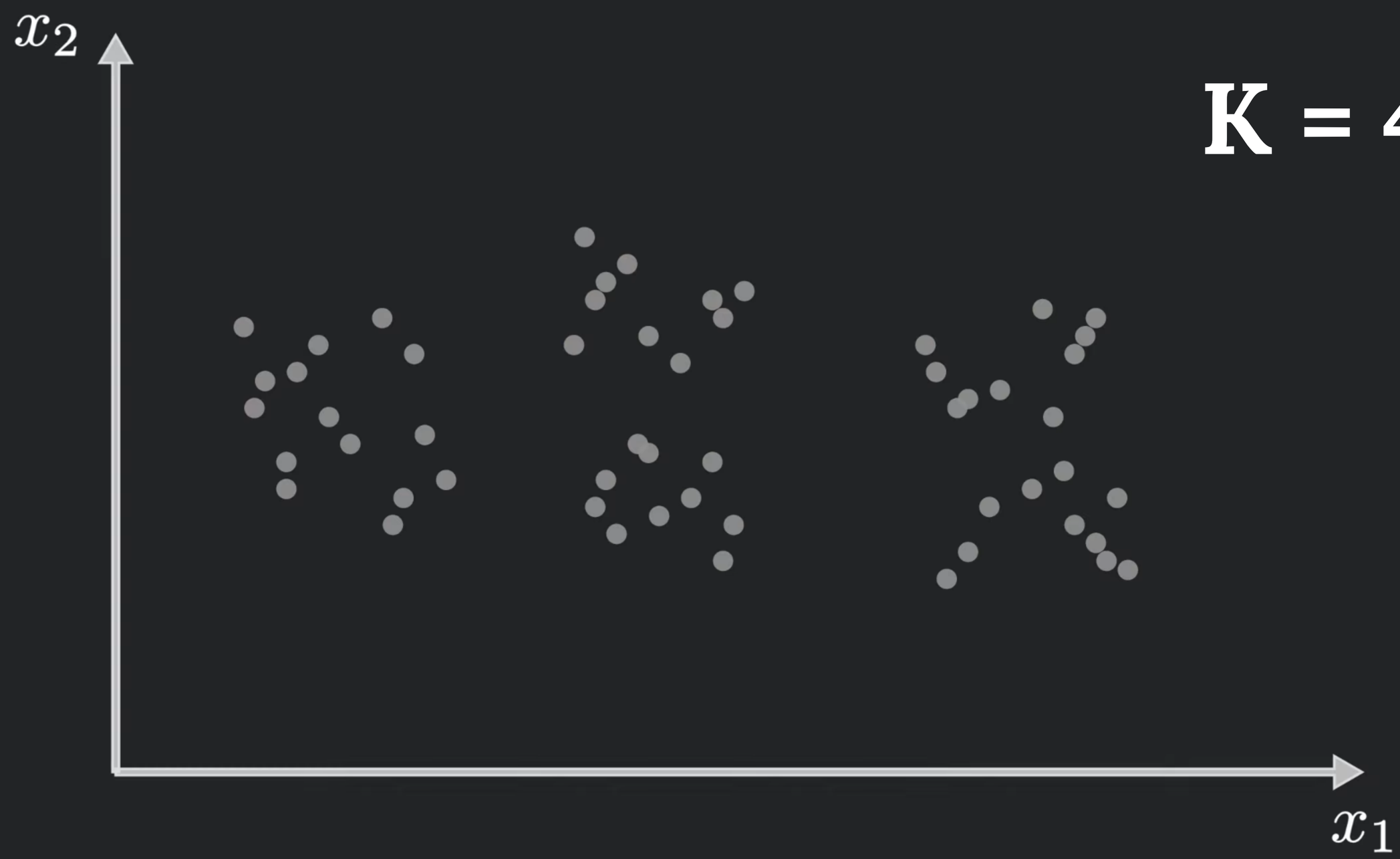
K = 2



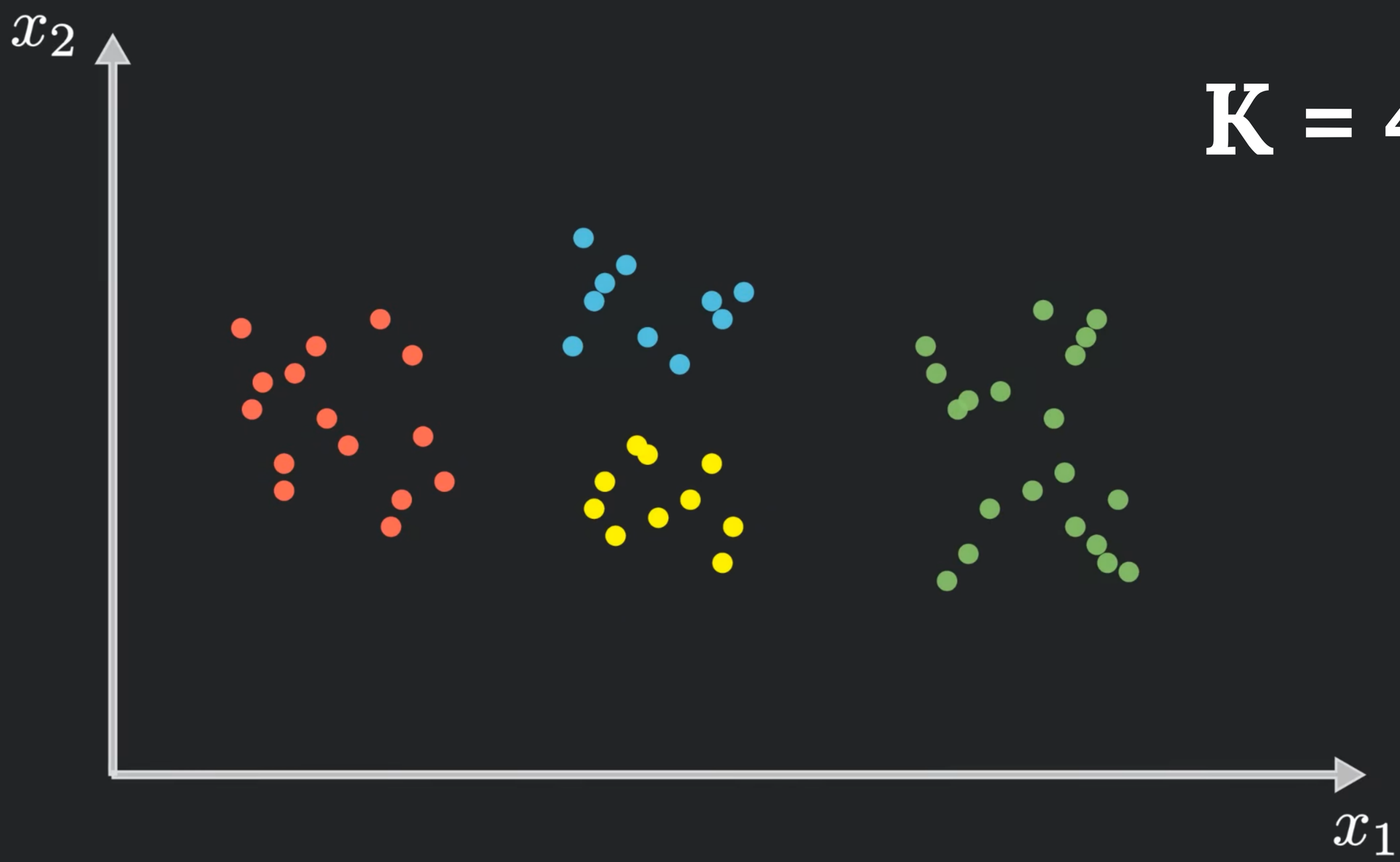
K = 3

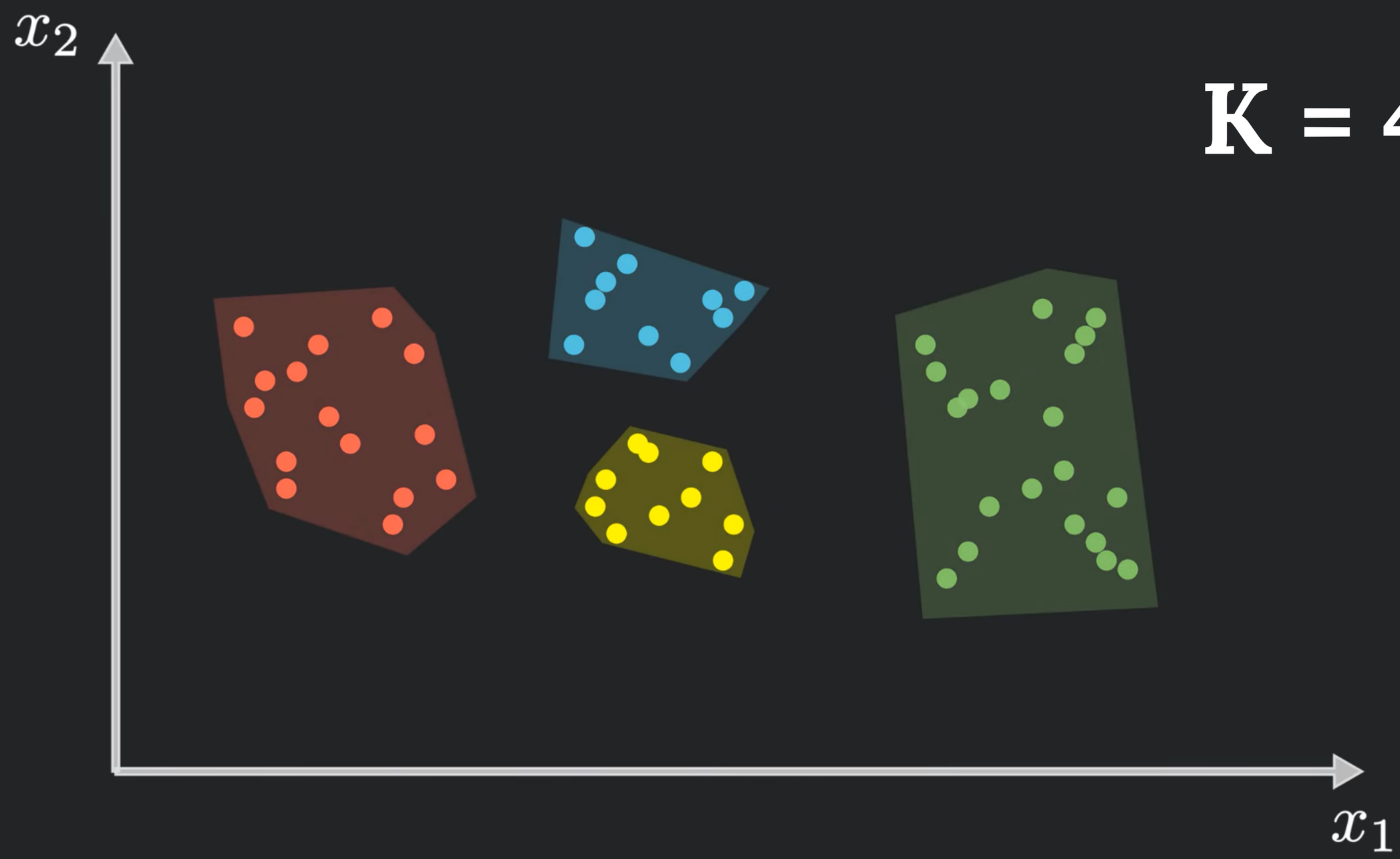

x_2
x_1
K = 3

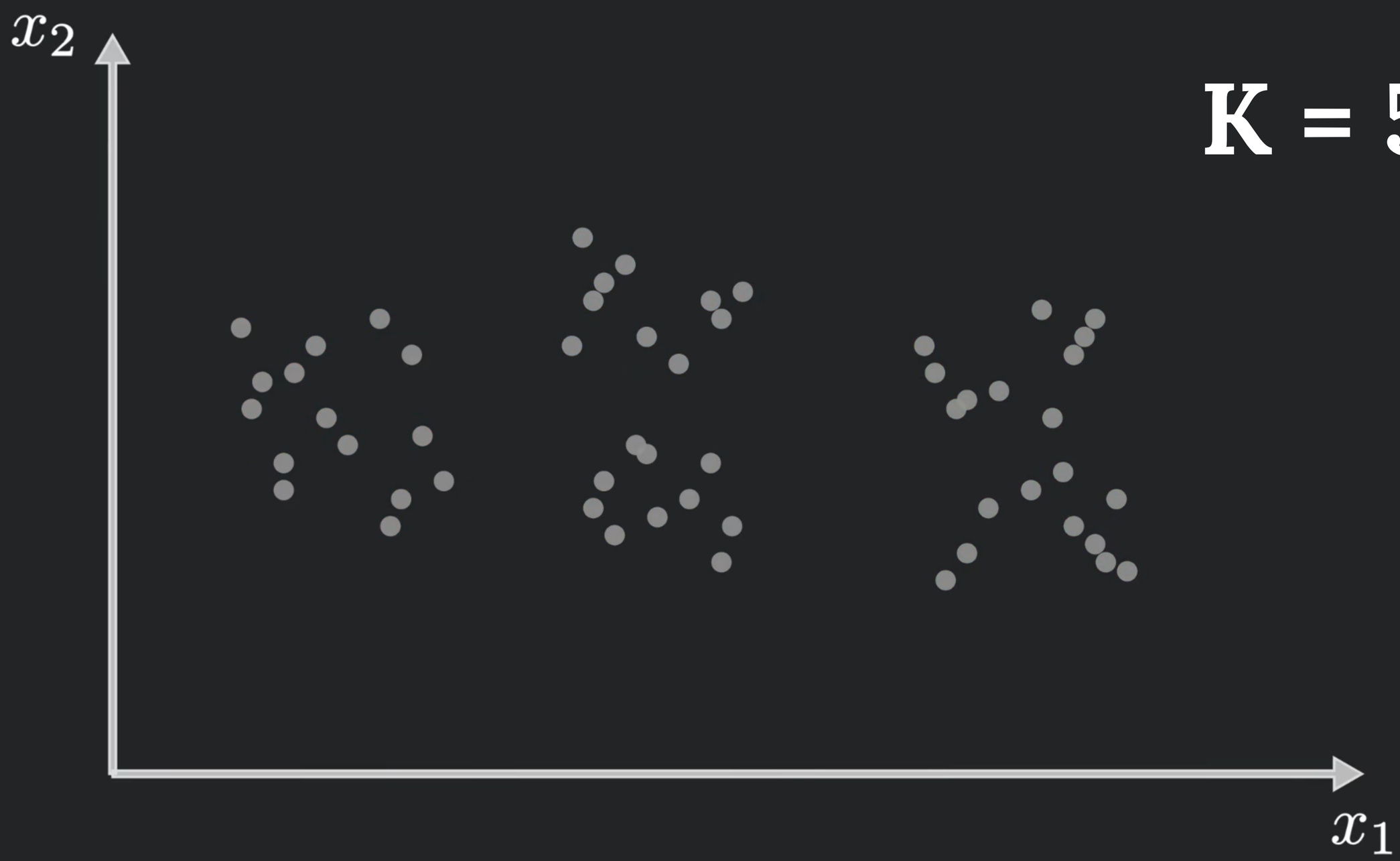

x_2
x_1
K = 3



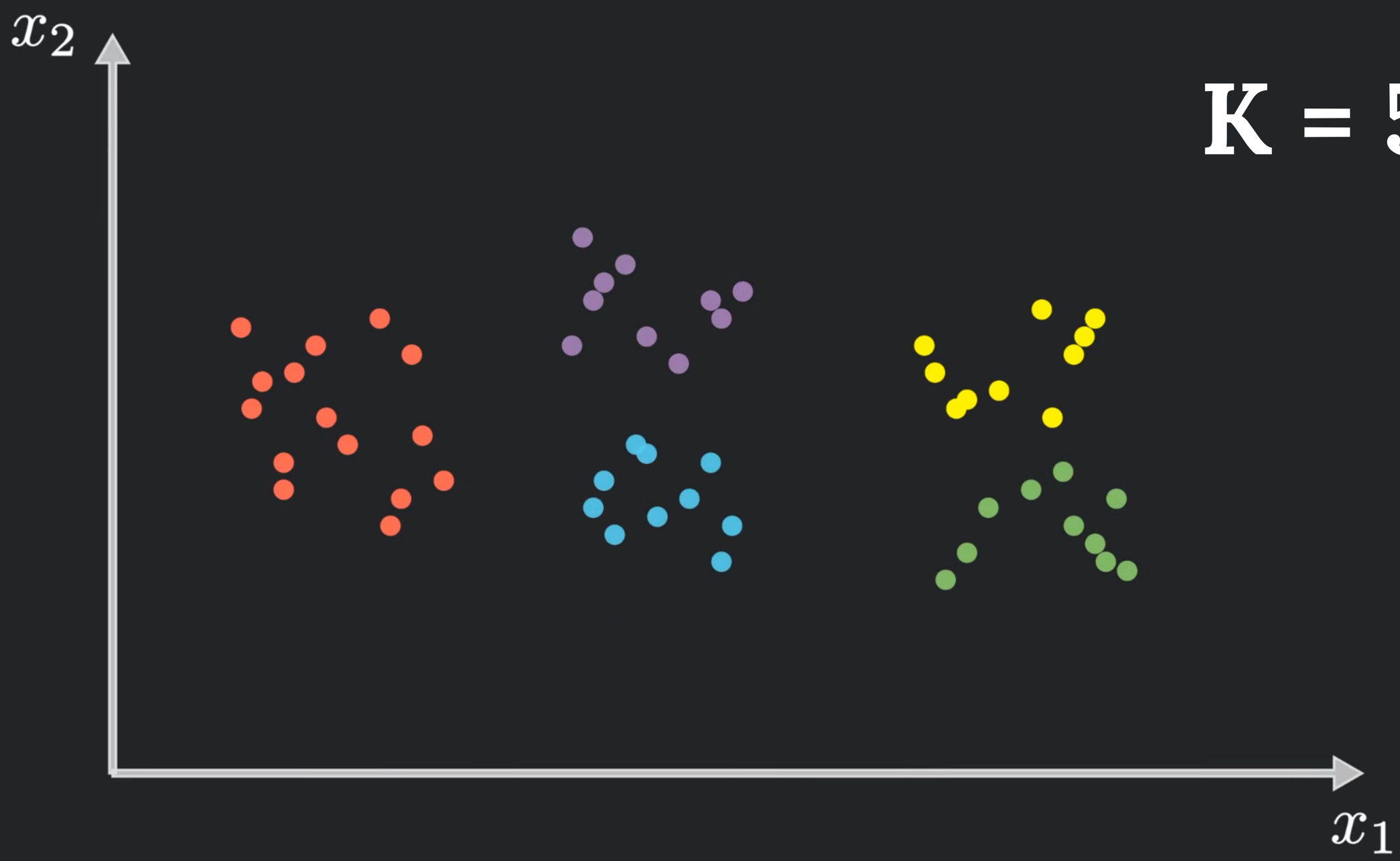
K = 4

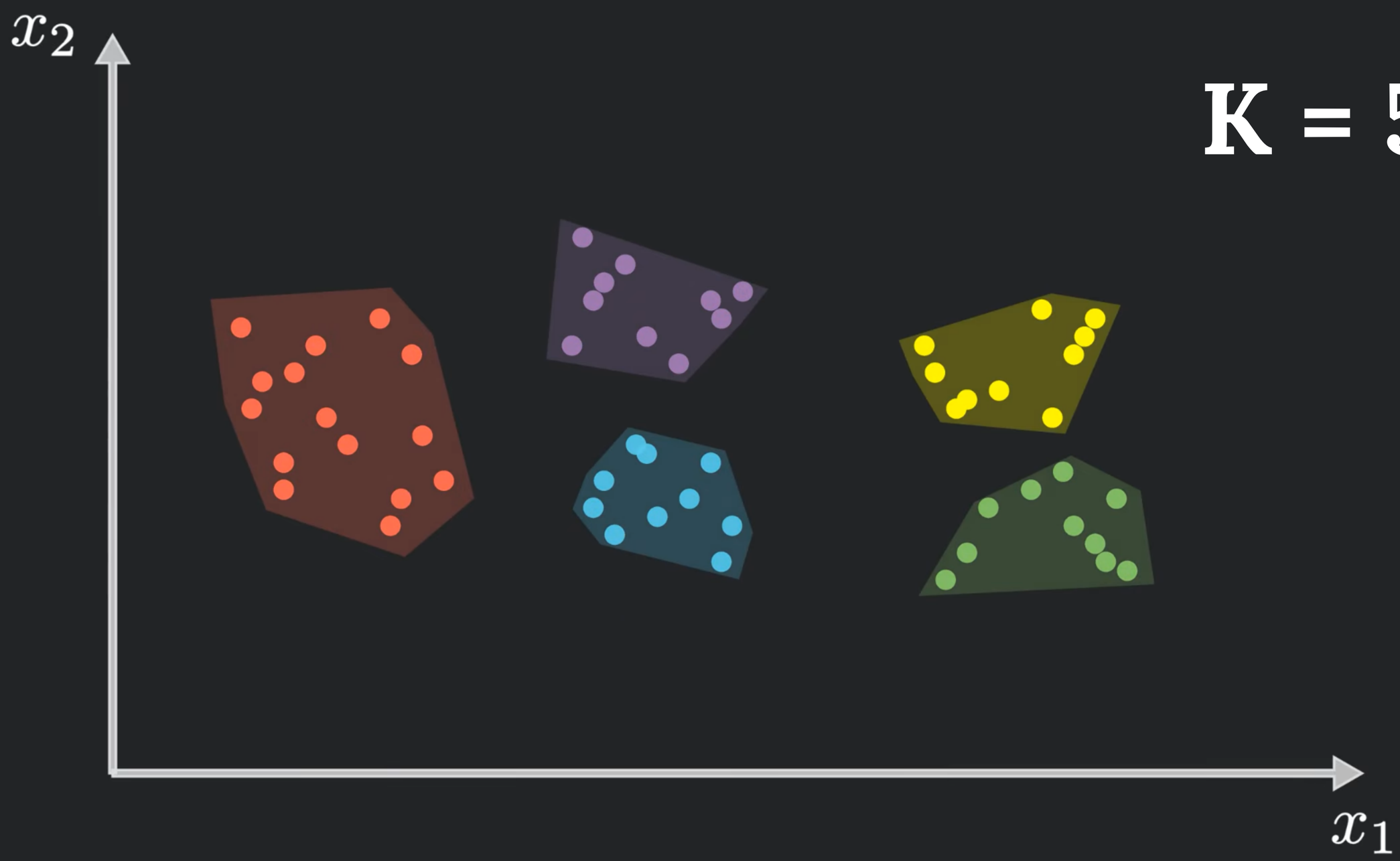

x_2
x_1
K = 4


x_2
x_1
K = 4

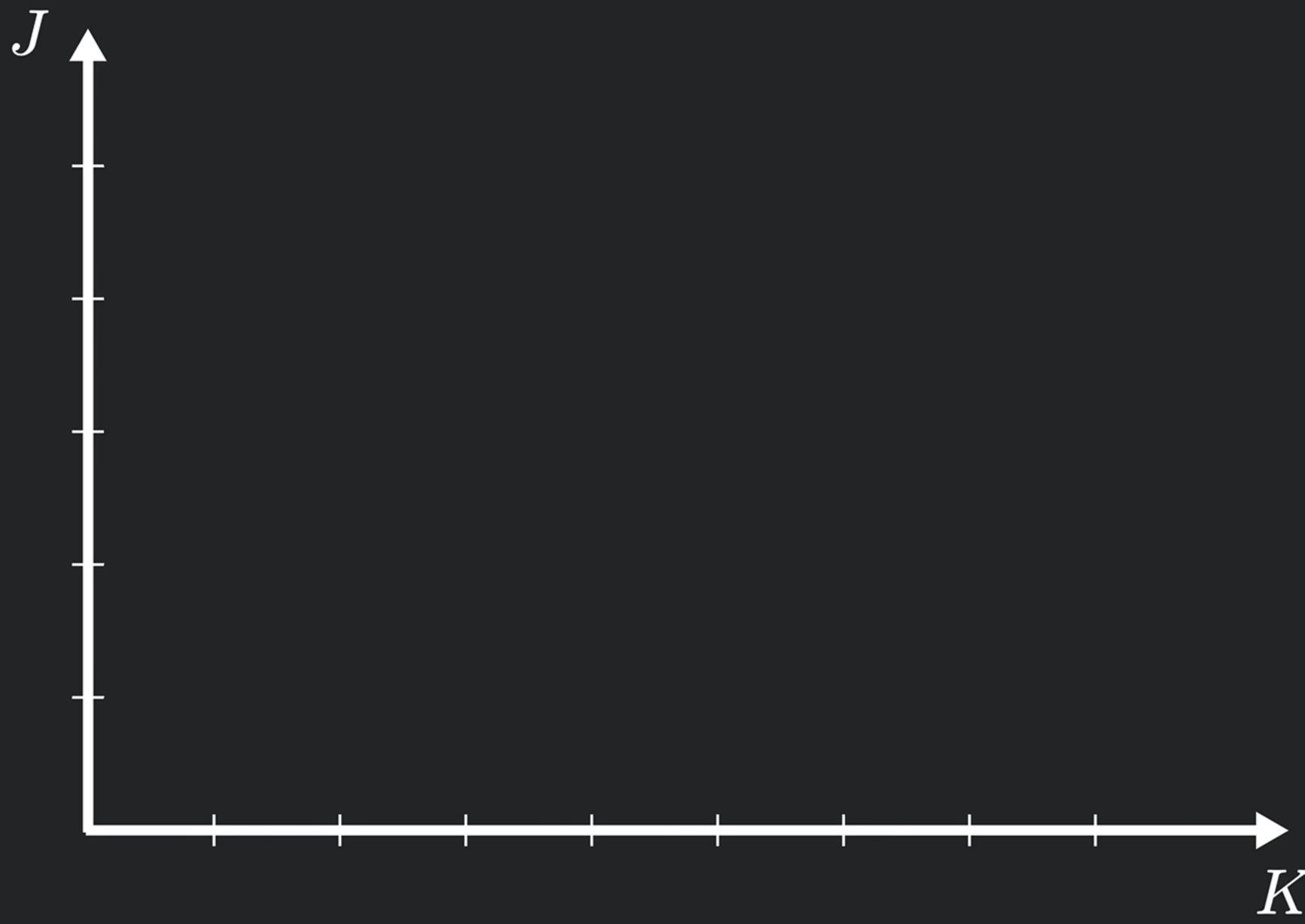


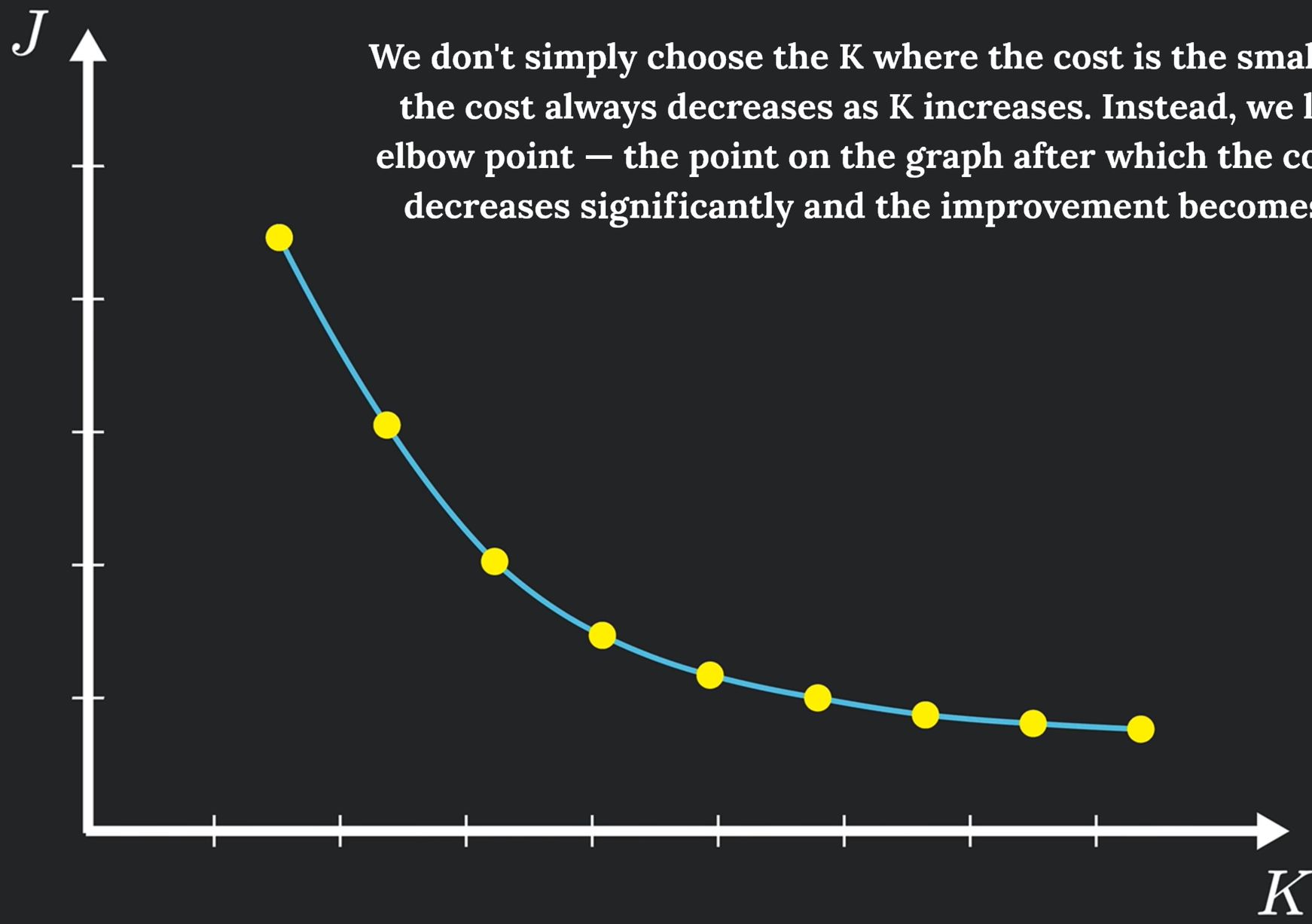
K = 5


x_2
x_1
K = 5

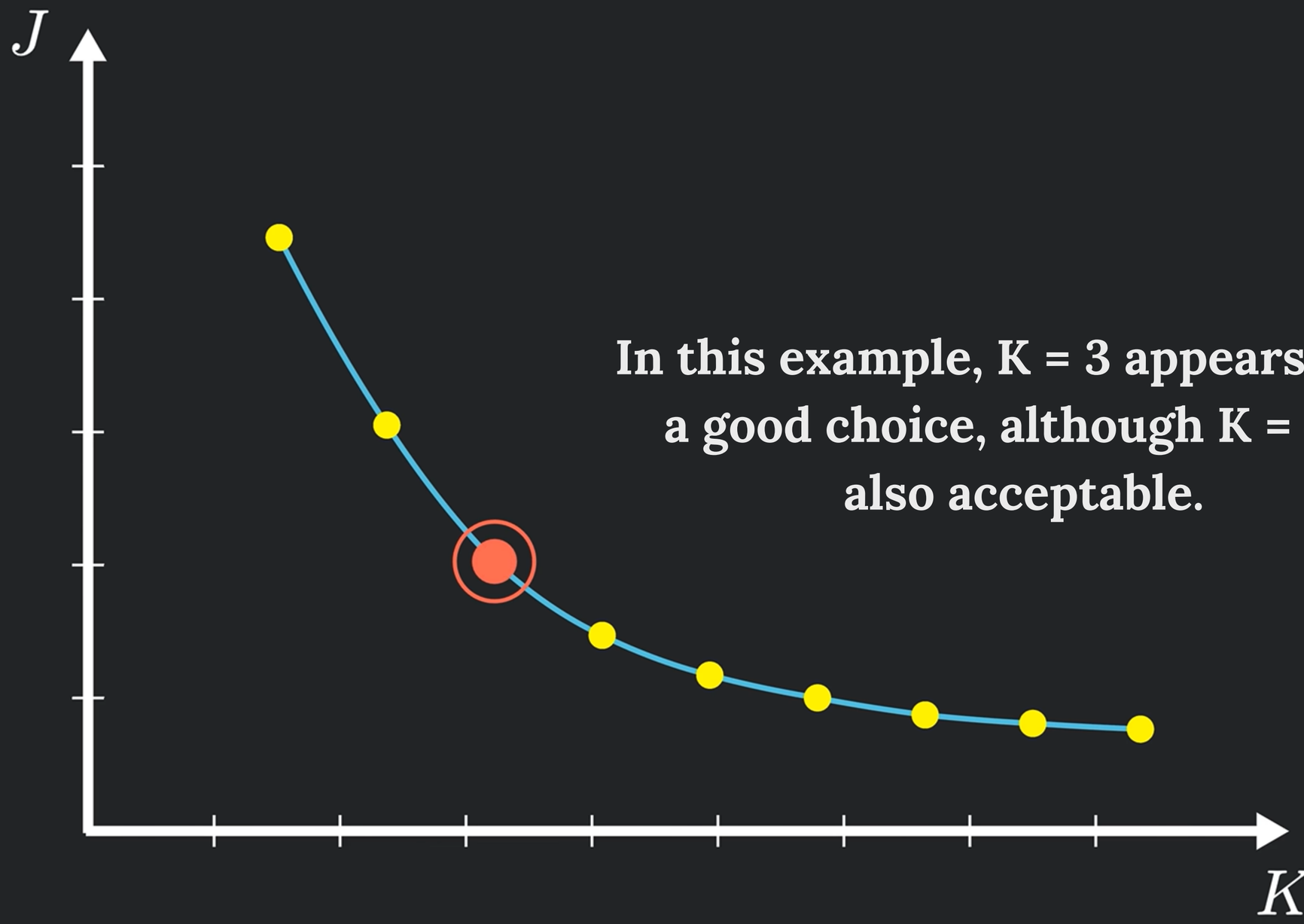

x_2
x_1
K = 5

One of the most widely used approaches to find a good value for K is the Elbow Method. In this method, we run K-Means for multiple values of K, and for each value we calculate the cost function J. Then we plot K against the cost.

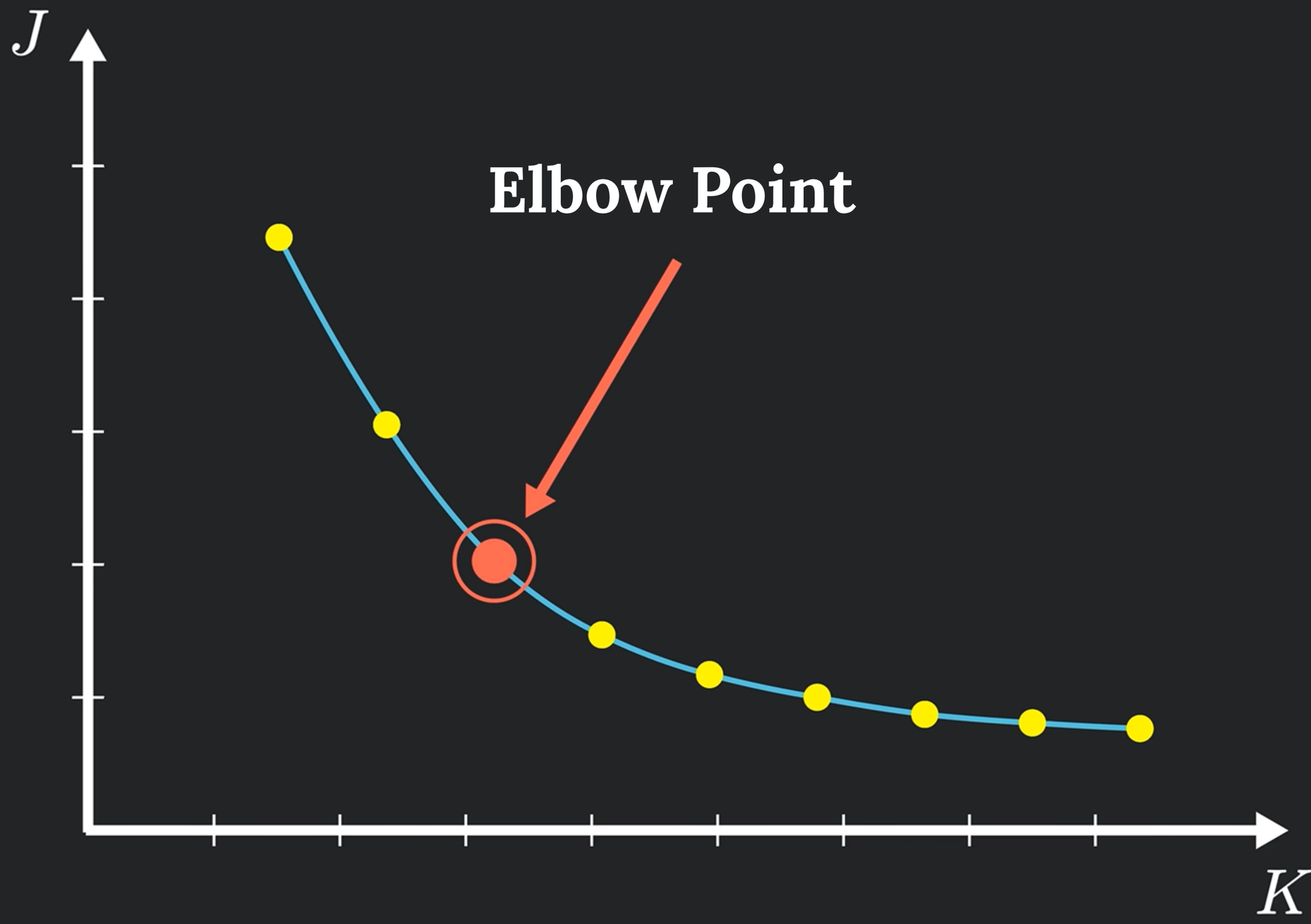

J
K

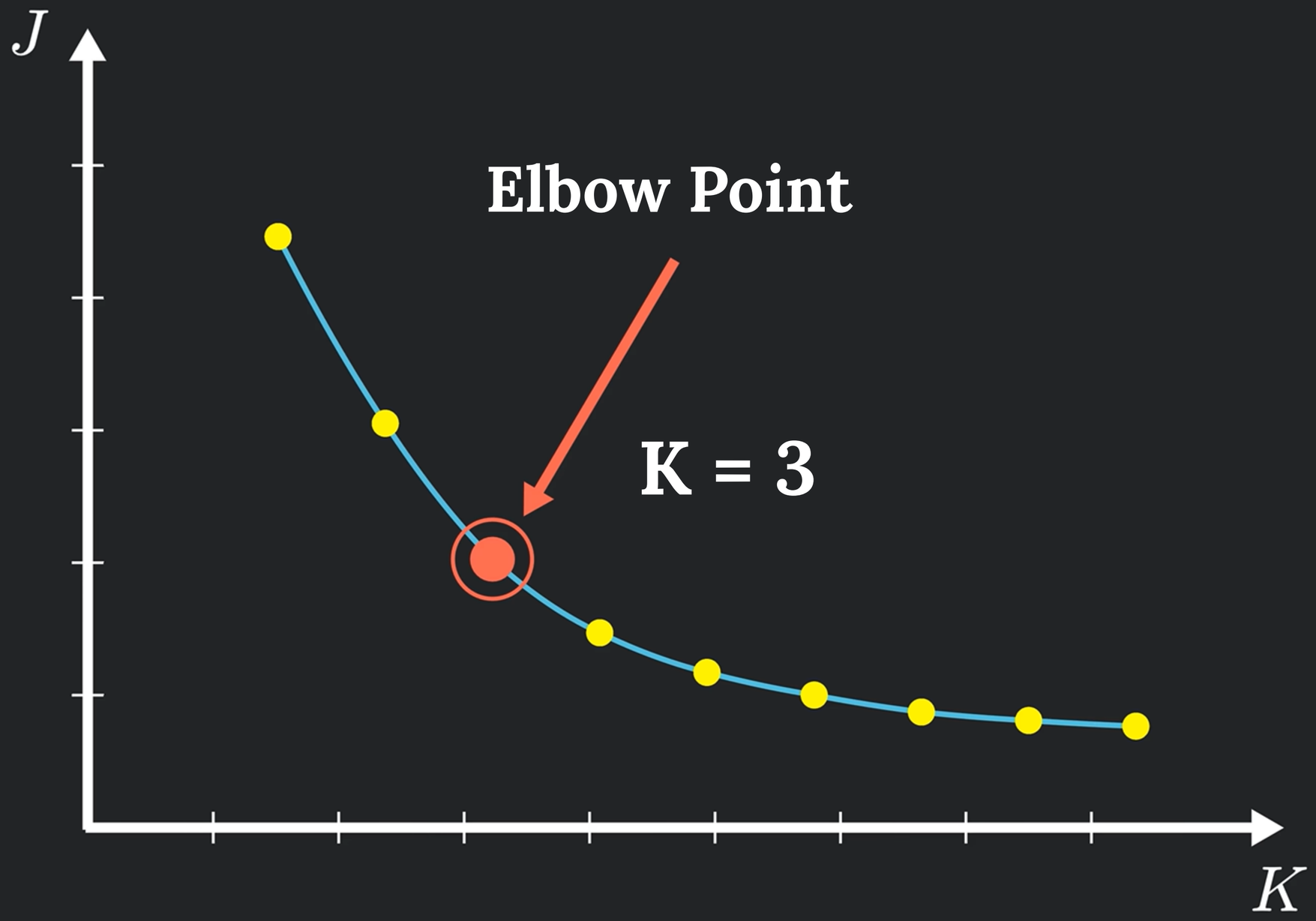


We don't simply choose the K where the cost is the smallest, because the cost always decreases as K increases. Instead, we look for the elbow point — the point on the graph after which the cost no longer decreases significantly and the improvement becomes marginal.



In this example, K = 3 appears to be a good choice, although K = 4 is also acceptable.


J
Elbow Point
K


J
Elbow Point
K = 3
K

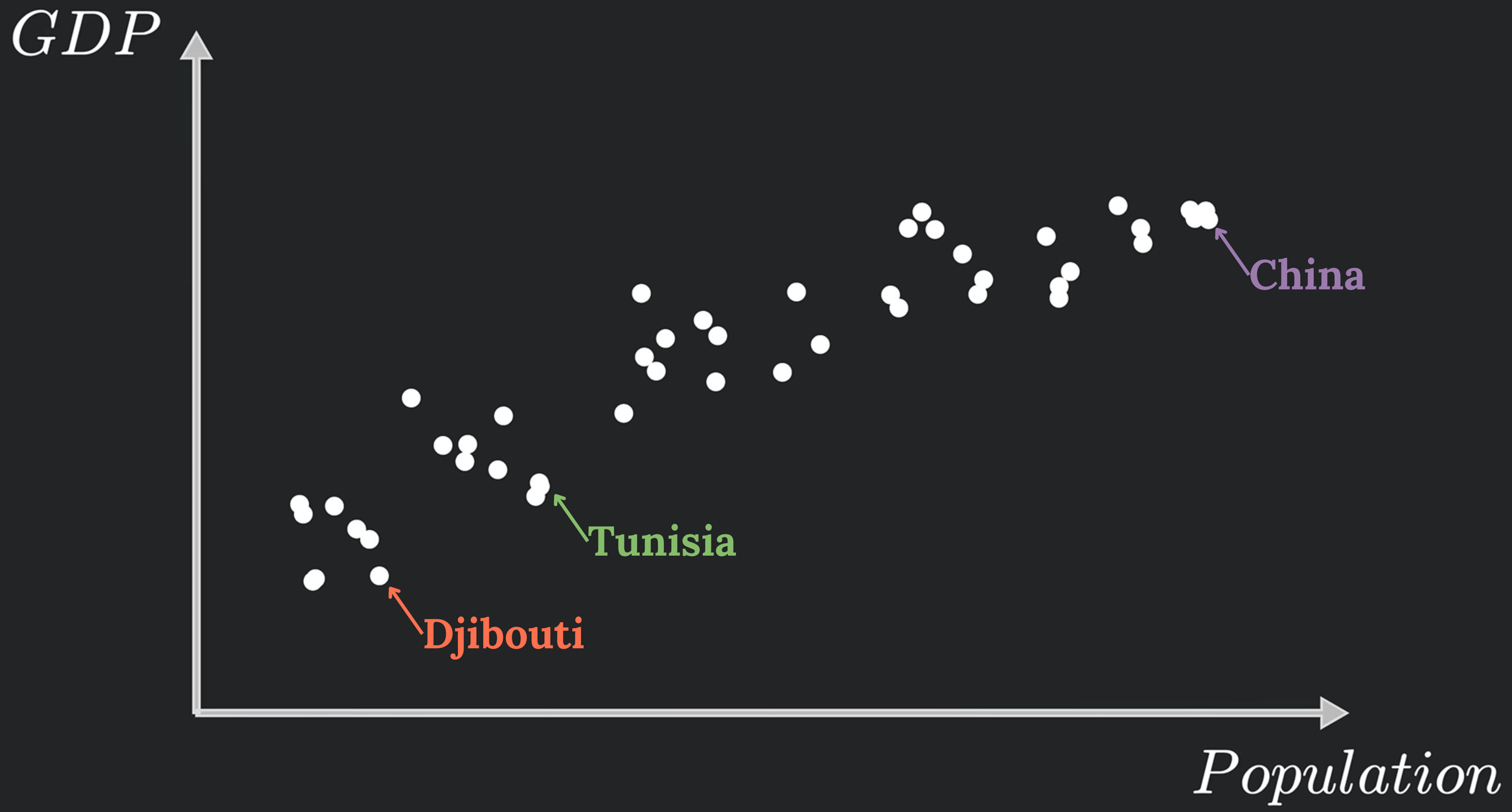
Let’s work on clustering countries by their GDP relative to their population size. The objective is to identify different groups — small, medium, and large economies — and assign each country to a cluster based on its position on a graph where GDP is a function of population.

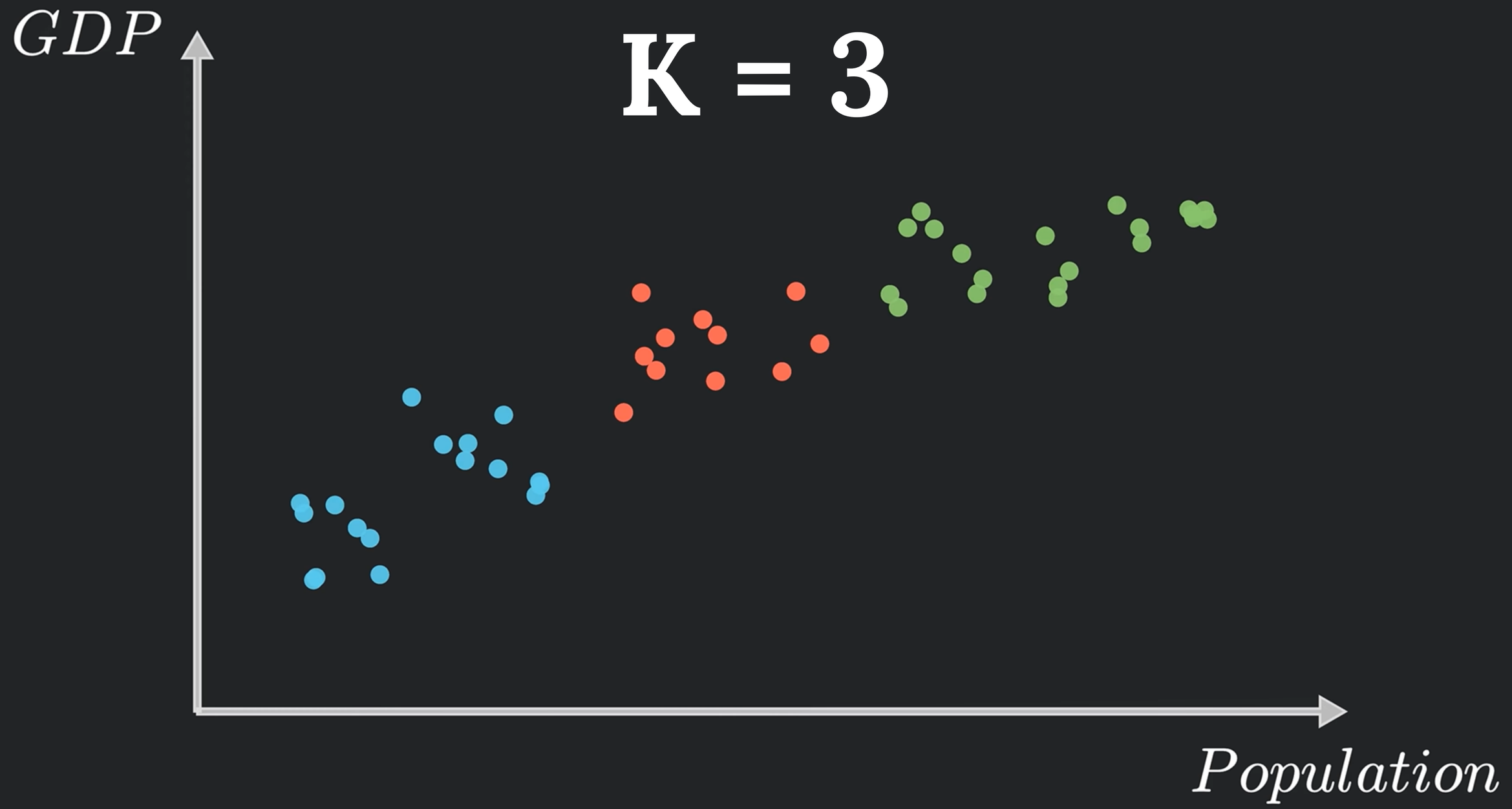
For example:

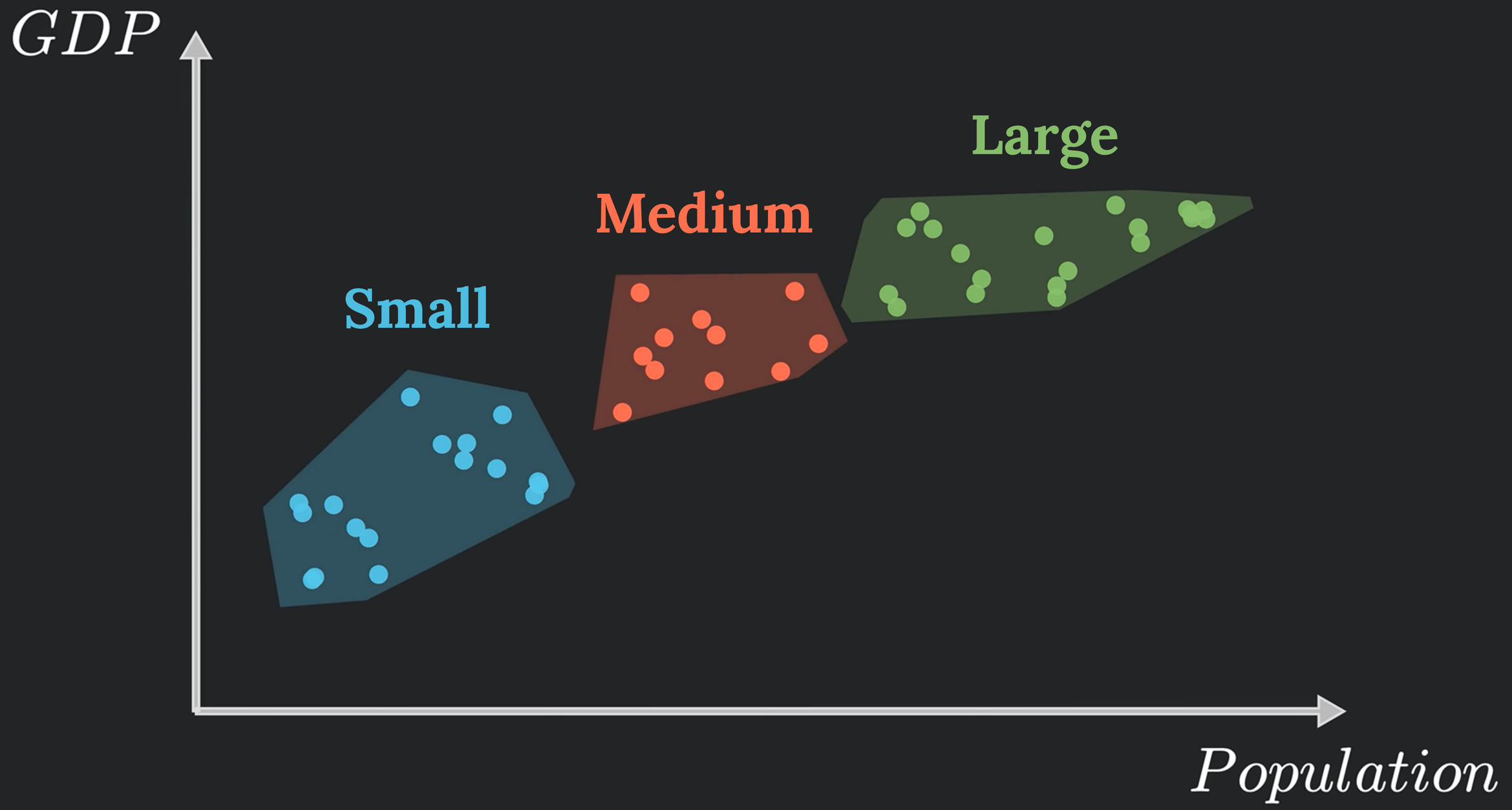
China: very high population, high GDP

Tunisia: medium population, moderate GDP

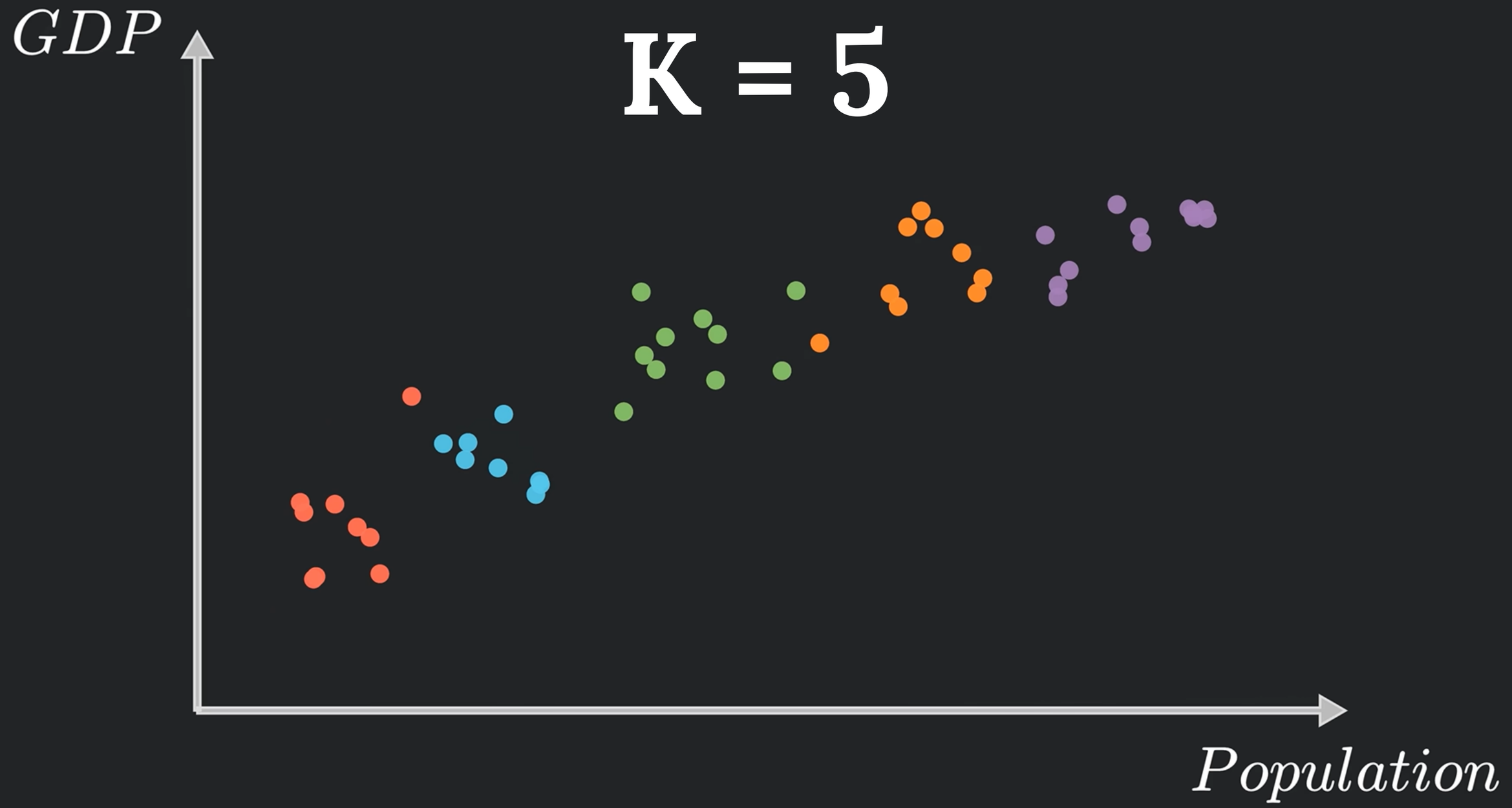
Djibouti: small population, low GDP

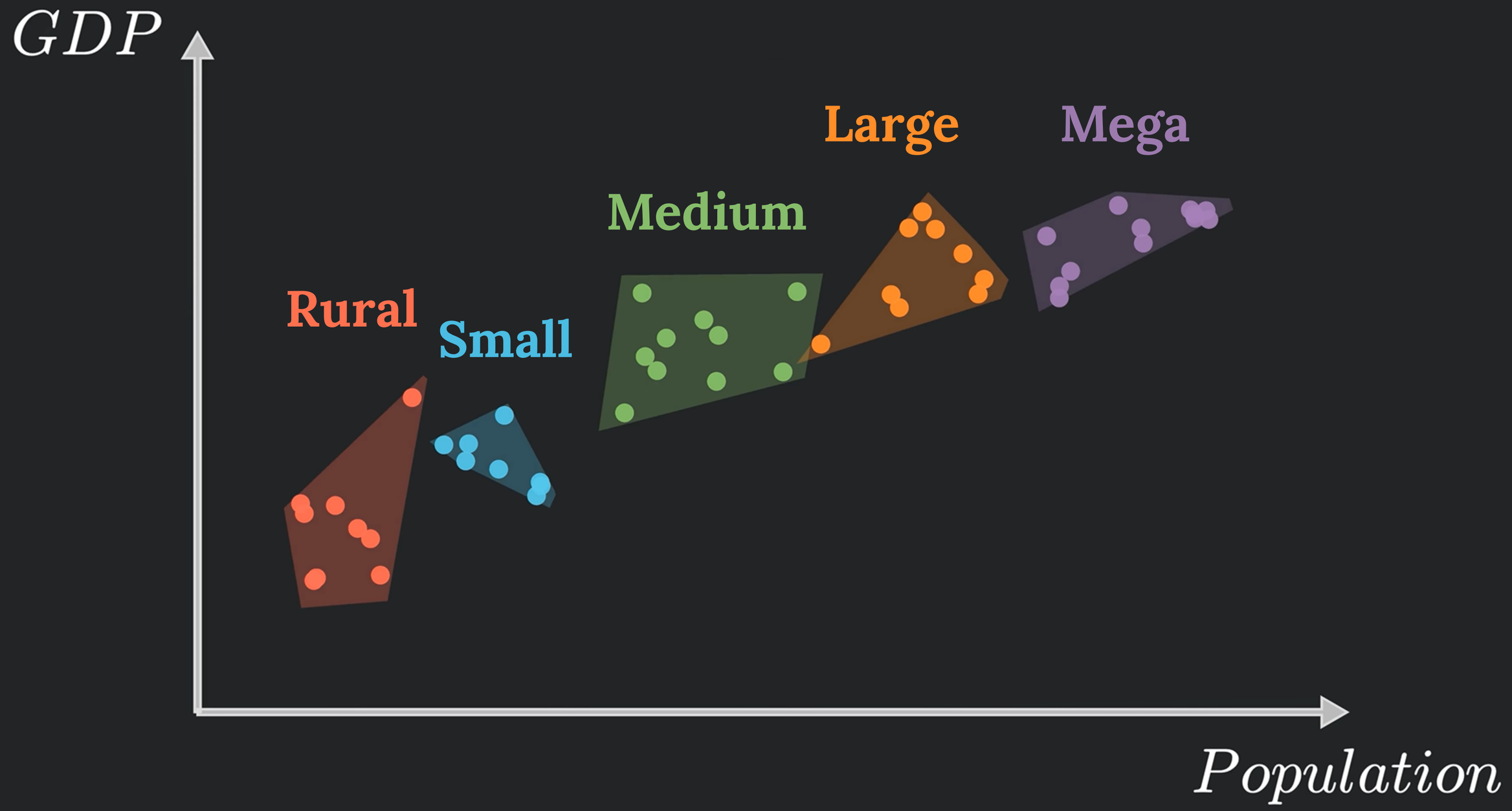

GDP
China
Tunisia
Djibouti
Population


GDP
K = 3
Population


GDP
Large
Medium
Small
Population

Remark: We should not rely only on mathematical methods to decide the optimal value of K. The number of clusters also depends on the practical goal of the analysis. In this example, if we choose K=3, the clusters might represent small, medium, and large cities. But if we choose K=5, the grouping could be more detailed — such as rural areas, small cities, medium cities, large cities, and megacities. Both interpretations are valid, depending on the purpose of the analysis. So while methods like the Elbow Method are helpful, we should also use common sense and domain knowledge when choosing the number of clusters.


GDP
K = 5
Population

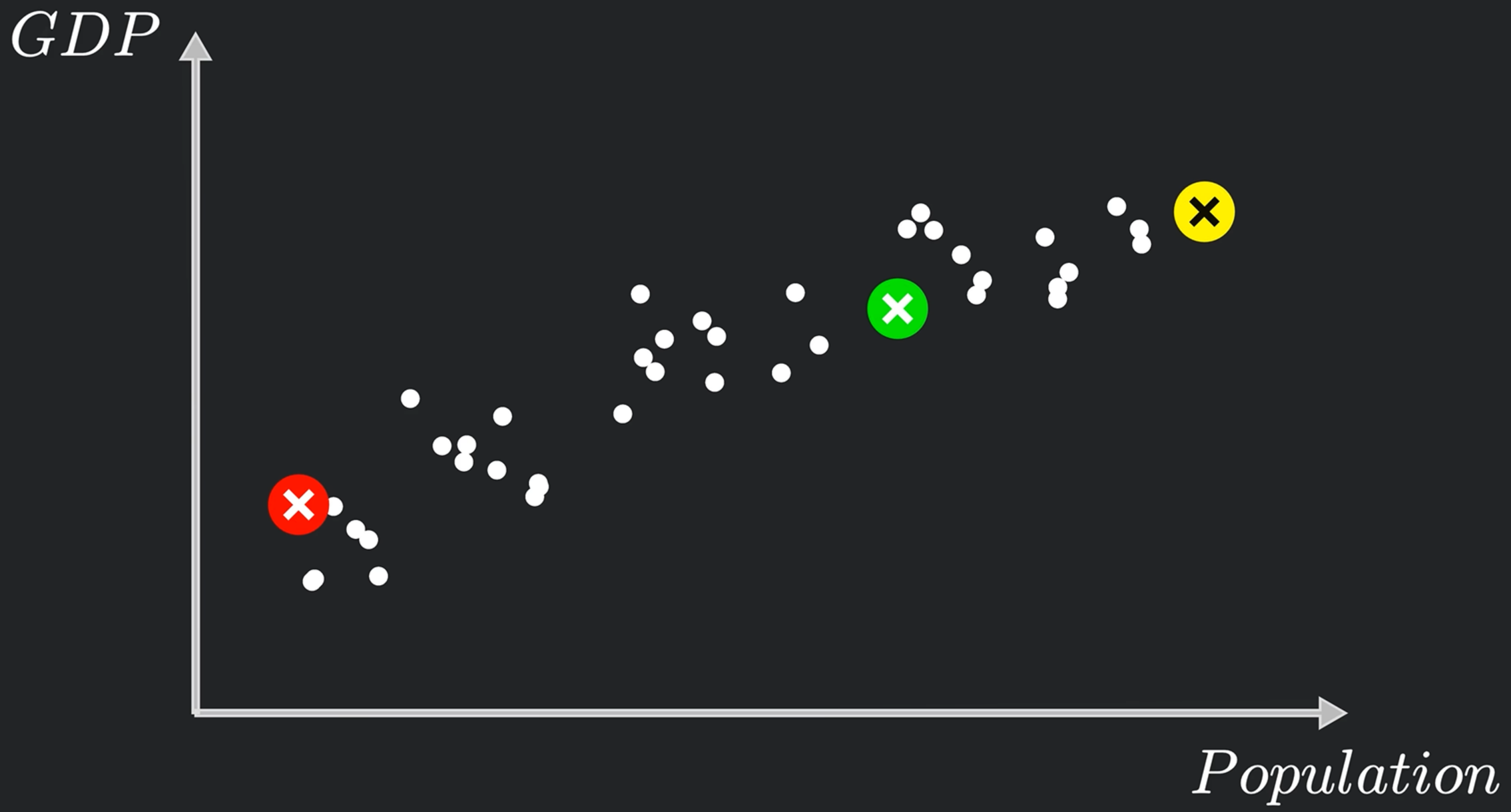

GDP
Rural
Small
Medium
Large
Mega
Population

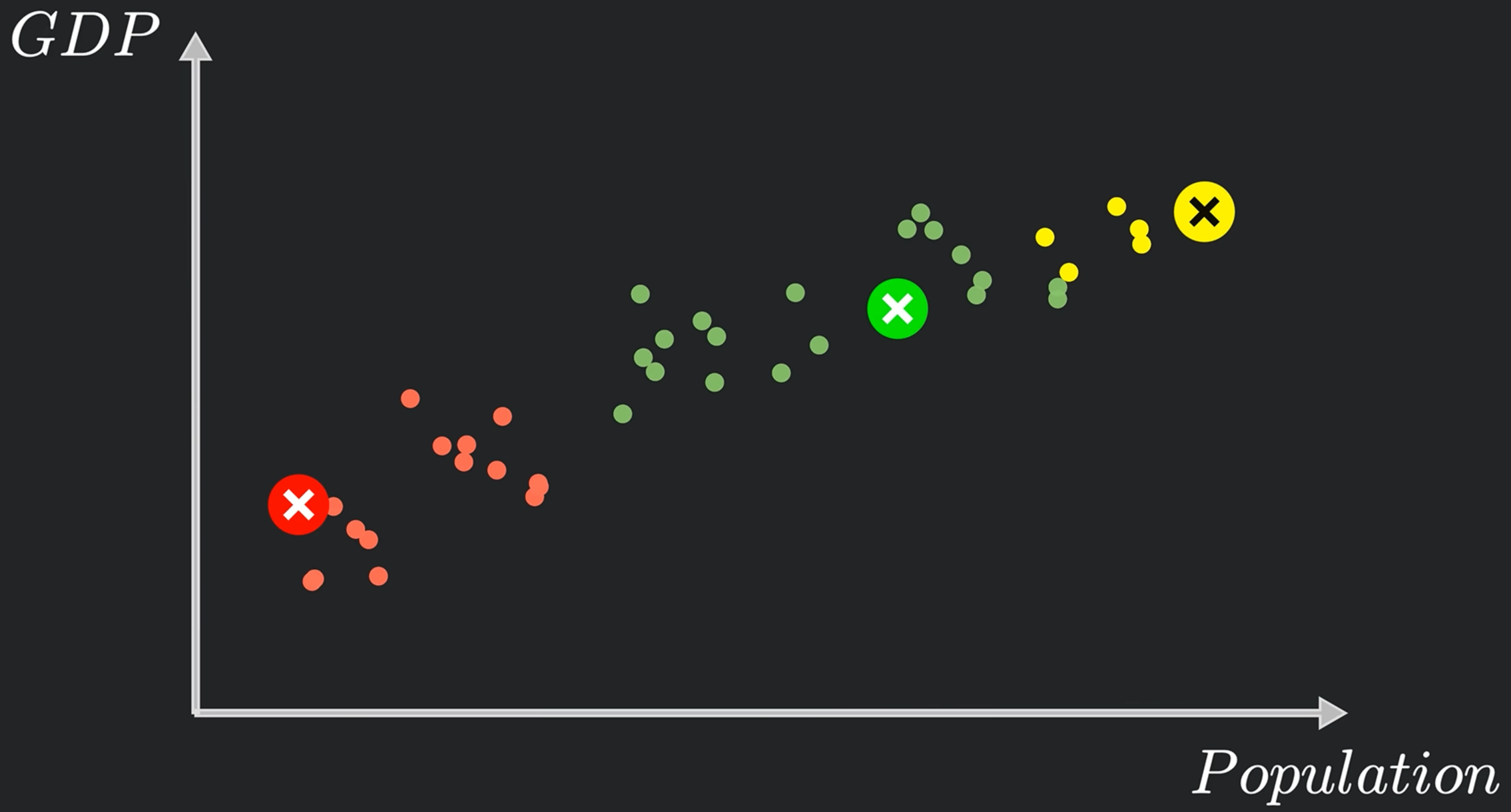
At the beginning, we placed the centroids anywhere on the screen just to demonstrate the concept. But in practice, we rarely do that. Instead, a common approach is to randomly select K data points directly from the dataset and use those as the initial centroids. This way, the centroids start at positions that already lie within the data distribution, which usually leads to faster convergence and more meaningful clusters compared to placing them completely at random.

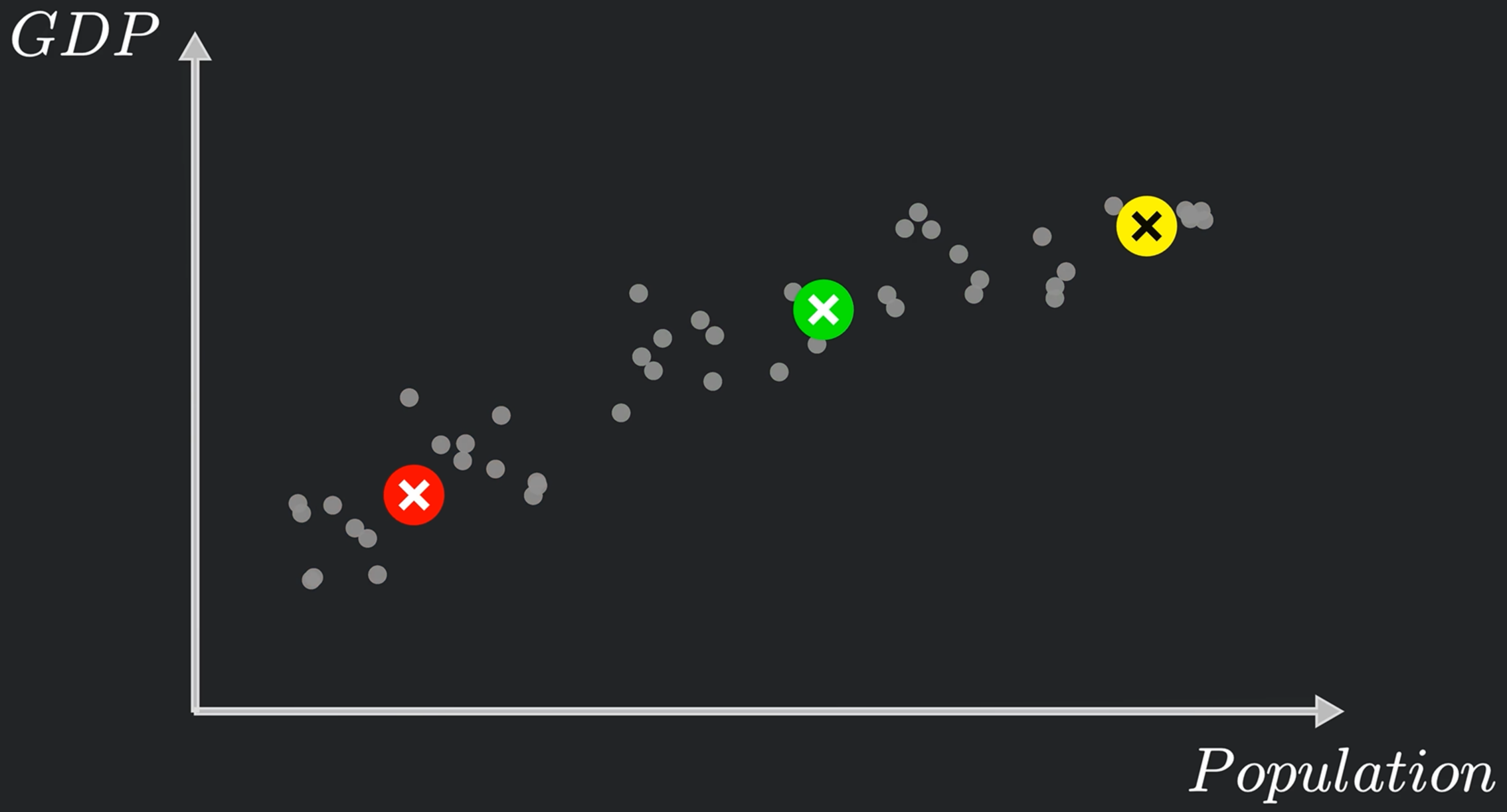
Let's see this in action. We will take the same GDP-versus-population dataset from earlier, and this time, suppose we want three clusters. We initialize three centroids by choosing three points from the dataset. Once that's done, the K-Means algorithm starts running as usual.

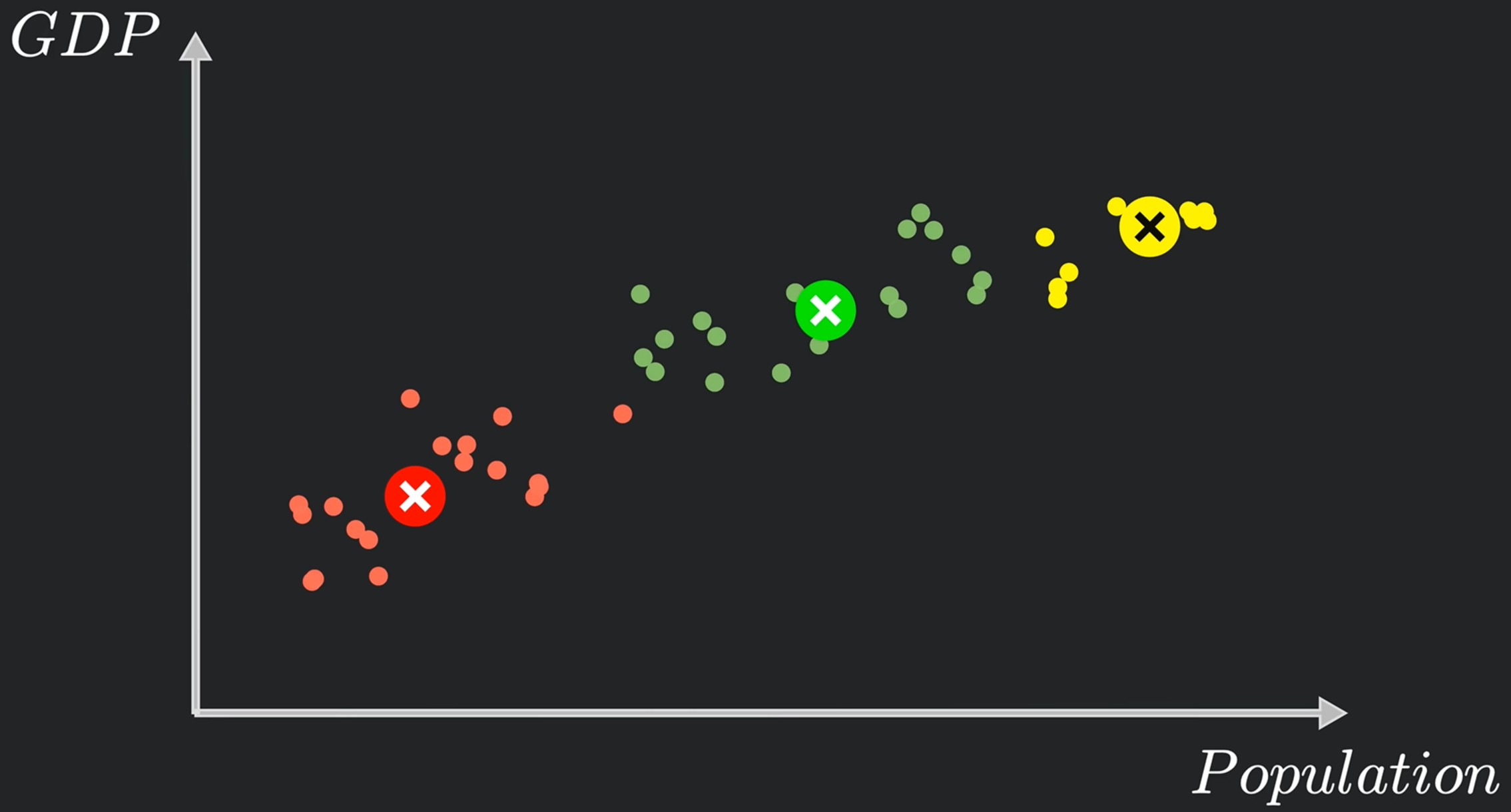
- Randomly pick K different points from the dataset
- Set these points as the initial centroids $\mu_1, \mu_2, \ldots, \mu_K$

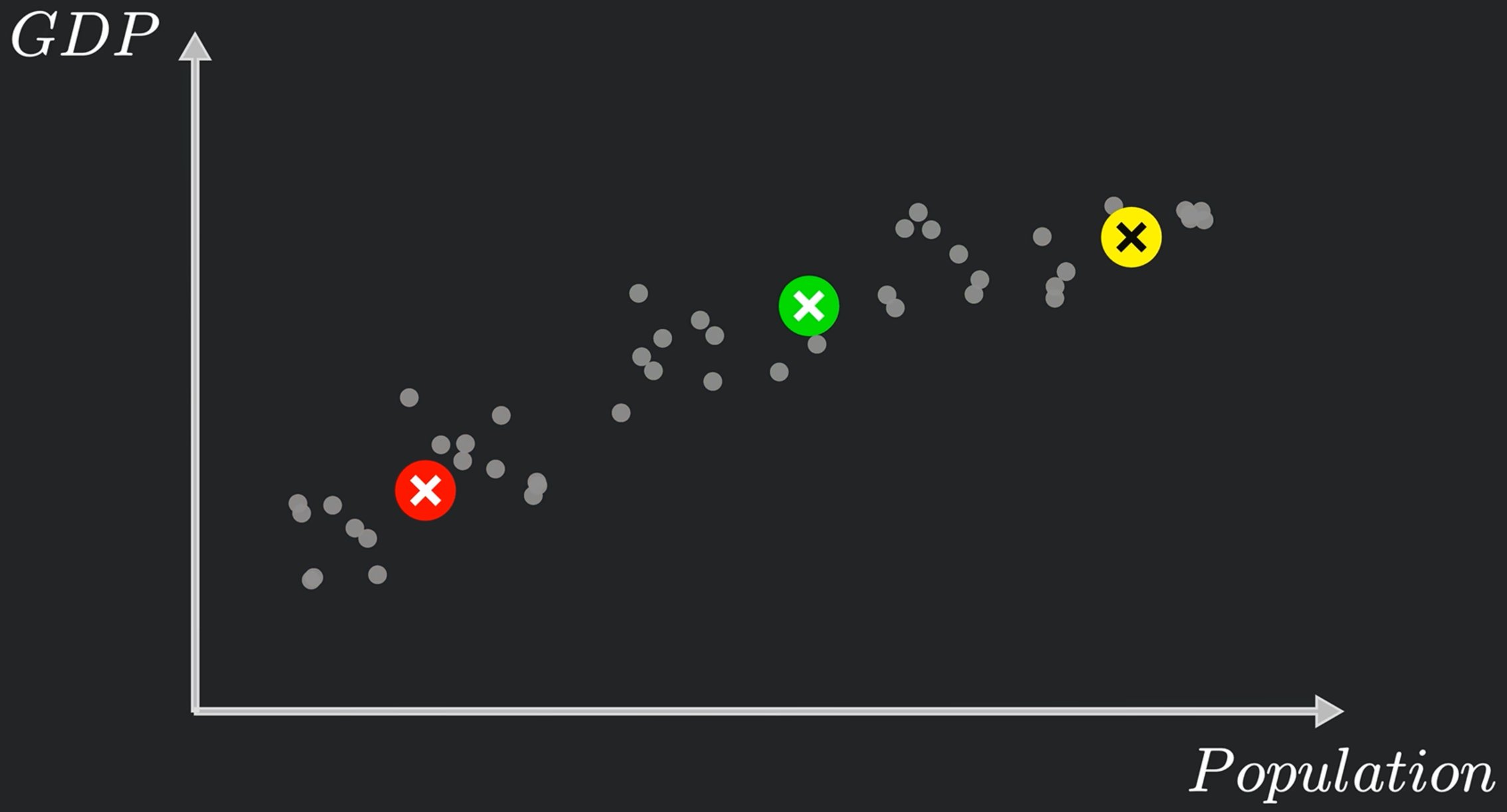

GDP
Population

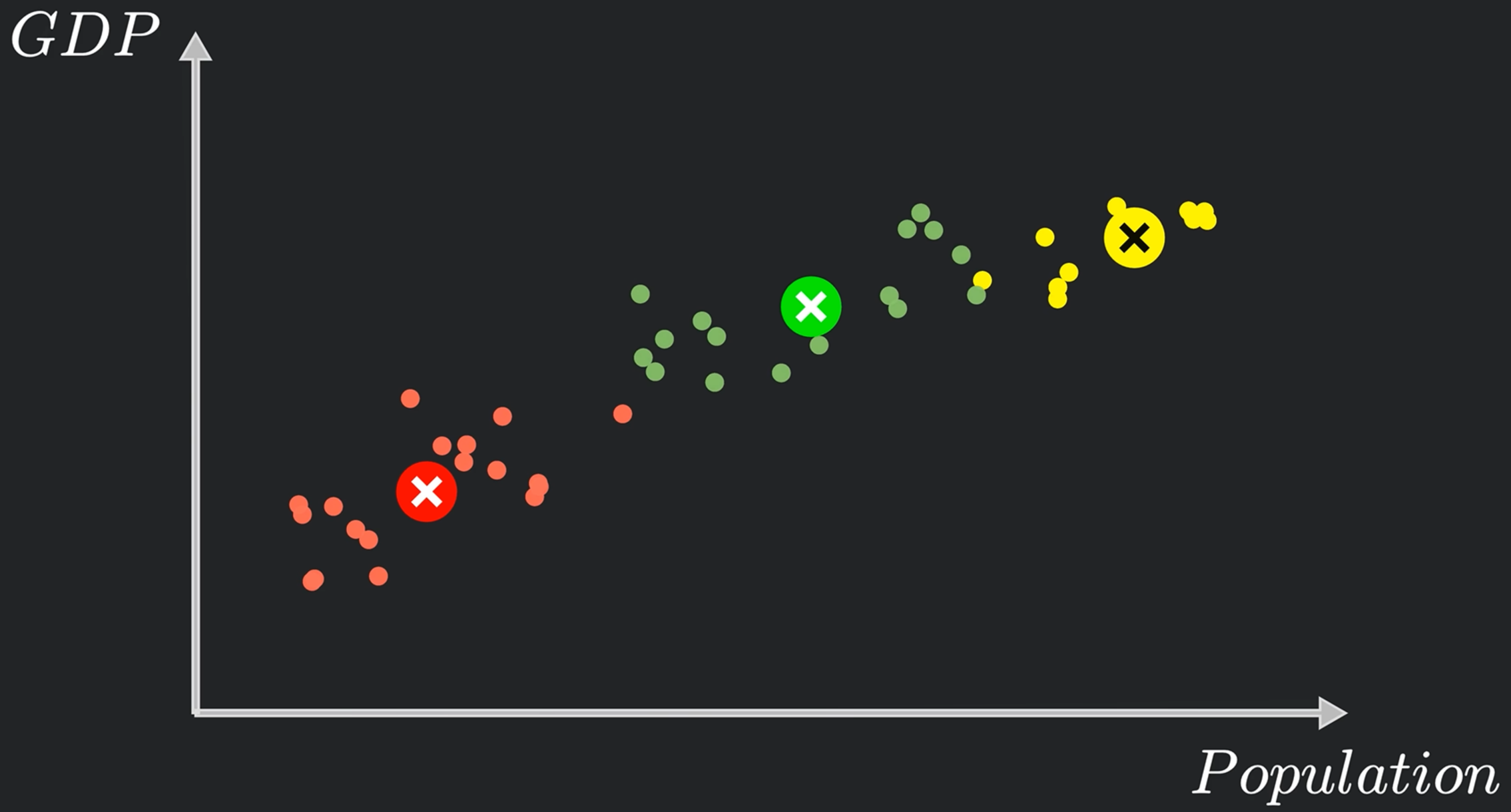

GDP
Population

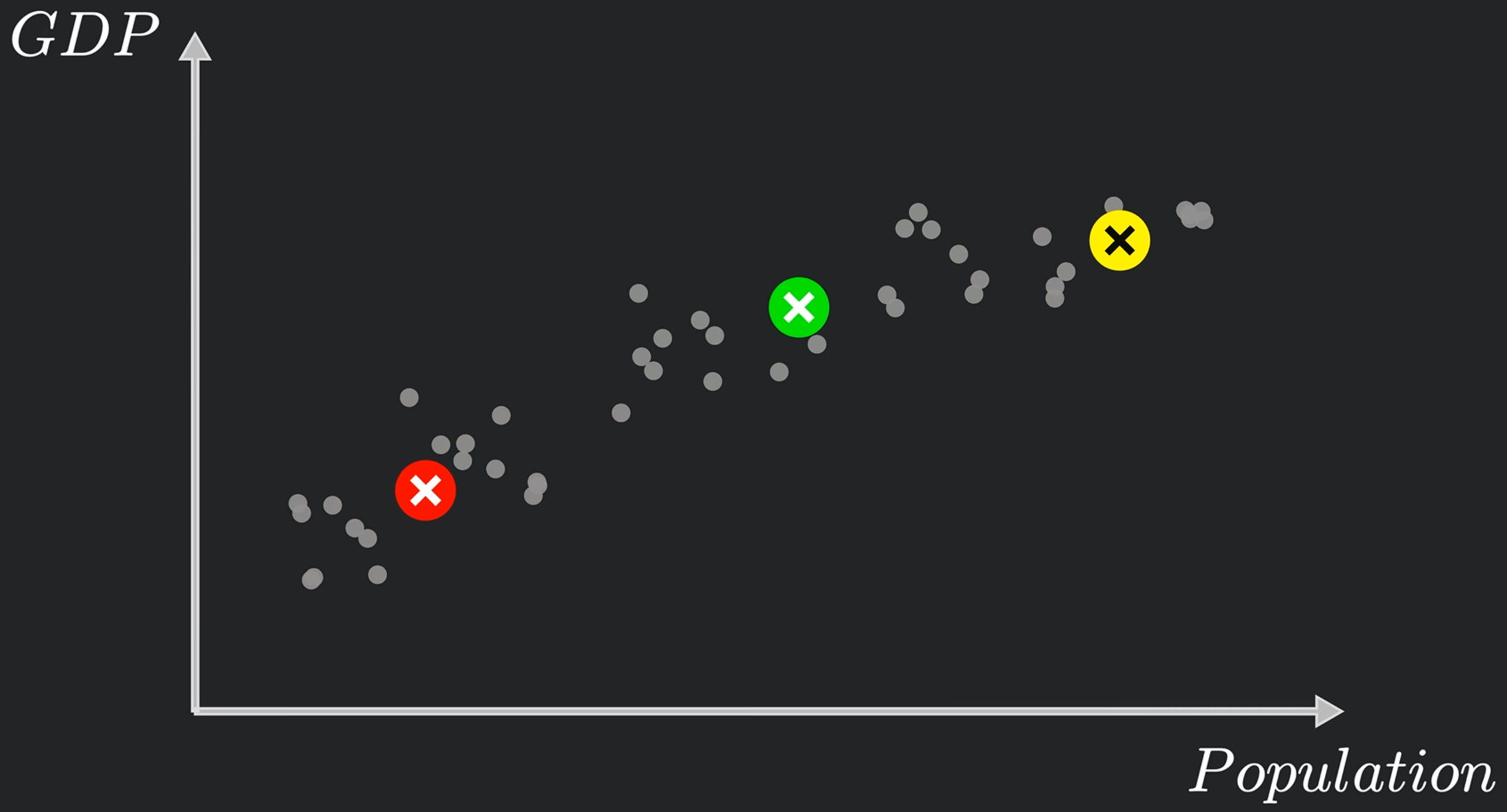

GDP
Population

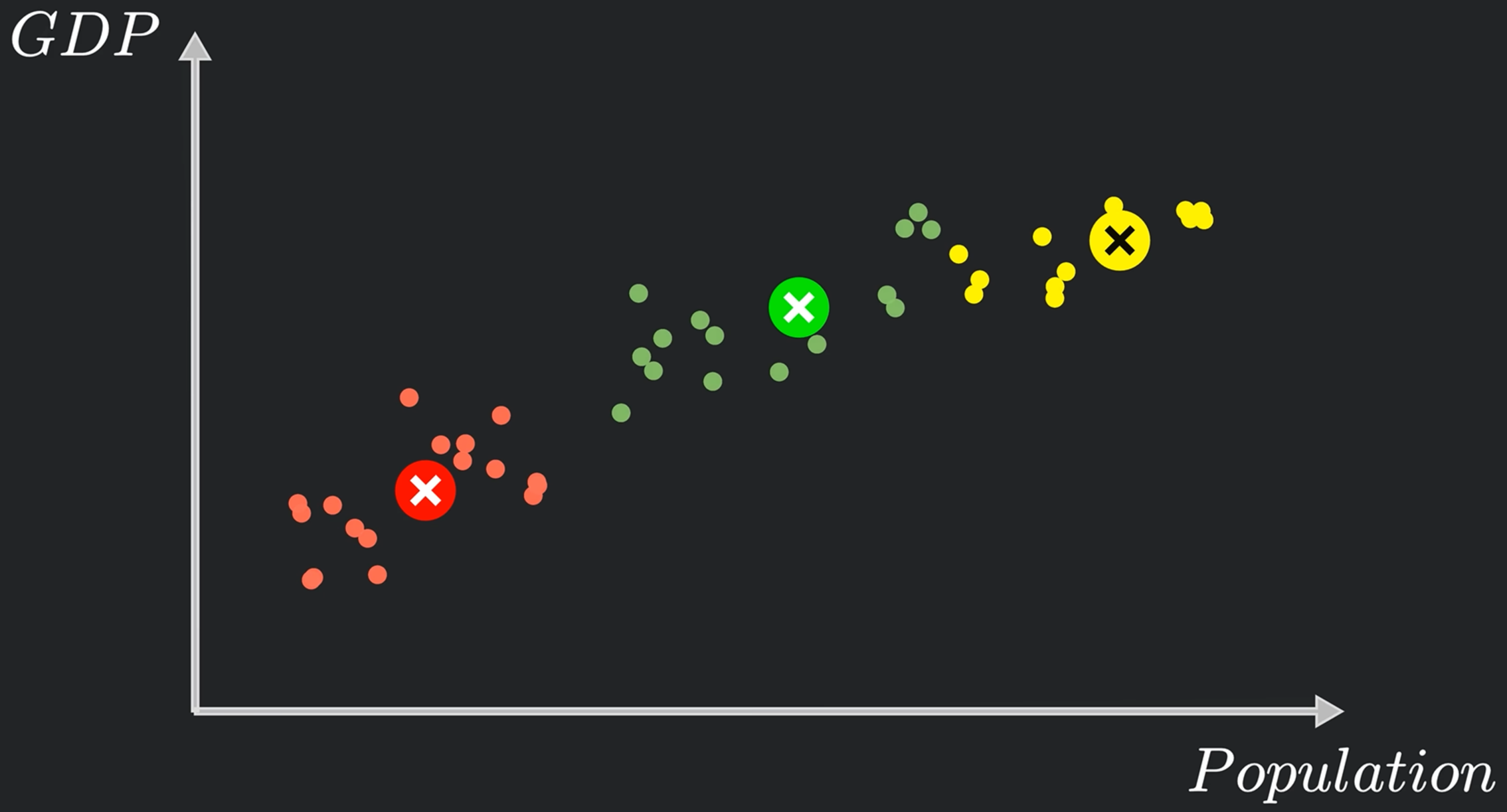

GDP
Population

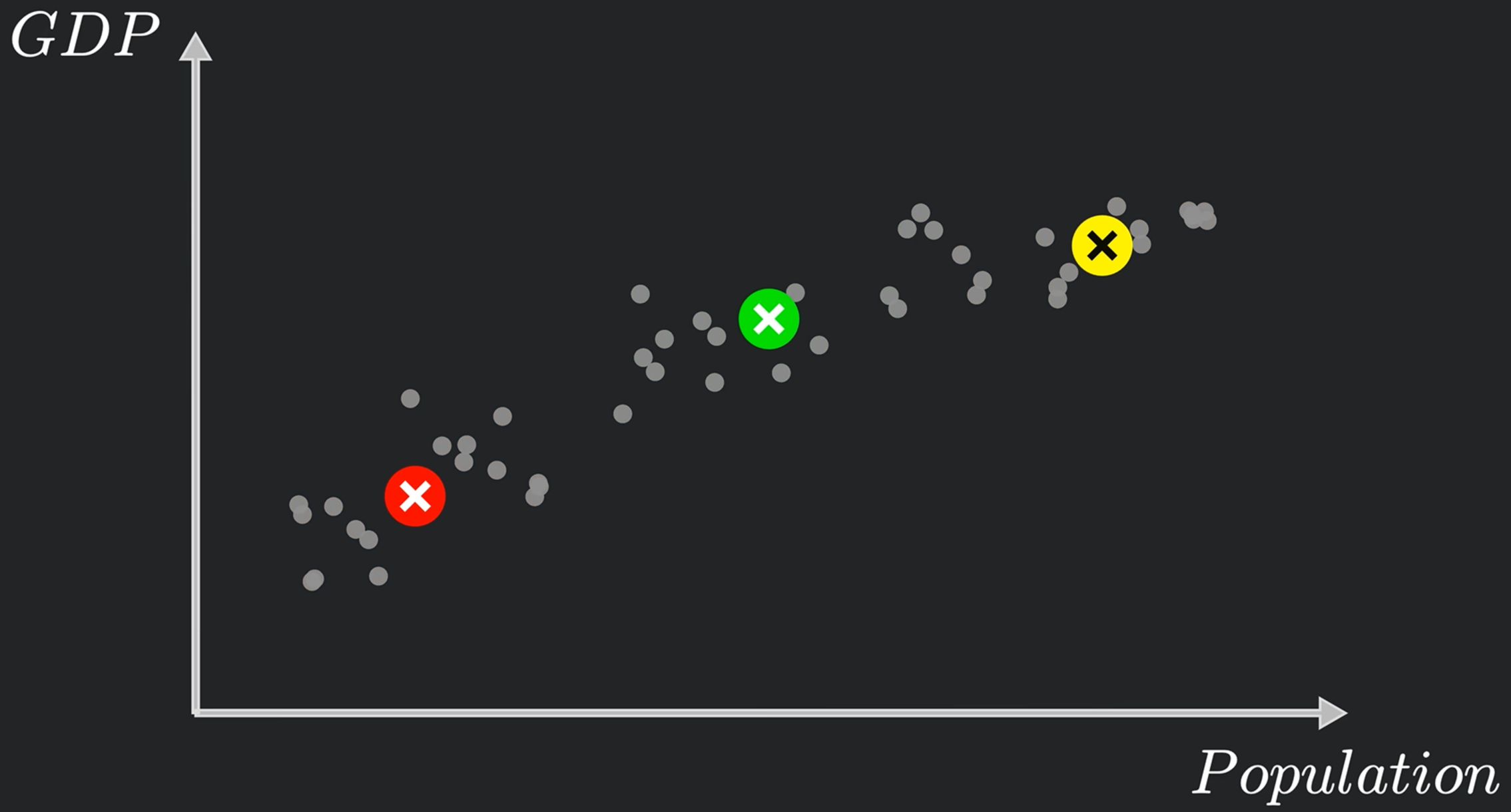

GDP
Population

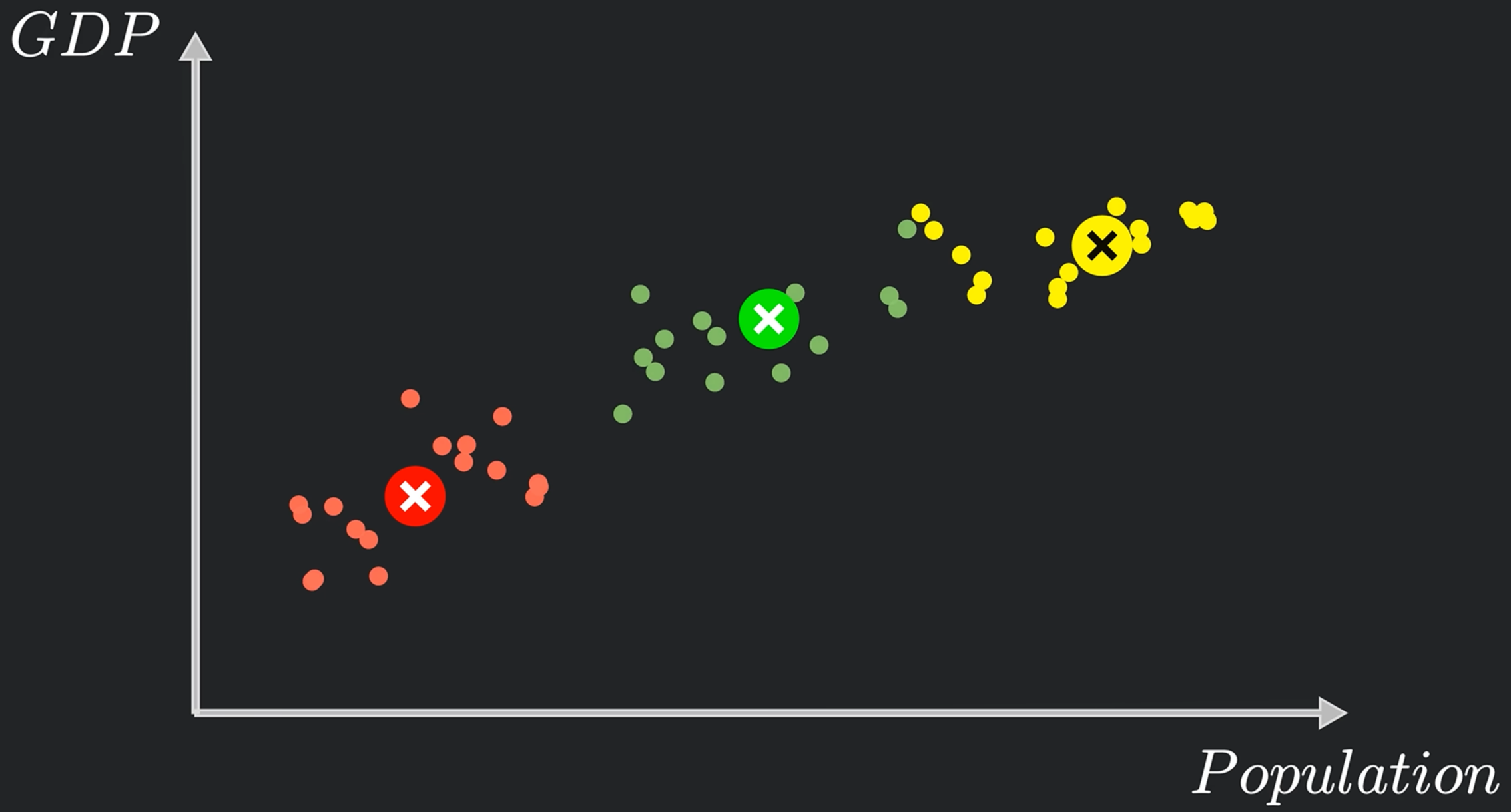

GDP
Population

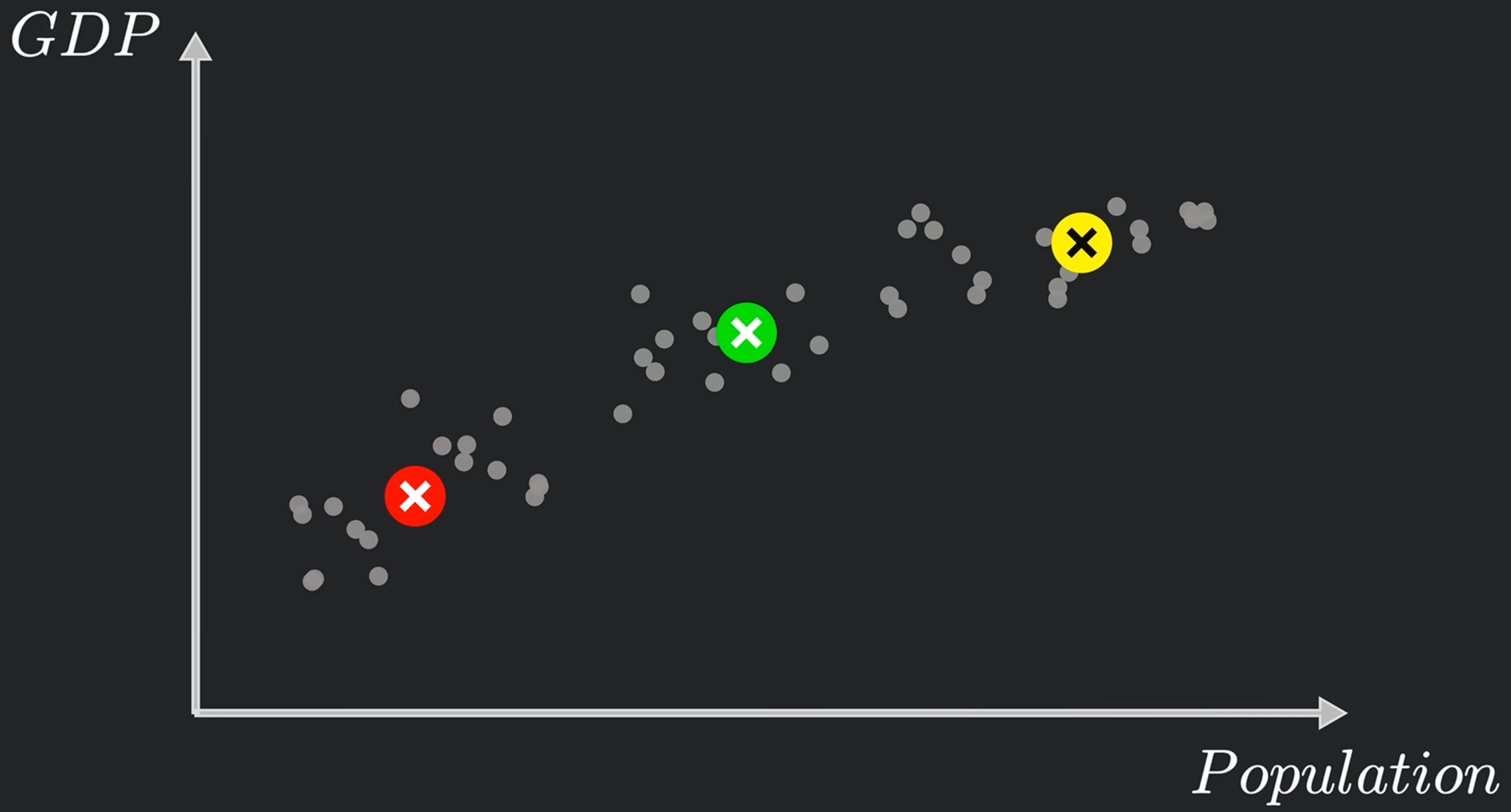

GDP
Population

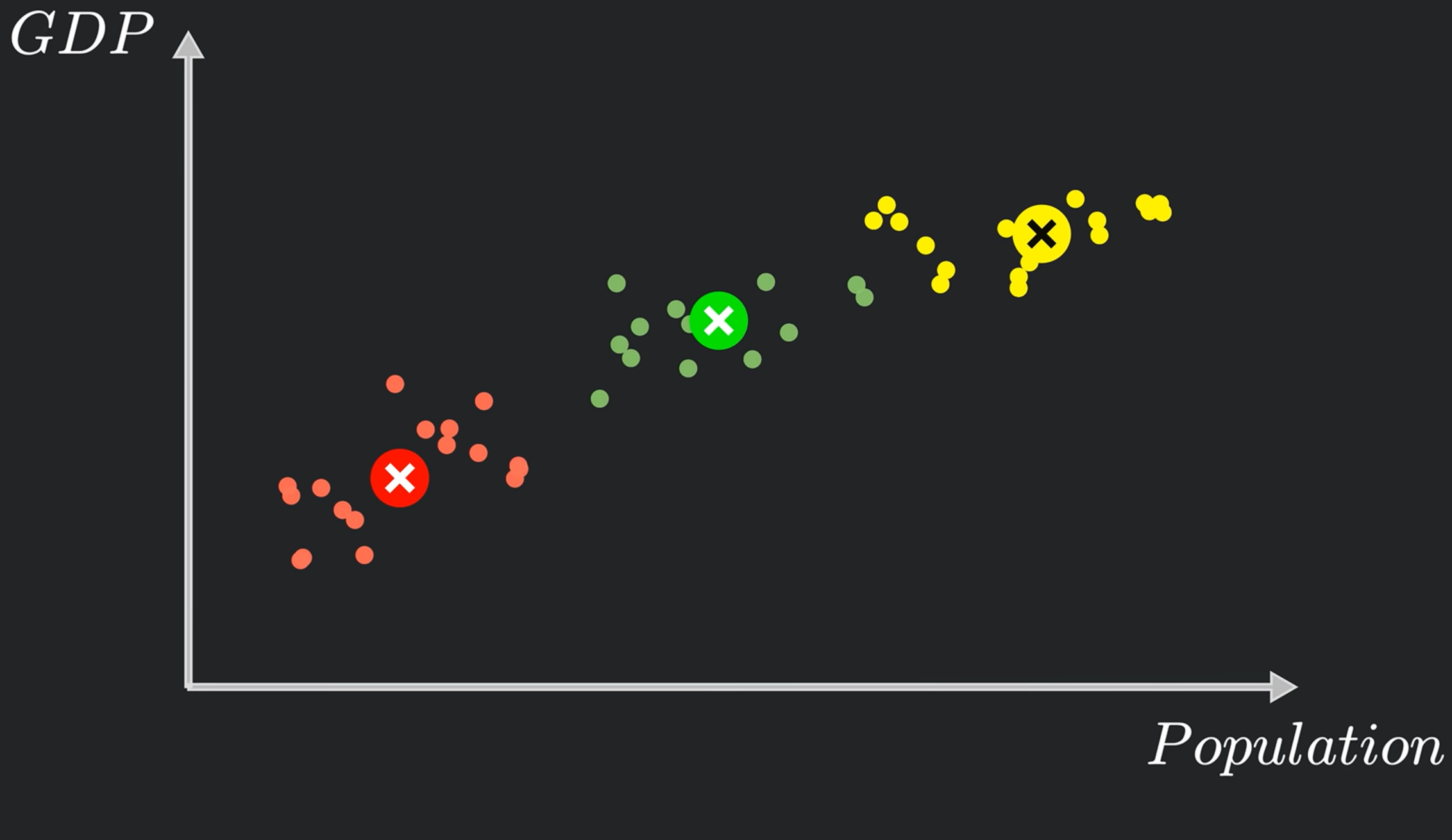

GDP
Population

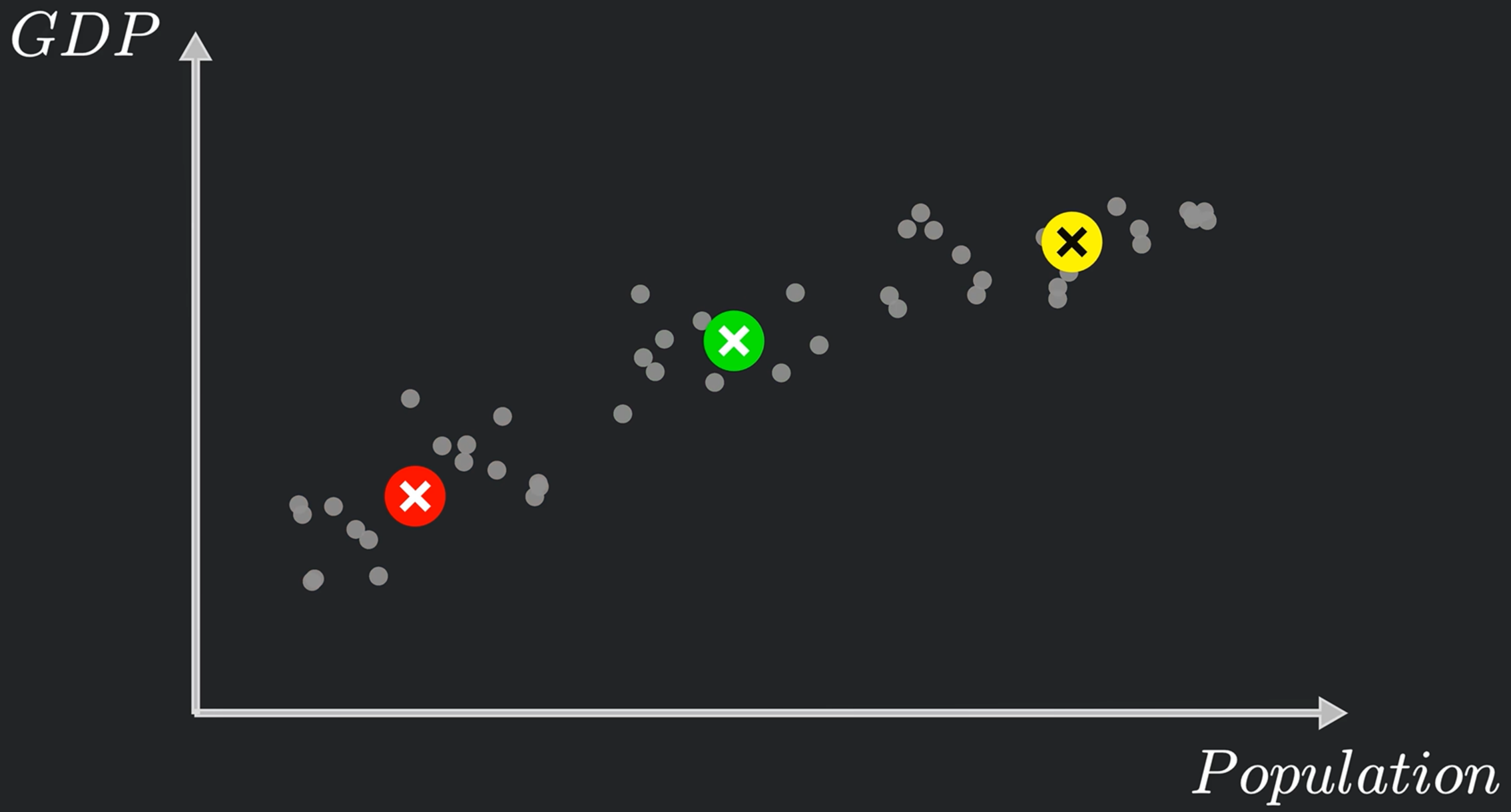

GDP
Population

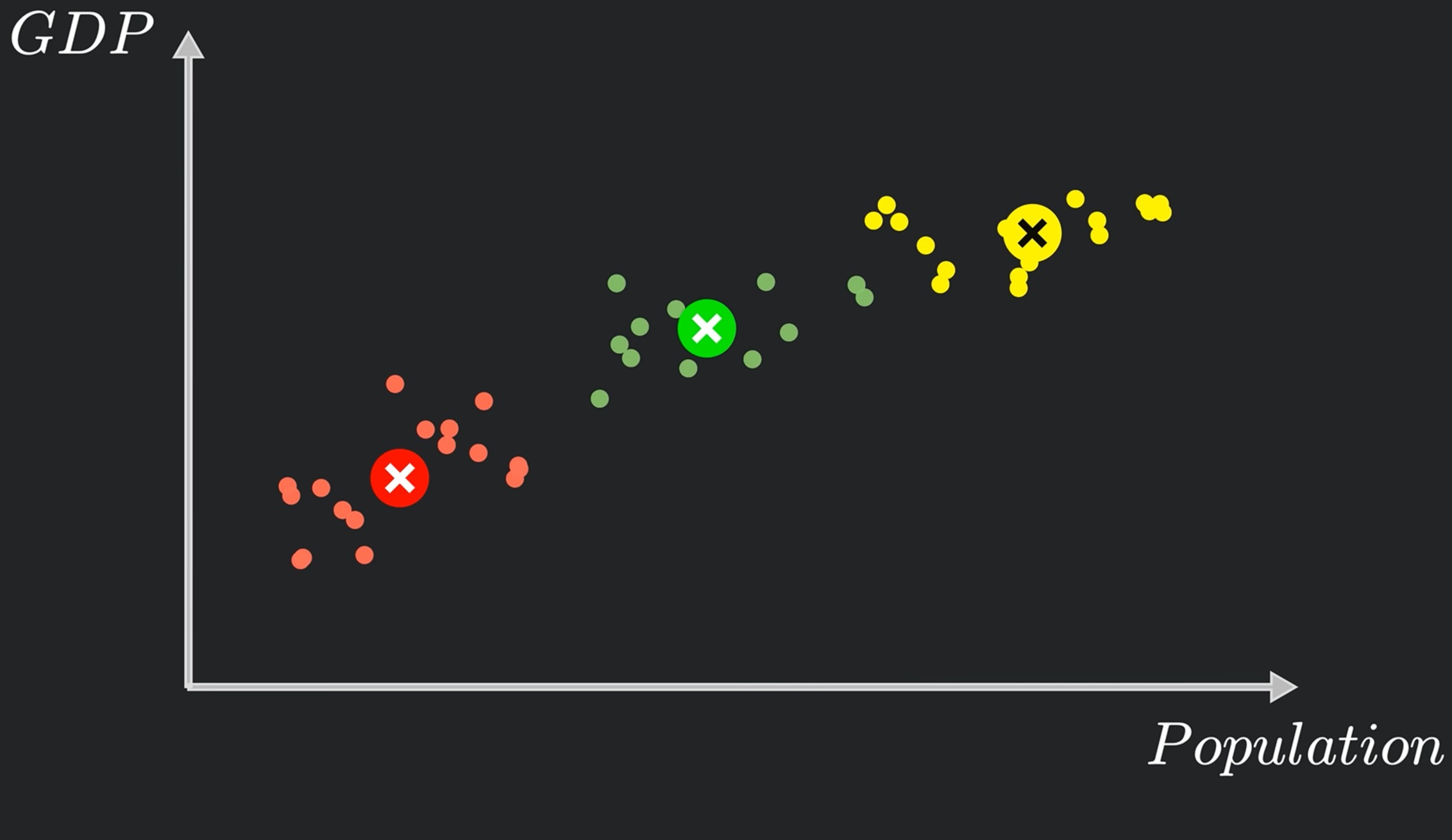

GDP
Population

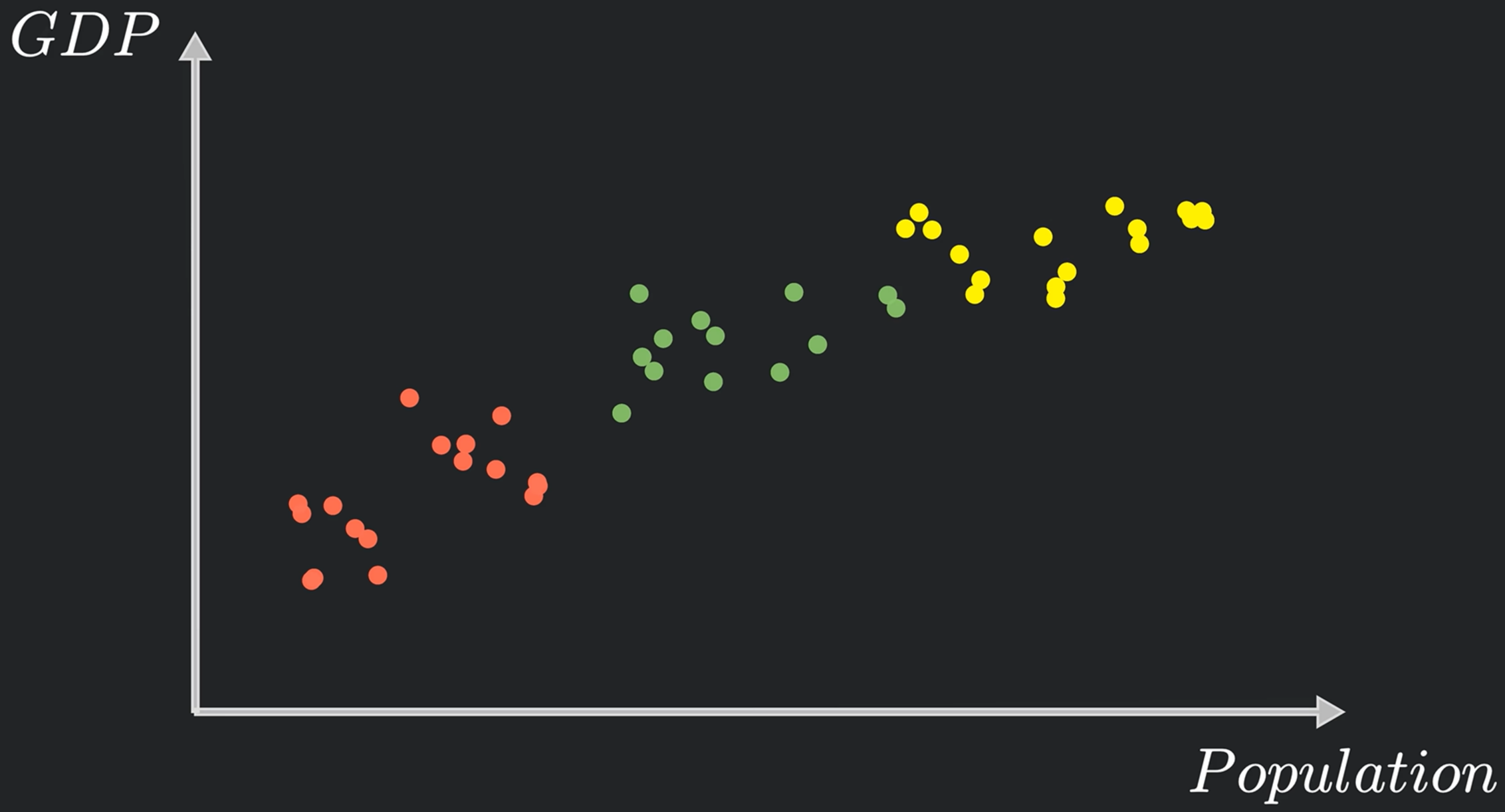

GDP
Population

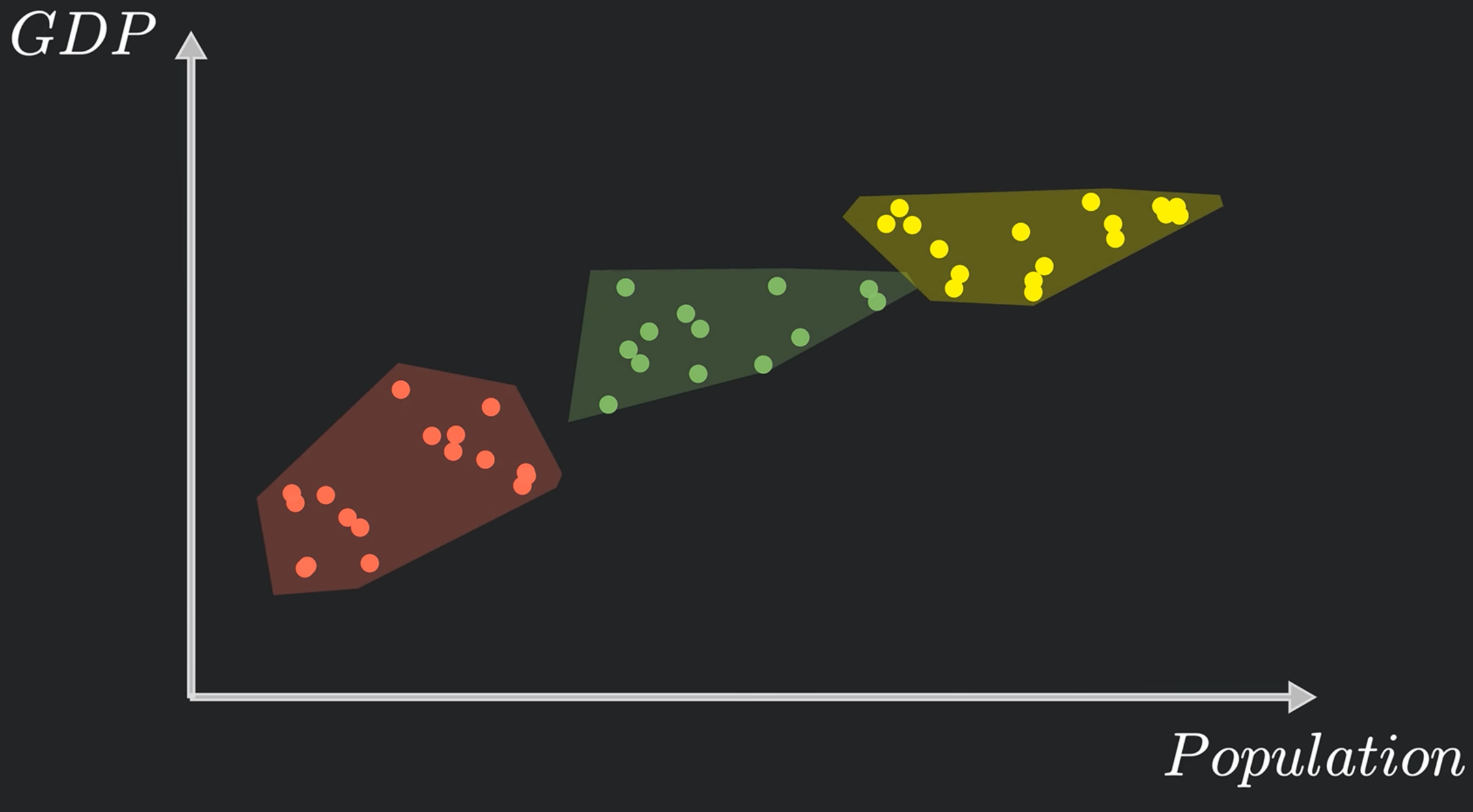

GDP
Population

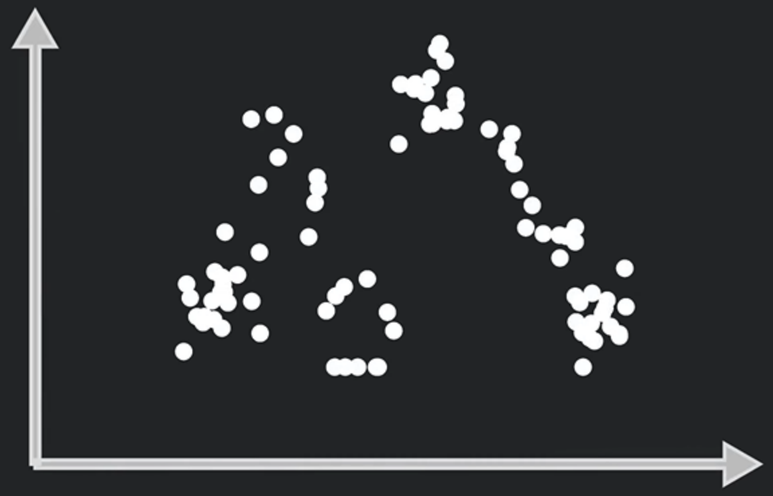

GDP
Population

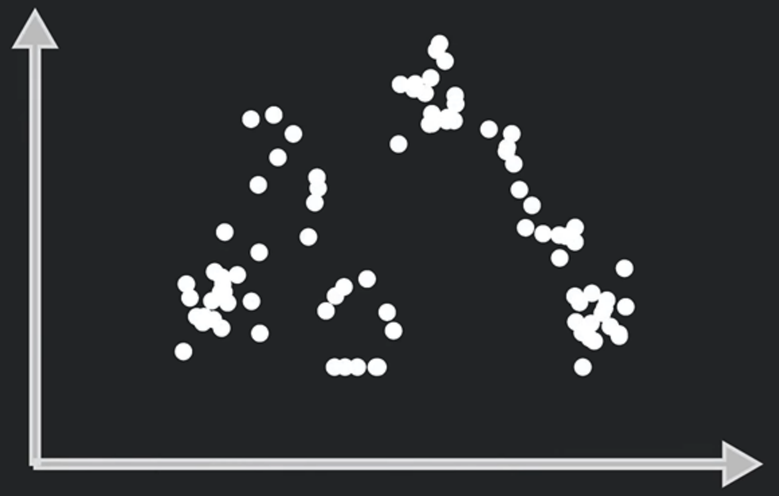

GDP
Population

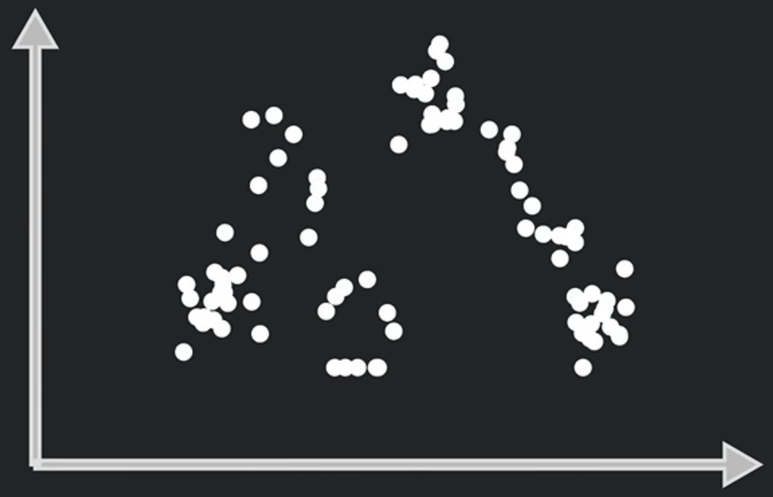

GDP
Population

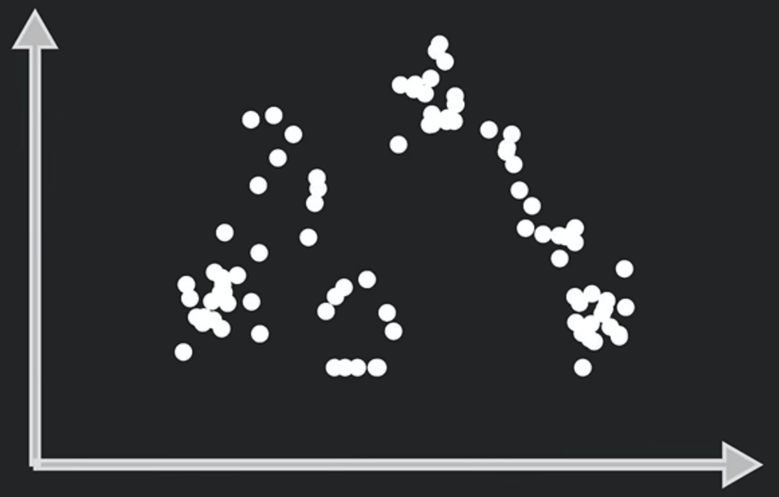

GDP
Population


GDP
Population



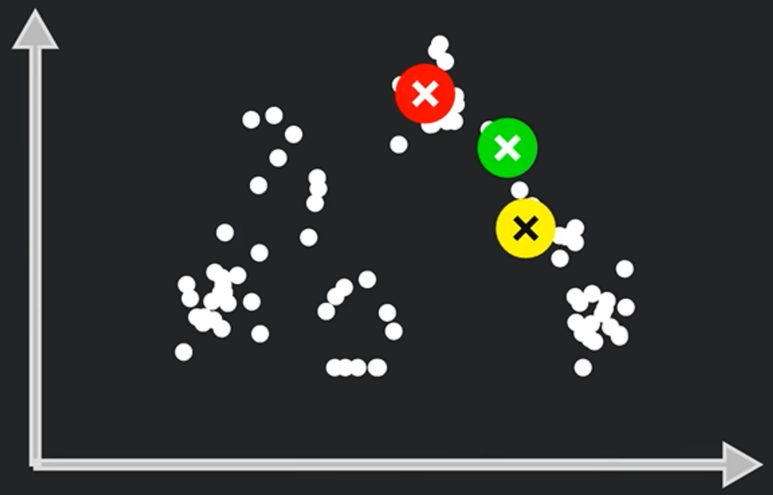






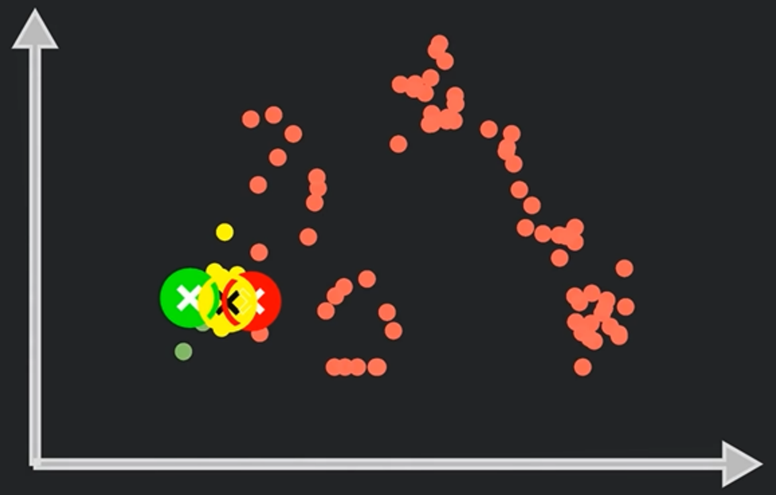




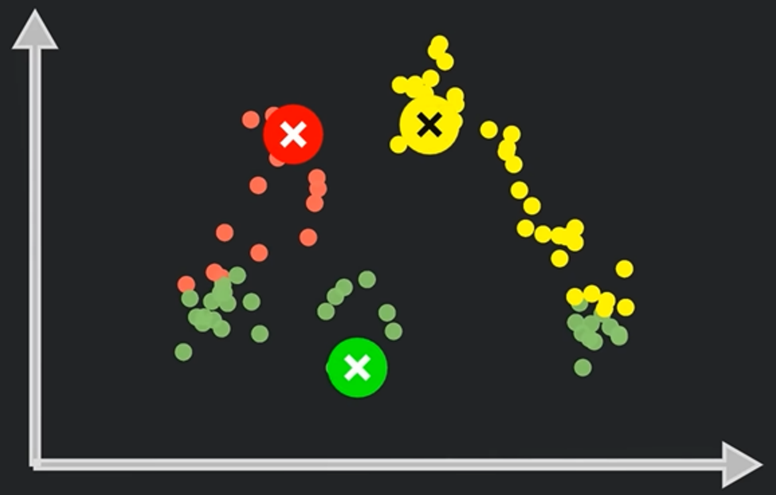


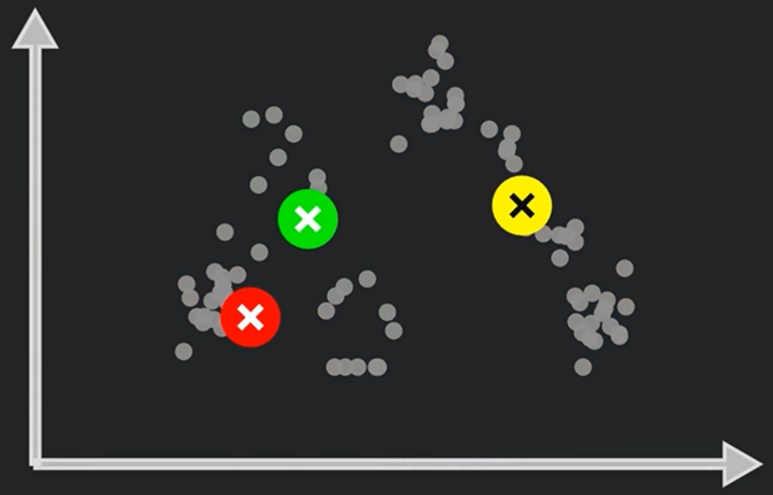


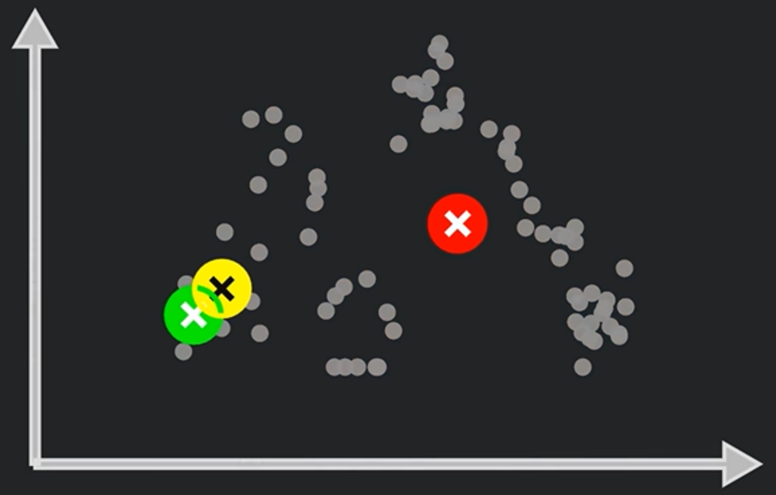










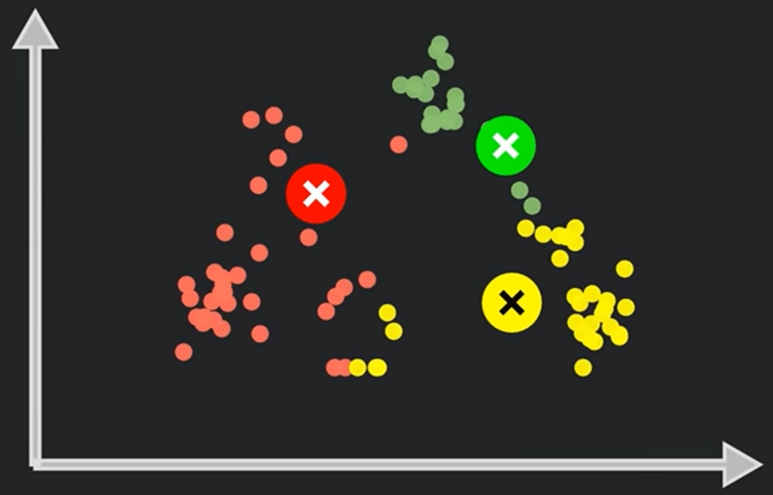




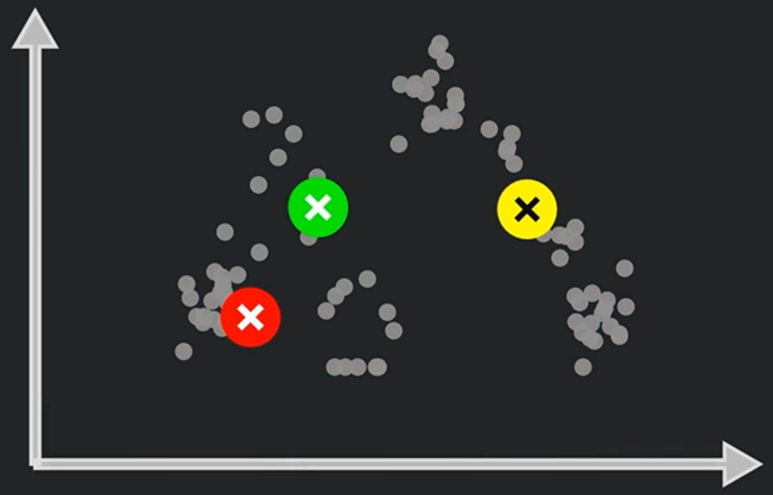


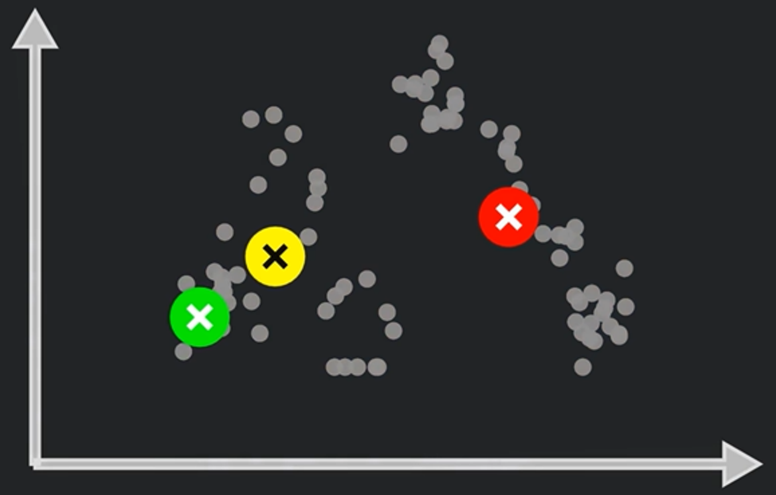


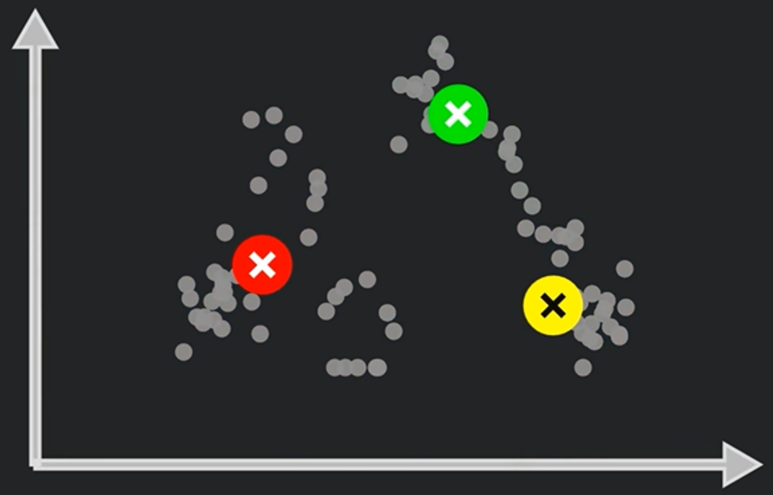




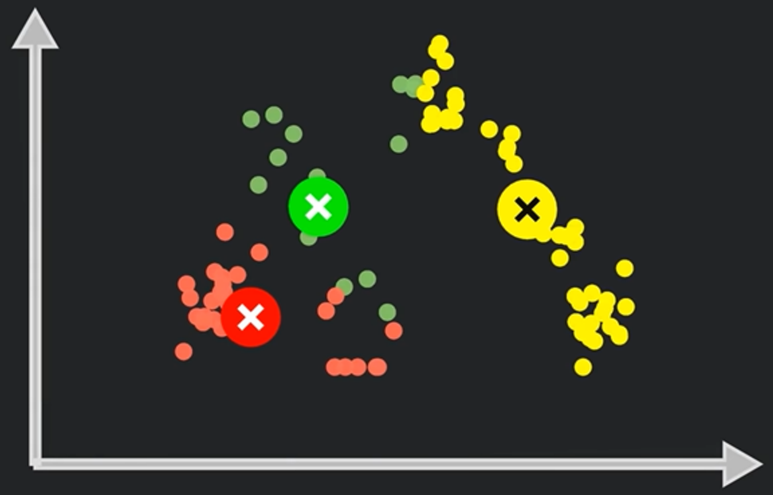


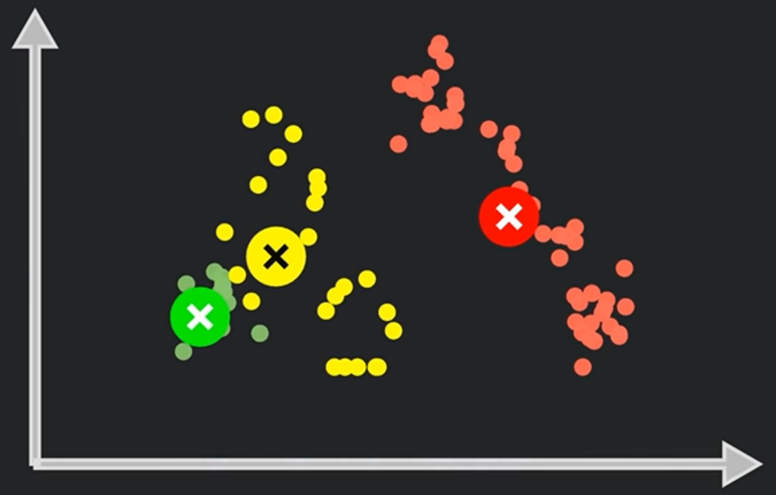


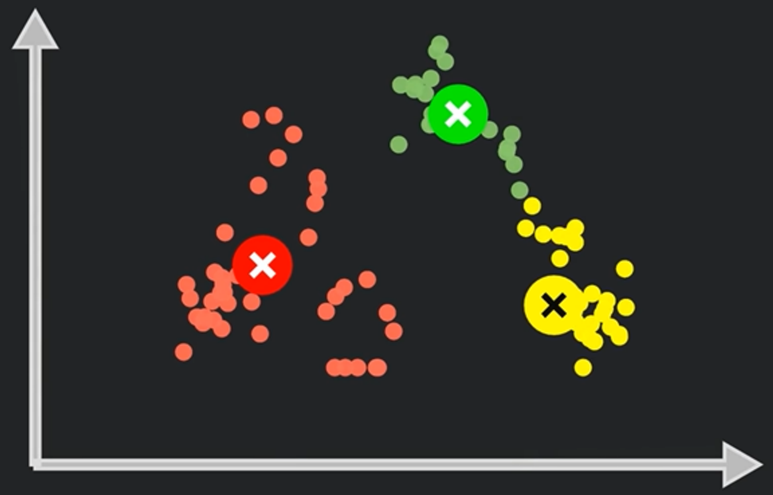




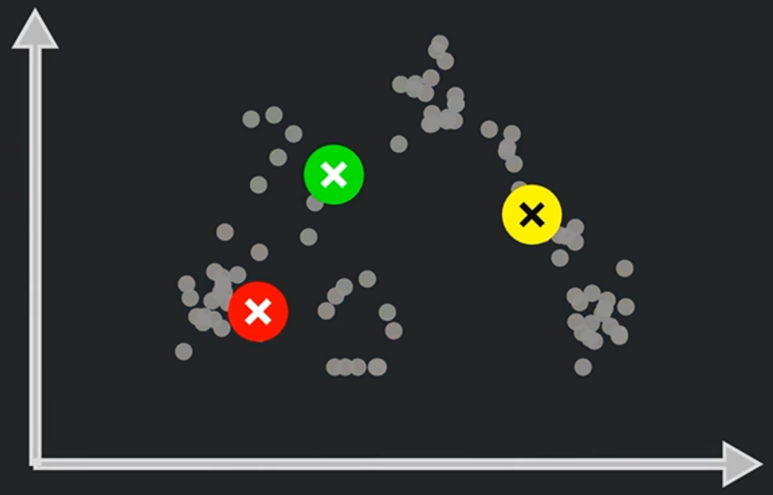


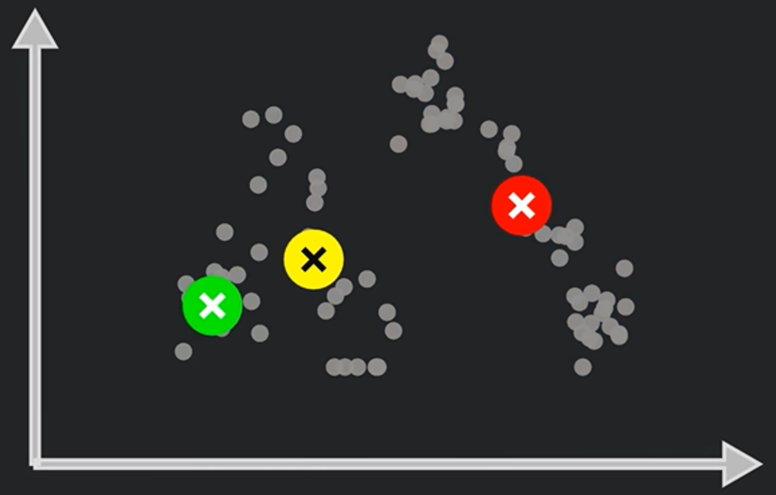


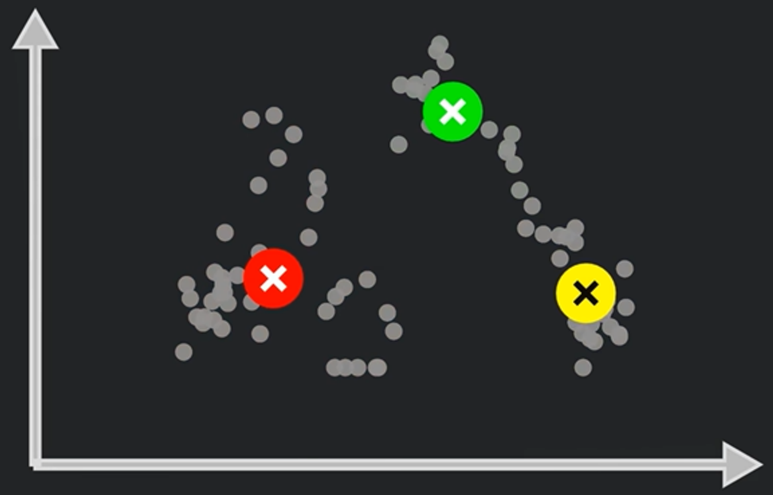


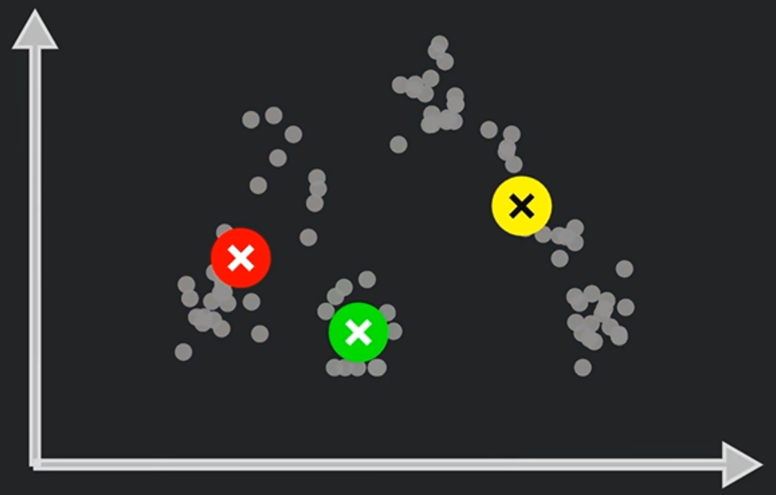






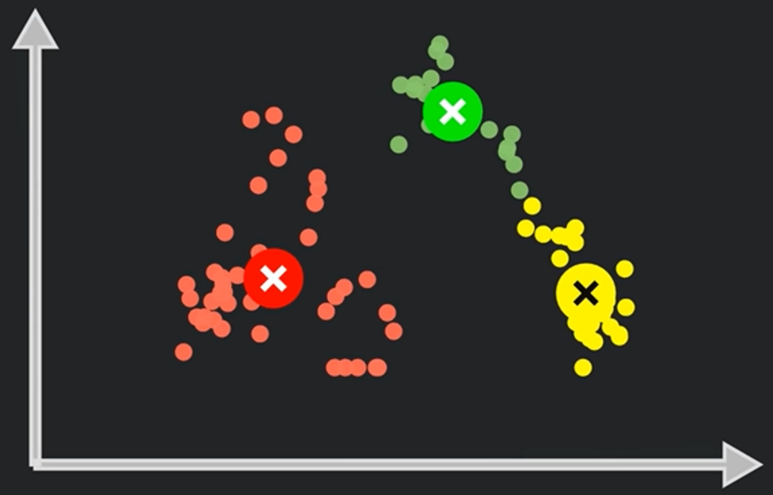


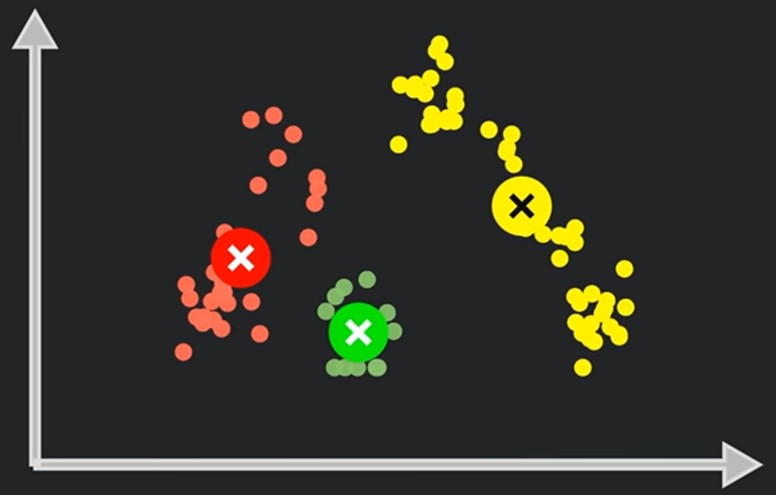


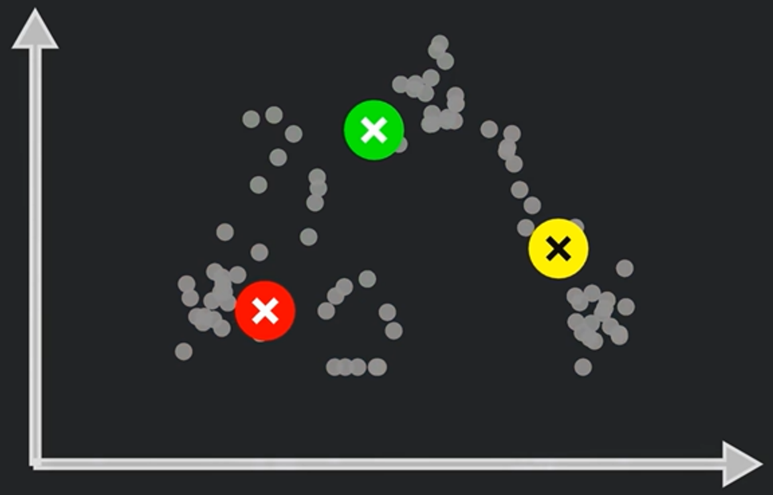




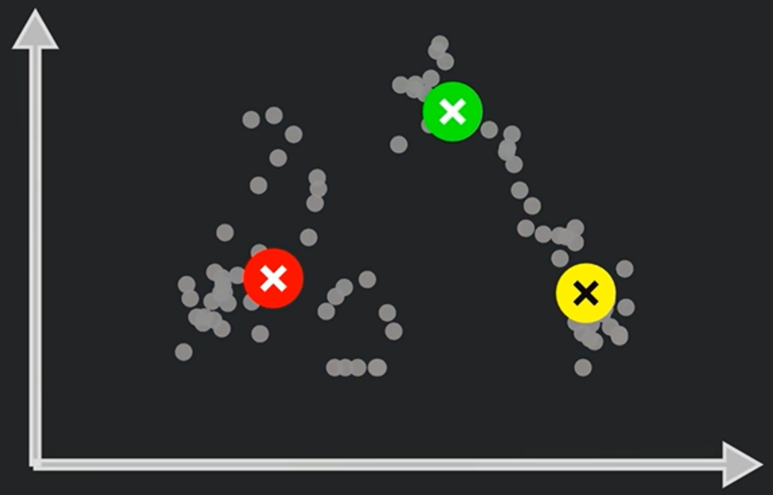


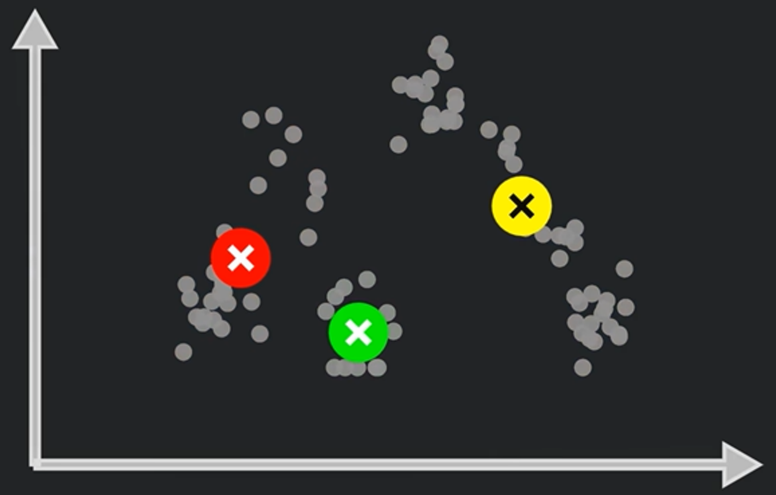


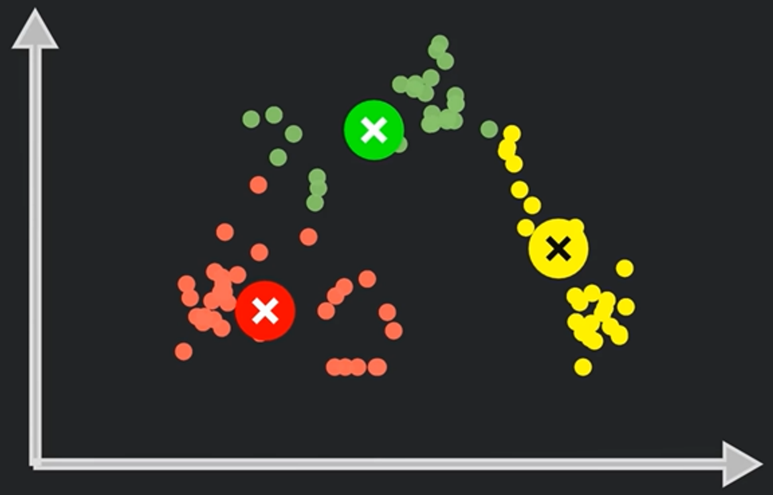




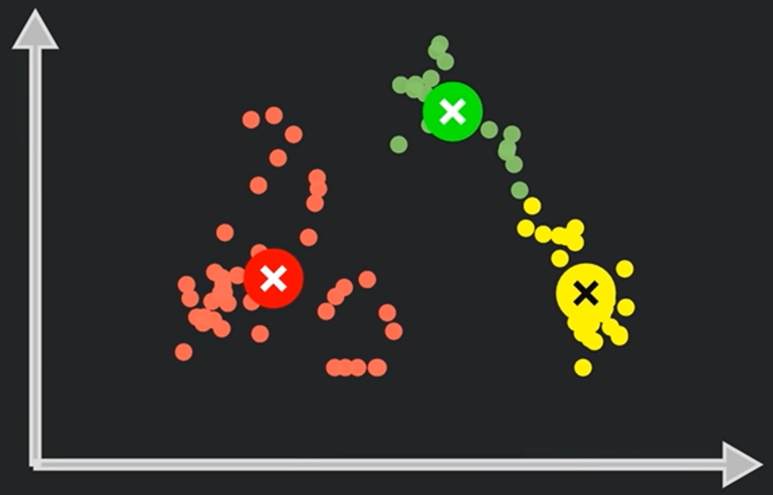


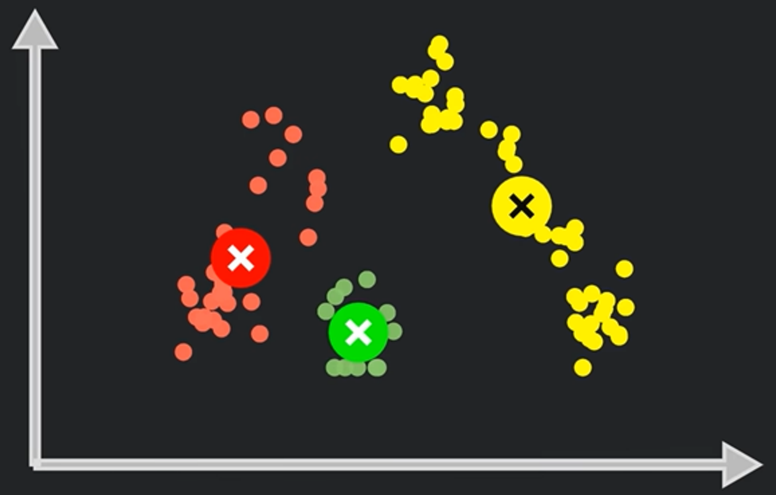


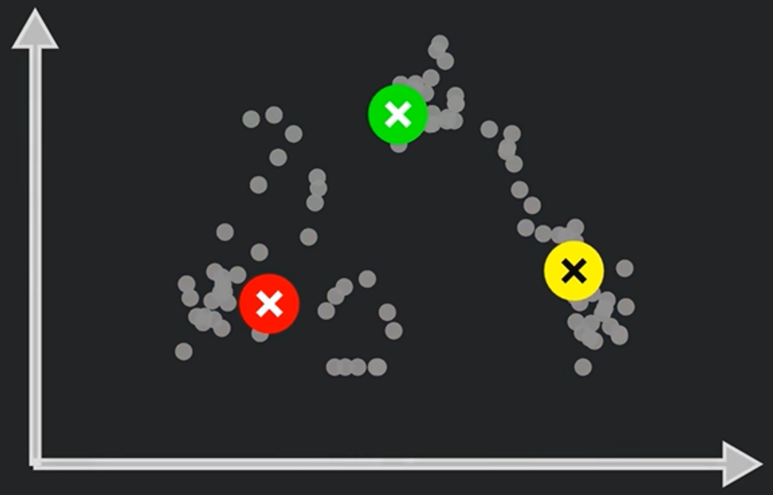




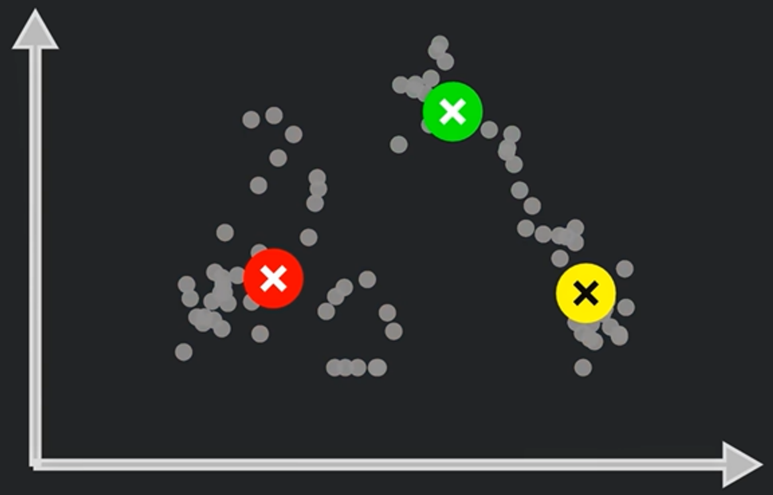


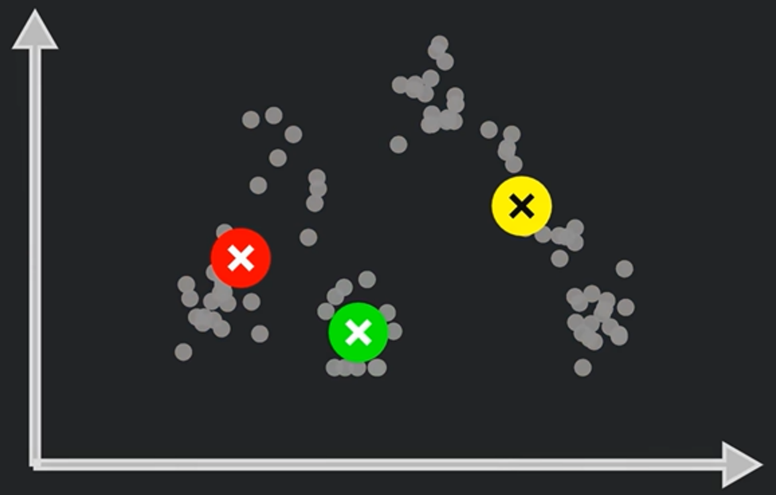






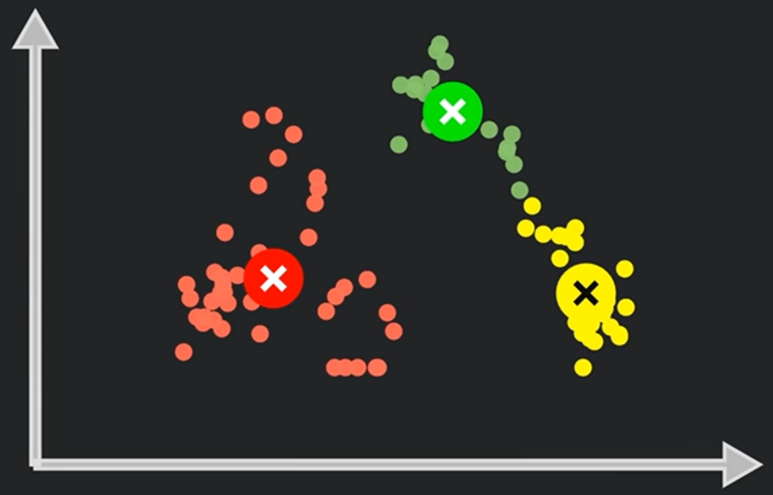


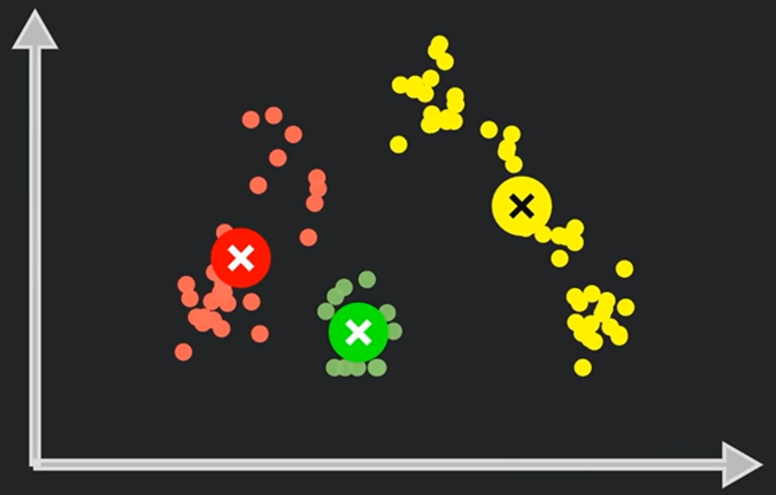


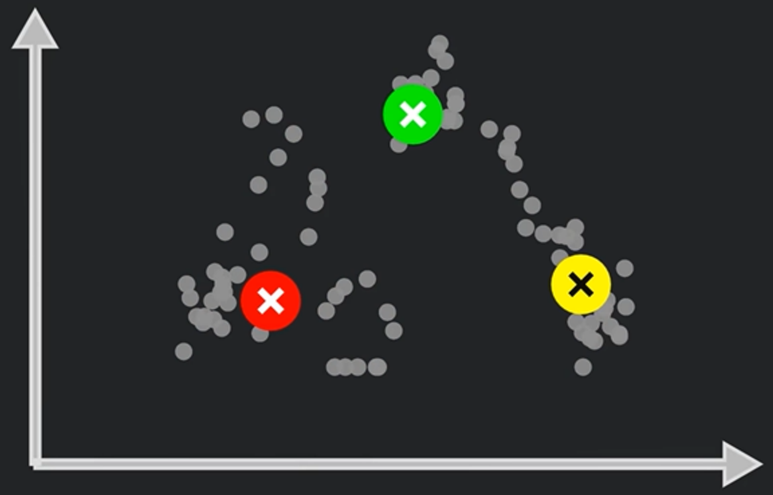


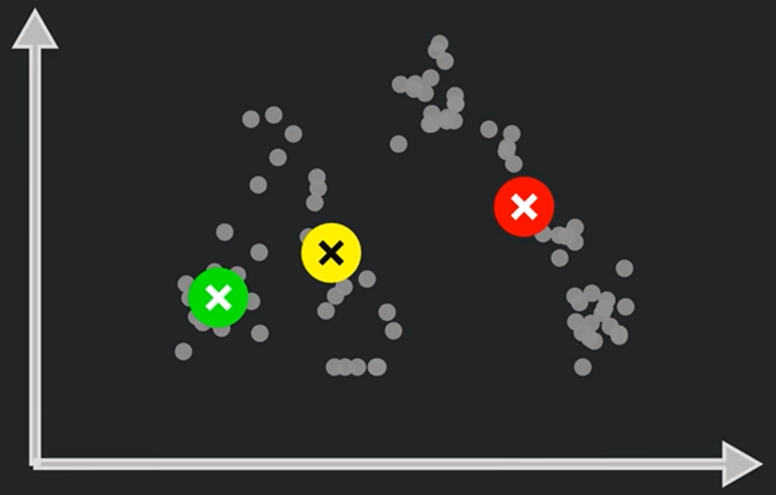


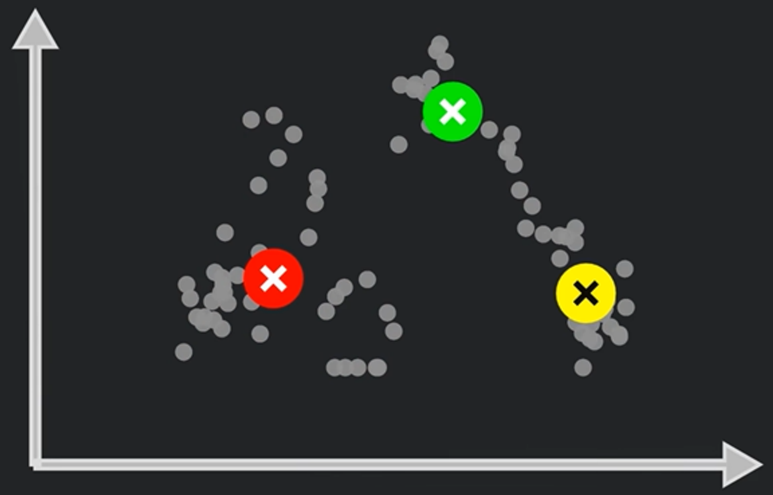


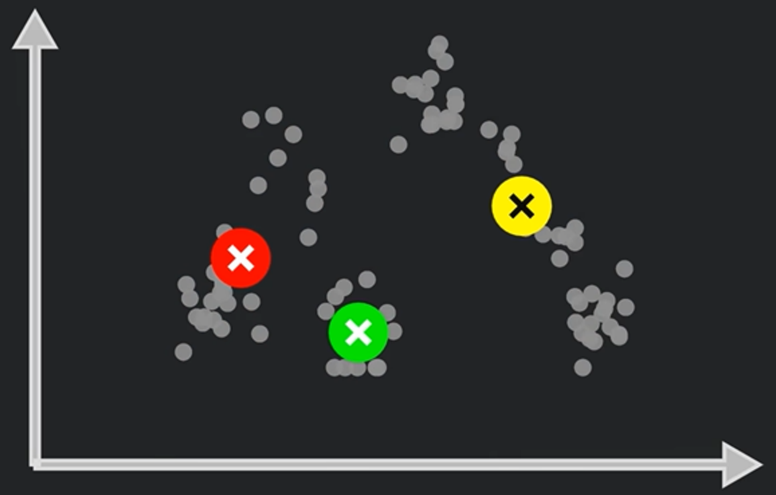


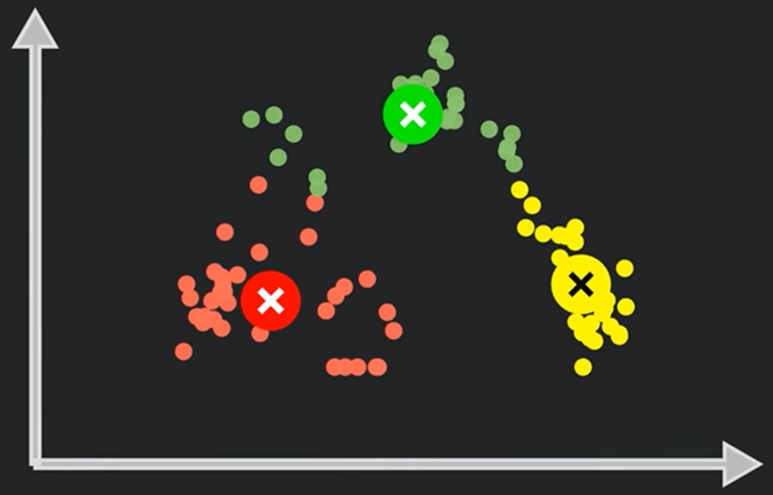




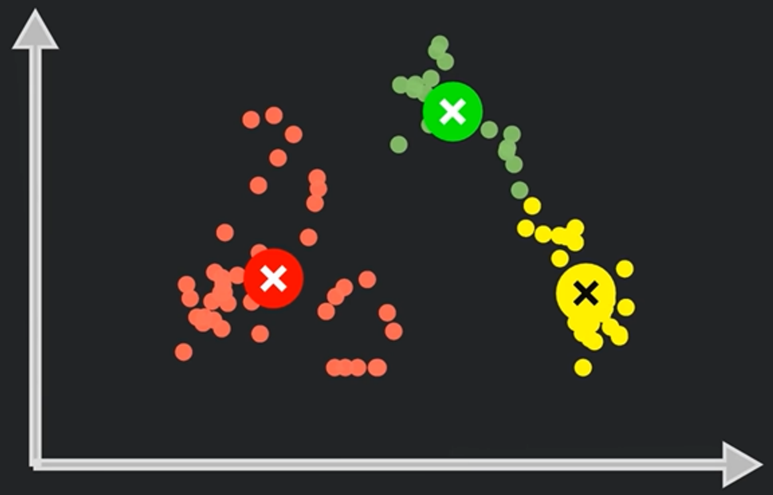


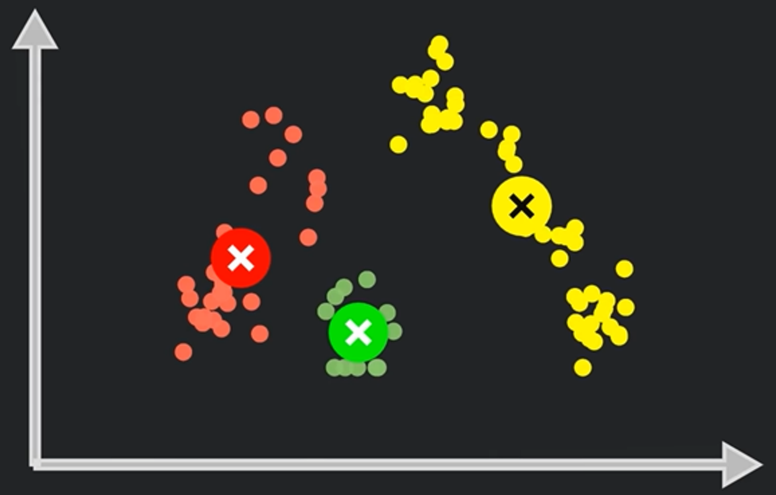


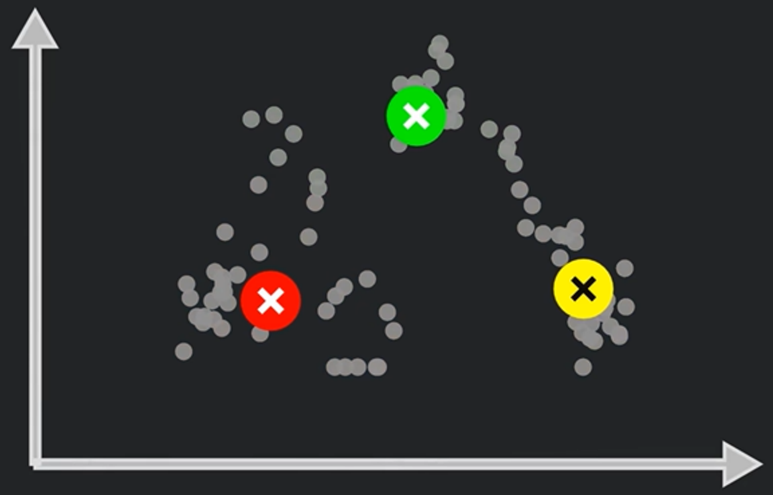




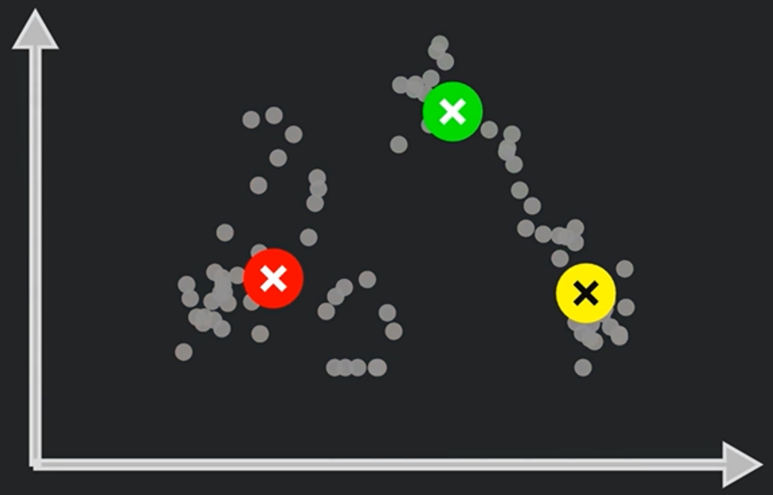


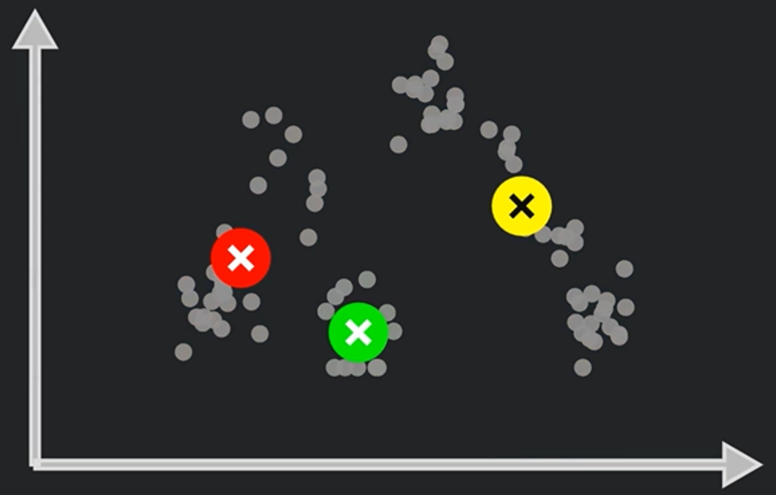


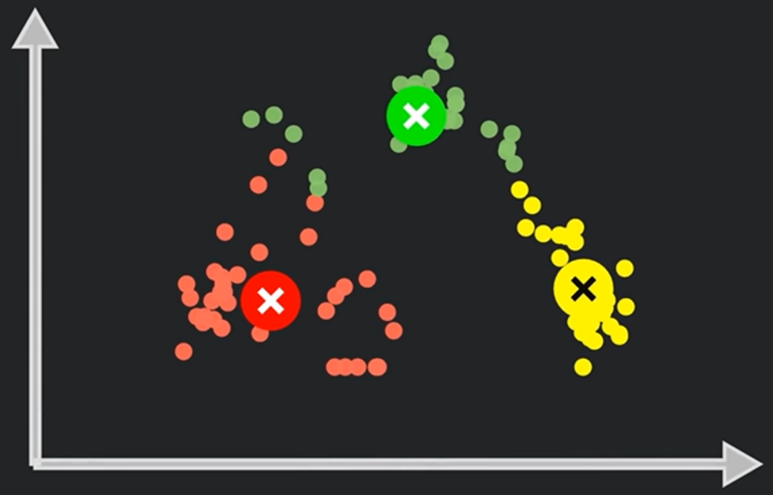




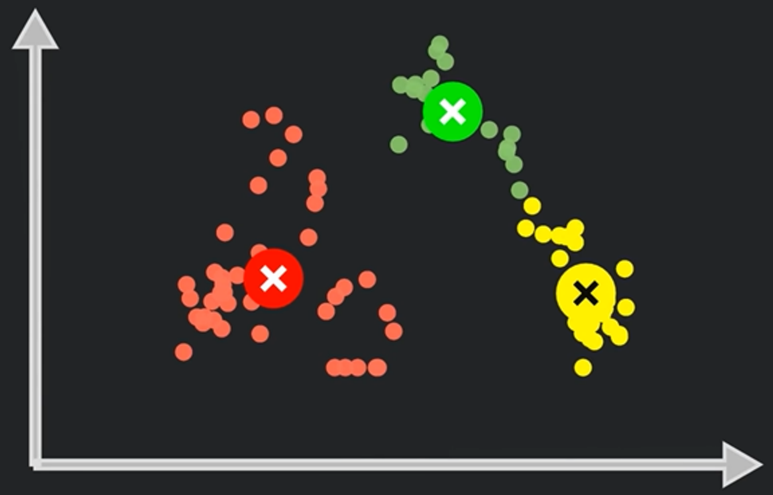


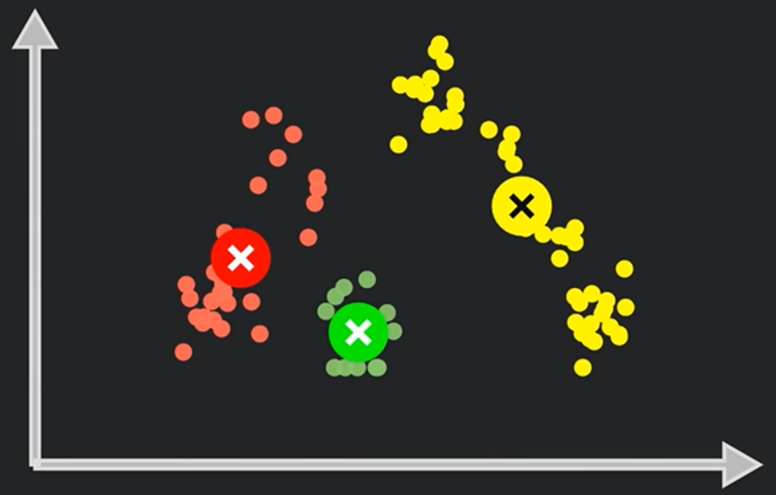


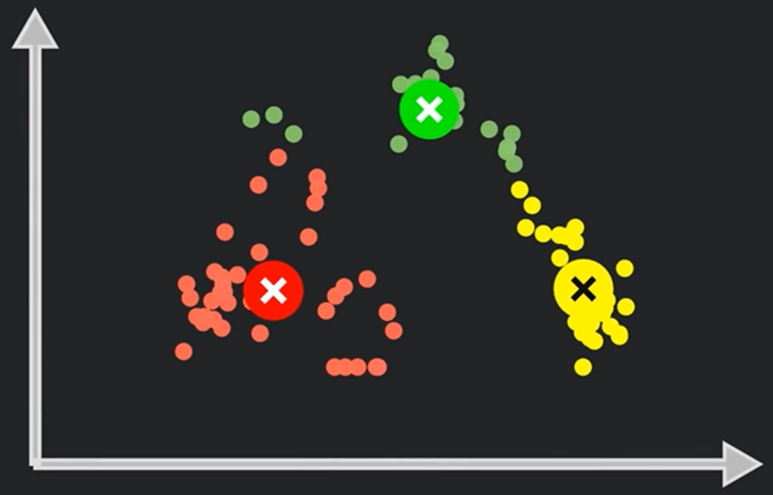




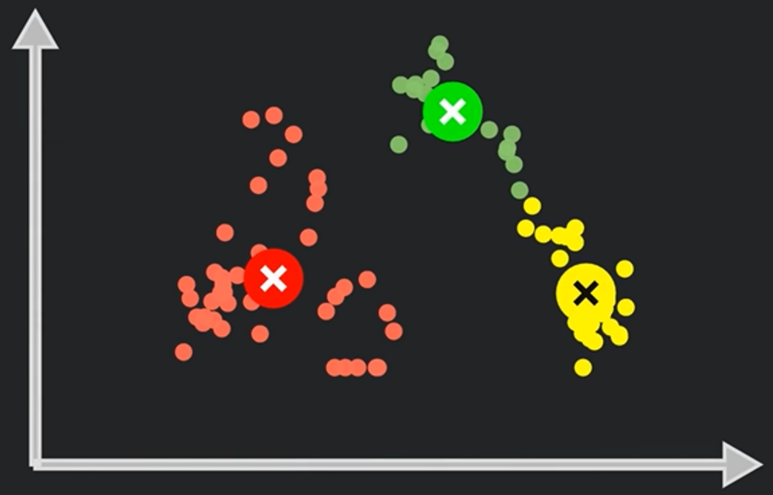


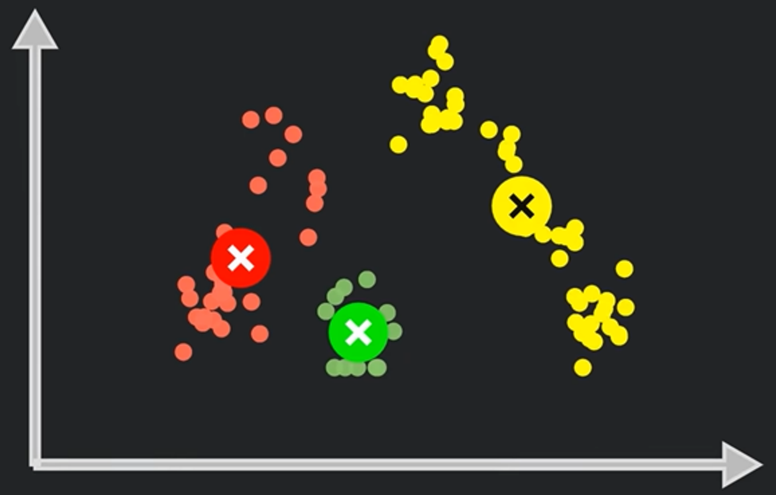


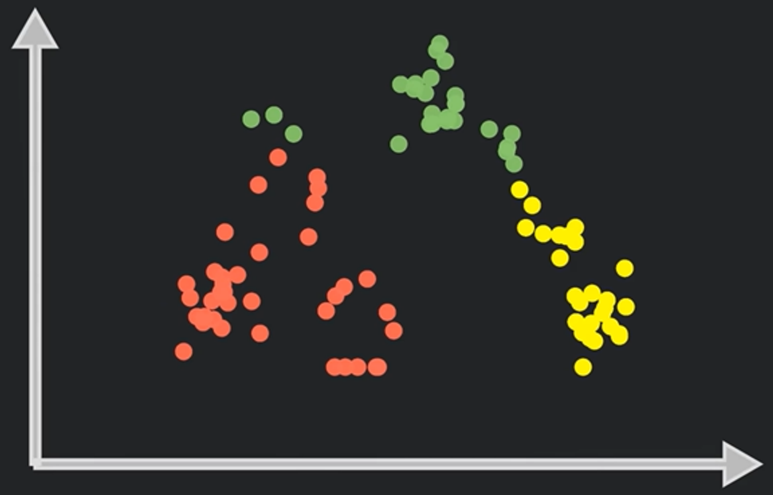


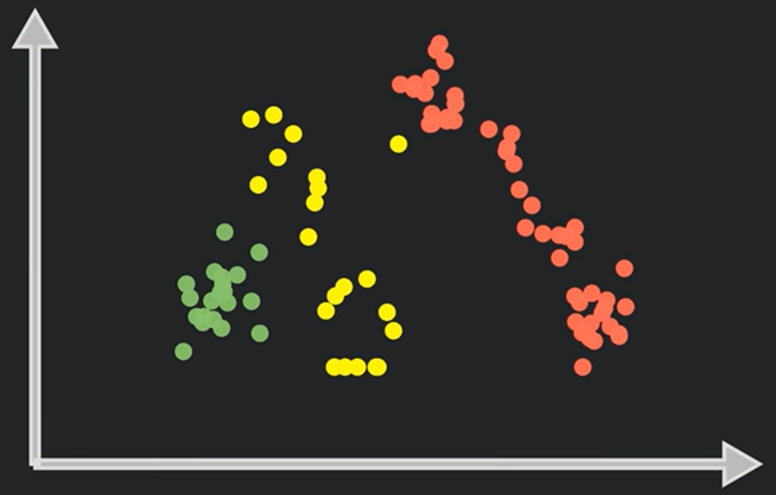


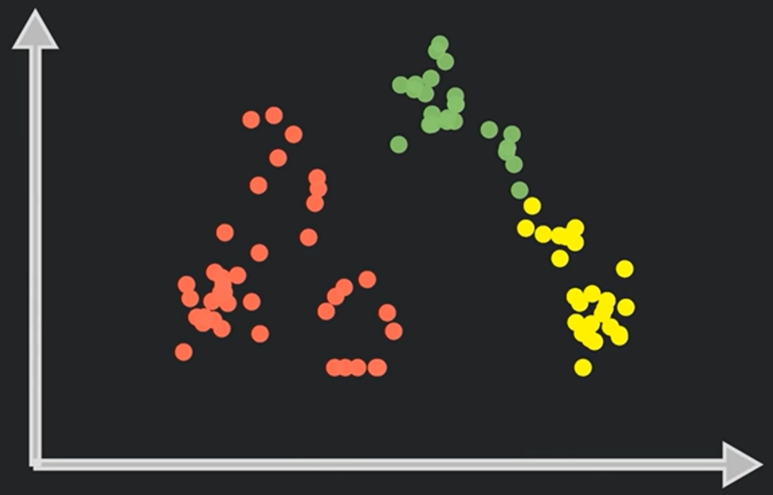


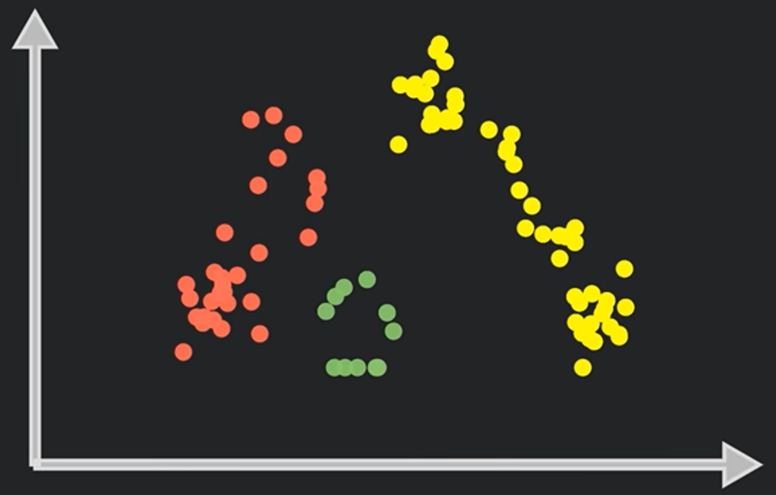


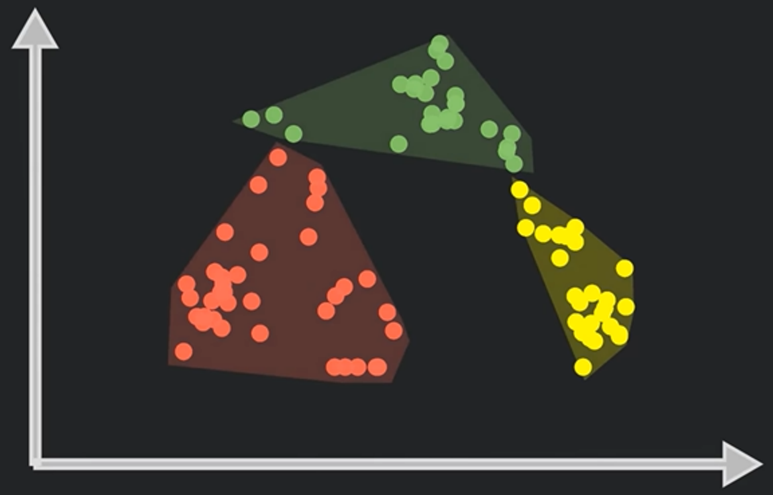


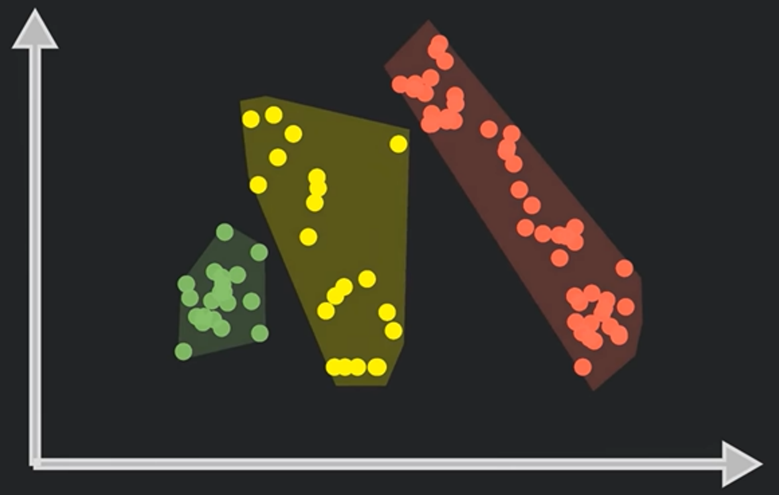


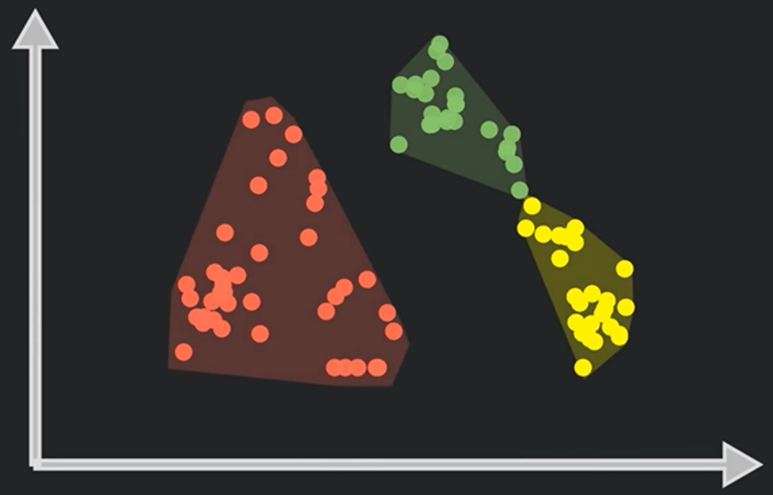


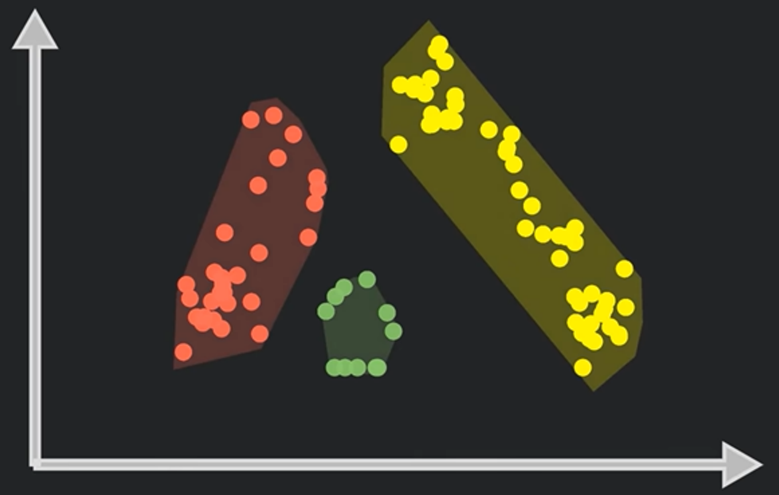


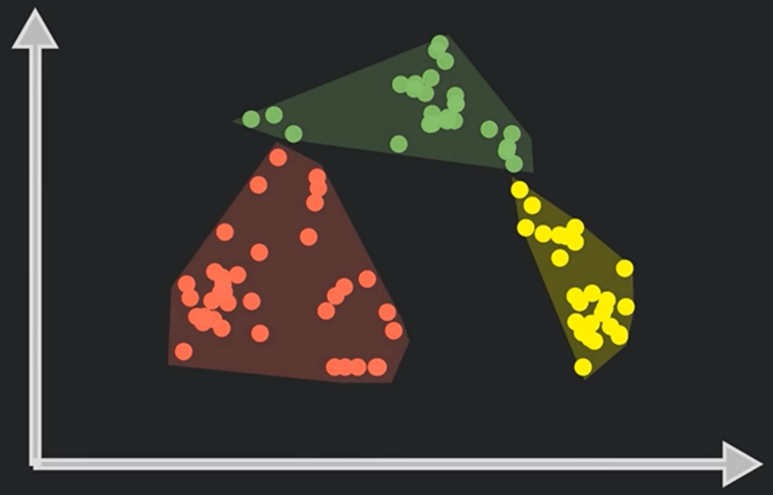




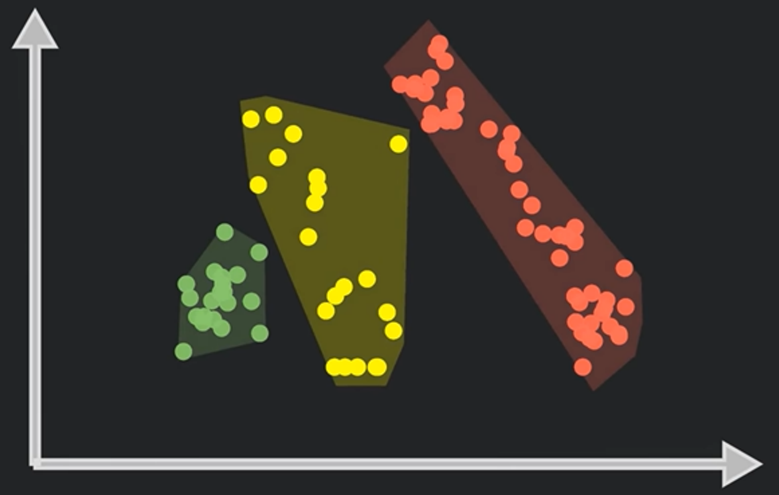




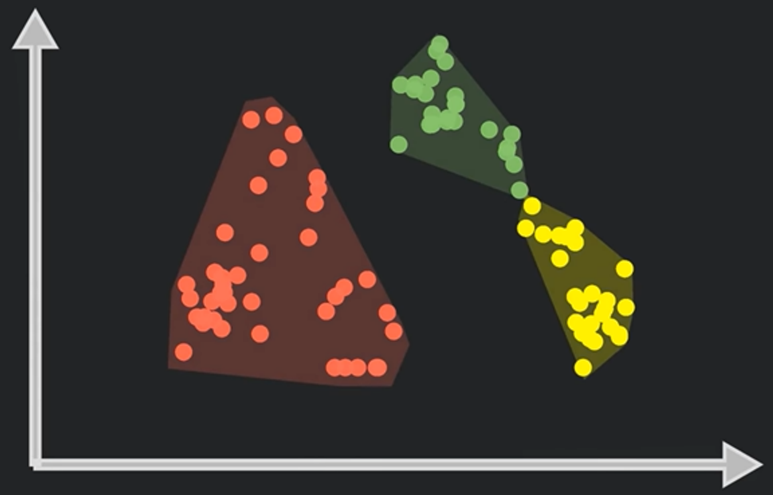




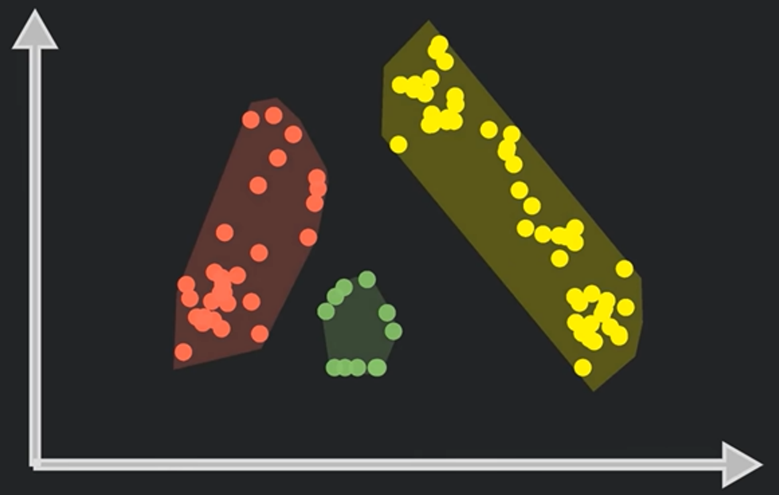
J: 14.9



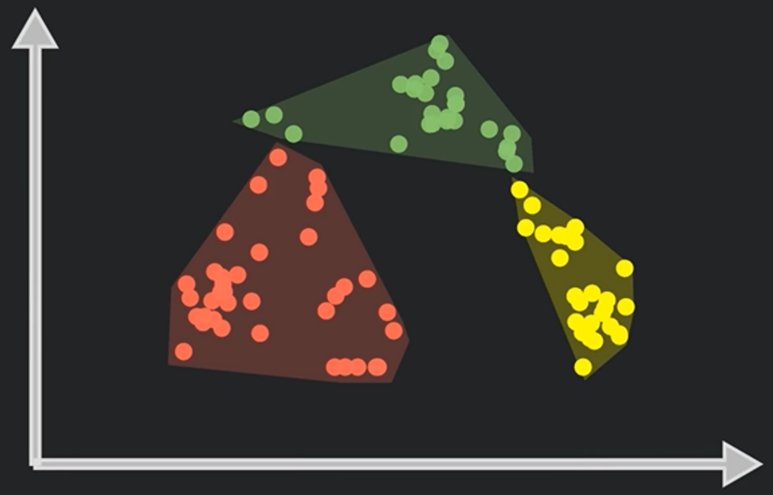
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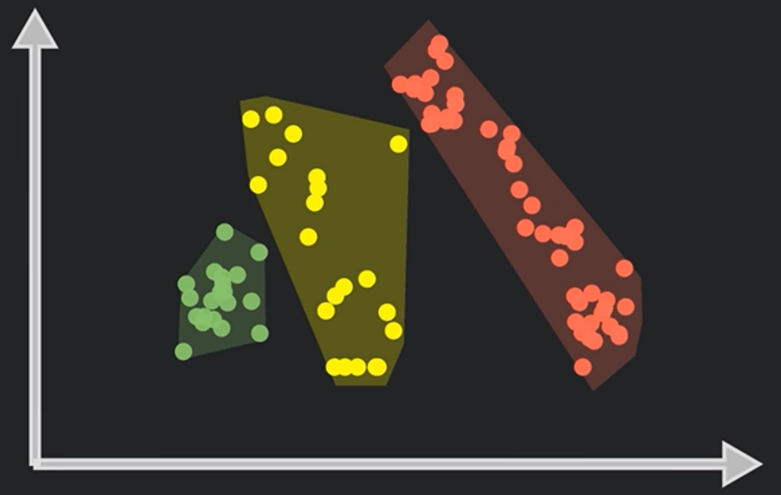
J: 14.7



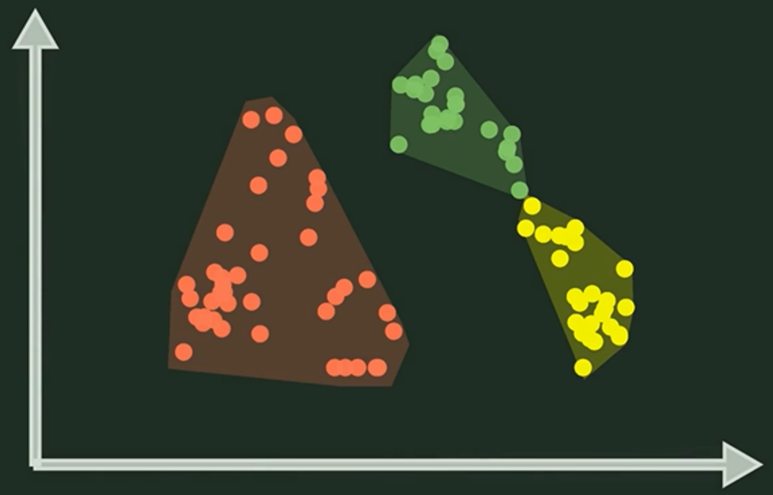
J: 29.2



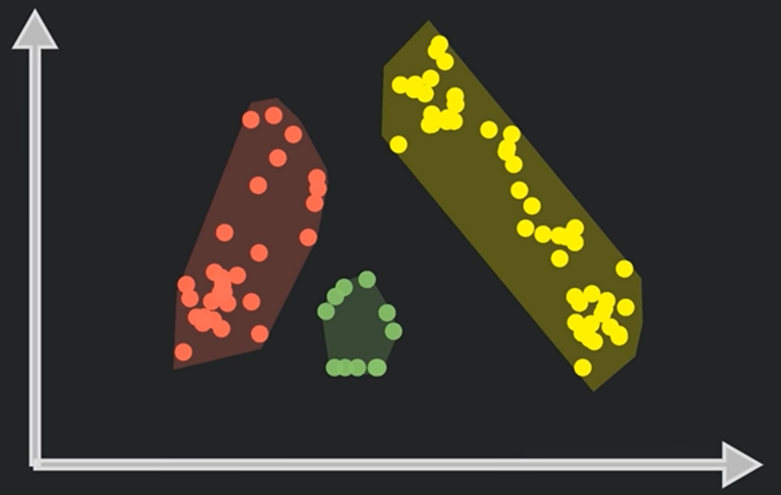
J: 14.9



J: 29.8



J: 14.7



J: 29.2

4. Advantages of K-Means:

1. **Simple and easy to implement**
   - The logic is straightforward: assign → update centroids → repeat.
2. **Fast and scalable**
   - Works efficiently with large datasets.
   - Example: segmenting millions of customers into groups based on purchase habits.
3. **Interpretable results**
   - Output clusters are easy to understand (centroid + grouped data).
   - Example: grouping students based on grades into high/medium/low performance.
4. **Guaranteed convergence**
   - It always reaches a result after several iterations (may not be the global best).
   - Example: after a few steps, the centroids stabilize and stop moving.

5. Limitations of K-Means:

1. **Requires predefining K (number of clusters)**
   - You must choose K before running the algorithm.
   - Example: You need to decide whether customers should be divided into 3 or 5 clusters.
2. **Sensitive to initialization**
   - Different starting centroids can lead to different results.
   - Example: Two runs may produce different clusters depending on the initial random points.
3. **Sensitive to outliers**
   - A single extreme point can pull the centroid far from the actual cluster center.
   - Example: In salary data, one person earning $1,000,000 distorts the cluster.
4. **Only works with numerical data**
   - Requires numerical distance calculation (Euclidean distance).
   - Example: It cannot cluster text labels like “red”, “blue”, “green” unless converted into numbers.

6. Conclusion

K-Means is one of the most fundamental clustering algorithms in machine learning.

It’s efficient, intuitive, and forms the basis for many advanced techniques in data science and AI.

Exercises

Q1. What does the centroid of a cluster represent?

A. The oldest data point in the cluster

B. The mean position of all points in the cluster

C. The data point farthest from the center

D. The most frequent value in the cluster

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A. The oldest data point in the cluster

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B

Q2. When does the K-Means algorithm stop updating centroids?

A. When the centroids no longer change much

B. After the first iteration

C. When there are no data points left

D. When new data arrives

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A

Q3. Which of the following techniques is often used to find a good value of K?

A. Gradient descent

B. Elbow method

C. Random forest

D. Cross-validation

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A. Gradient descent

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B

Q4. In K-Means, what does “K” represent?

A. The number of iterations

B. The number of clusters

C. The number of features in the dataset

D. The total number of data points

Q4. In K-Means, what does “K” represent?

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B

Q5. In the K-Means algorithm, which of the following can still occur even after many iterations?

A. A data point changes its assigned cluster in every iteration indefinitely

B. Two centroids can end up at the same location, effectively merging clusters

C. The total cost function J may temporarily increase between iterations

D. A centroid may move outside the convex hull of the points in its cluster

Q5. In the K-Means algorithm, which of the following can still occur even after many iterations?

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B

Exercise 1:

1. Plot the data points (X vs. Y) on a scatter chart to visualize their distribution.
2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2) with initial centroids: C_1 = (3.5 , 4.5) and C_2 = (3.5 , 5).

DATASET

Point	X	Y
A	1	1
B	1.5	2
C	2.5	4
D	5	7
E	3.5	5
F	4.5	5
G	3.5	4.5

EUCLIDEAN DISTANCE

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i C_j} x_i$$

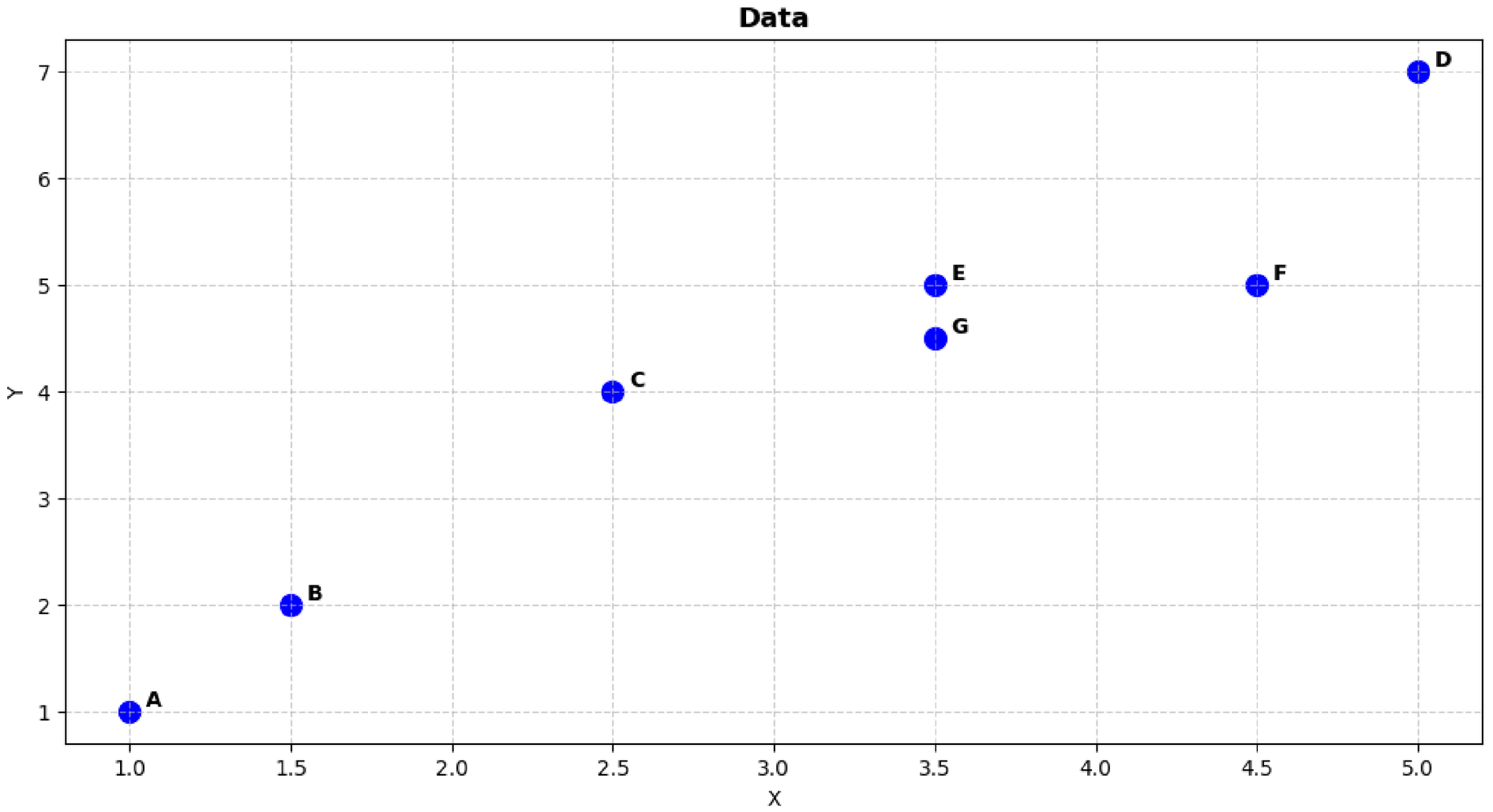
μ_j : new centroid (mean vector) of cluster

C_j : set of all points assigned to cluster j

$|C_j|$: number of points in cluster j

x_i: data point belonging to cluster j

1. Plot the data points (X vs. Y) on a scatter chart to visualize their distribution.



2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

Step 1: Choose initial centroids

$C_1 = (3.5 , 4.5)$

$C_2 = (3.5 , 5)$

Step 2: Compute distances to initial centroids (Iteration 1)

Point	$C_1 = (3.5 , 4.5)$	$C_2 = (3.5 , 5)$
A(1,1)	4.3	4.71
B(1.5,2)	3.2	3.6
C(2.5,4)	1.11	1.41
D(5,7)	2.91	2.5
E(3.5,5)	0.5	0
F(4.5,5)	1.11	1
G(3.5,4.5)	0	0.5

2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

Step 3: Update Centroids

Cluster 1:
A(1 , 1) , B(1.5 , 2) , C(2.5 , 4) , G(3.5,4.5)

$$C_1(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$$C_2(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$C_1(X) = \frac{1}{4}(1 + 1.5 + 2.5 + 3.5) = 2.12$

$C_1(Y) = \frac{1}{4}(1 + 2 + 4 + 4.5) = 2.87$

$C_1(X,Y) = (2.12 , 2.87)$

2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

Step 3: Update Centroids

Cluster 2:
D(5 , 7) , E(3.5 , 5) , F(4.5 , 5)

$$C_1(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$$C_2(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$C_2(X)$ = $\frac{1}{3}(5 + 3.5 + 4.5) = 4.3$

$C_2(Y)$ = $\frac{1}{3}(7 + 5 + 5) = 5.6$

$C_2(X,Y)$ =(4.3 , 5.6)

2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

new Centroids

C_1 = (2.12 , 2.87)

C_2 = (4.3 , 5.6)

Step 4: Compute distances to initial centroids (Iteration 2)

Point	C_1 = (2.12 , 2.87)	C_2 = (4.3 , 5.6)
A(1,1)	2.19	5.73
B(1.5,2)	1.07	4.63
C(2.5,4)	1.18	2.47
D(5,7)	5.01	1.57
E(3.5,5)	2.58	1
F(4.5,5)	3.19	0.63
G(3.5,4.5)	2.13	1.36

2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

Step 5: Update Centroids

Cluster 1:
A(1 , 1) , B(1.5 , 2) , C(2.5 , 4)

$$C_1(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$$C_2(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$C_1(X)$ = $\frac{1}{3}(1 + 1.5 + 2.5) = 1.6$

$C_1(Y)$ = $\frac{1}{3}(1 + 2 + 4) = 2.3$

$C_1(X,Y)$ =(1.6 , 2.3)

2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

Step 5: Update Centroids

Cluster 2:
G(3.5 , 4.5) , D(5 , 7) , E(3.5 , 5) , F(4.5 , 5)

$$C_1(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$$C_2(X, Y) = \left(\frac{\sum_{i=1}^{n} X_i}{n}, \frac{\sum_{i=1}^{n} Y_i}{n} \right)$$

$C_2(X)$ = $\frac{1}{4}(3.5+5+3.5+4.5) = 4.12$

$C_2(Y)$ = $\frac{1}{4}(4.5+7+5+5) = 5.37$

$C_2(X,Y)$ =(4.12 , 5.37)

2. Apply the K-Means algorithm to cluster the data into 2 groups (K = 2).

New centroids:

C_1 = (1.6, 2.3)

C_2 = (4.12 , 5.37)

Step 6: Compute distances to initial centroids (Iteration 3)

Point	C_1 = (1.6, 2.3)	C_2 = (4.12 , 5.37)
A(1,1)	1.49	5.37
B(1.5,2)	0.32	4.28
C(2.5,4)	1.86	2.13
D(5,7)	5.74	1.86
E(3.5,5)	3.26	0.72
F(4.5,5)	3.97	0.53
G(3.5,4.5)	2.89	1.07

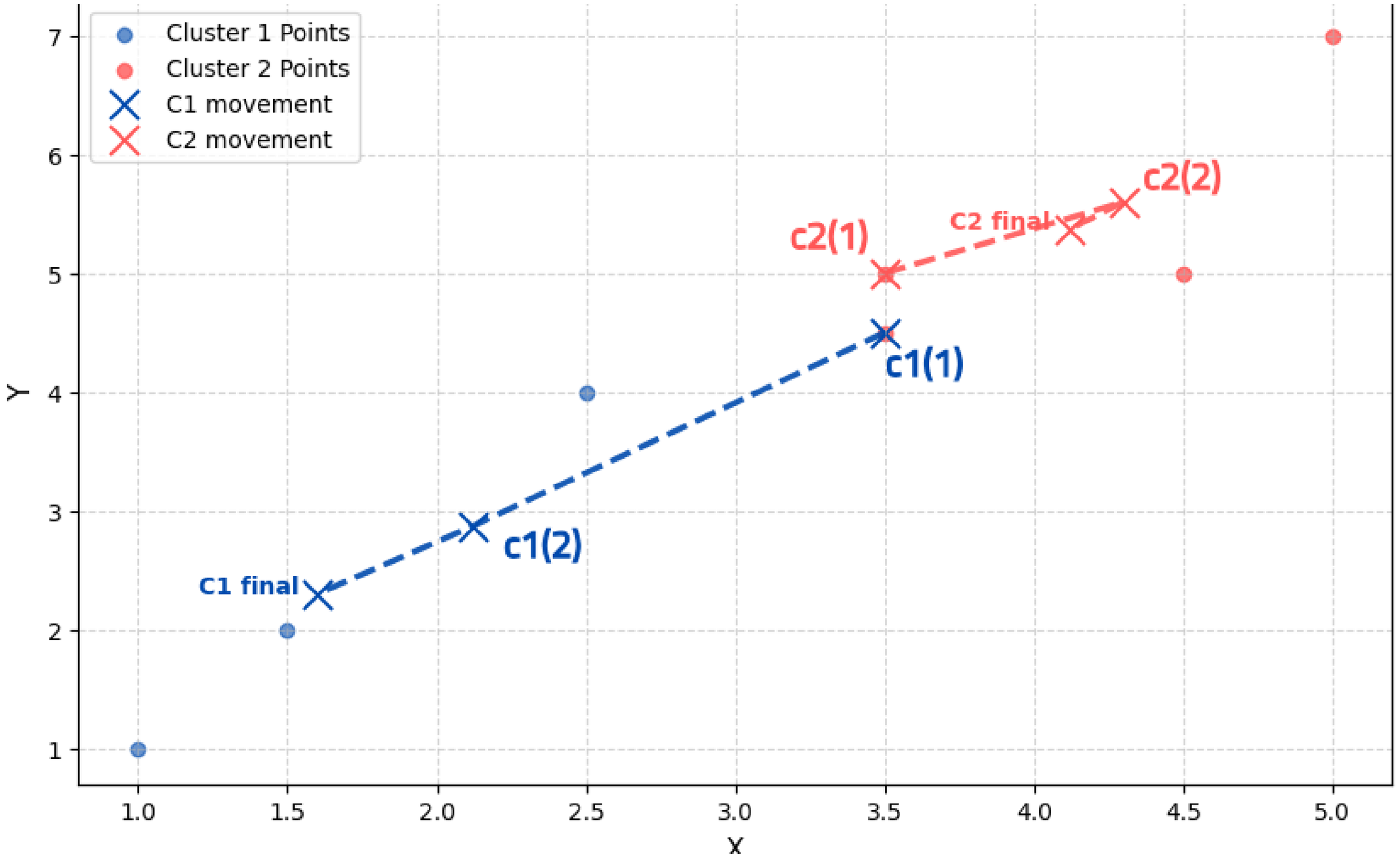
Centroid movement

Iteration	C_1	C_2
1	(3.5 , 4.5)	(3.5 , 5)
2	(2.12 , 2.87)	(4.3 , 5.6)
3	(1.6 , 2.3)	(4.12 , 5.37)

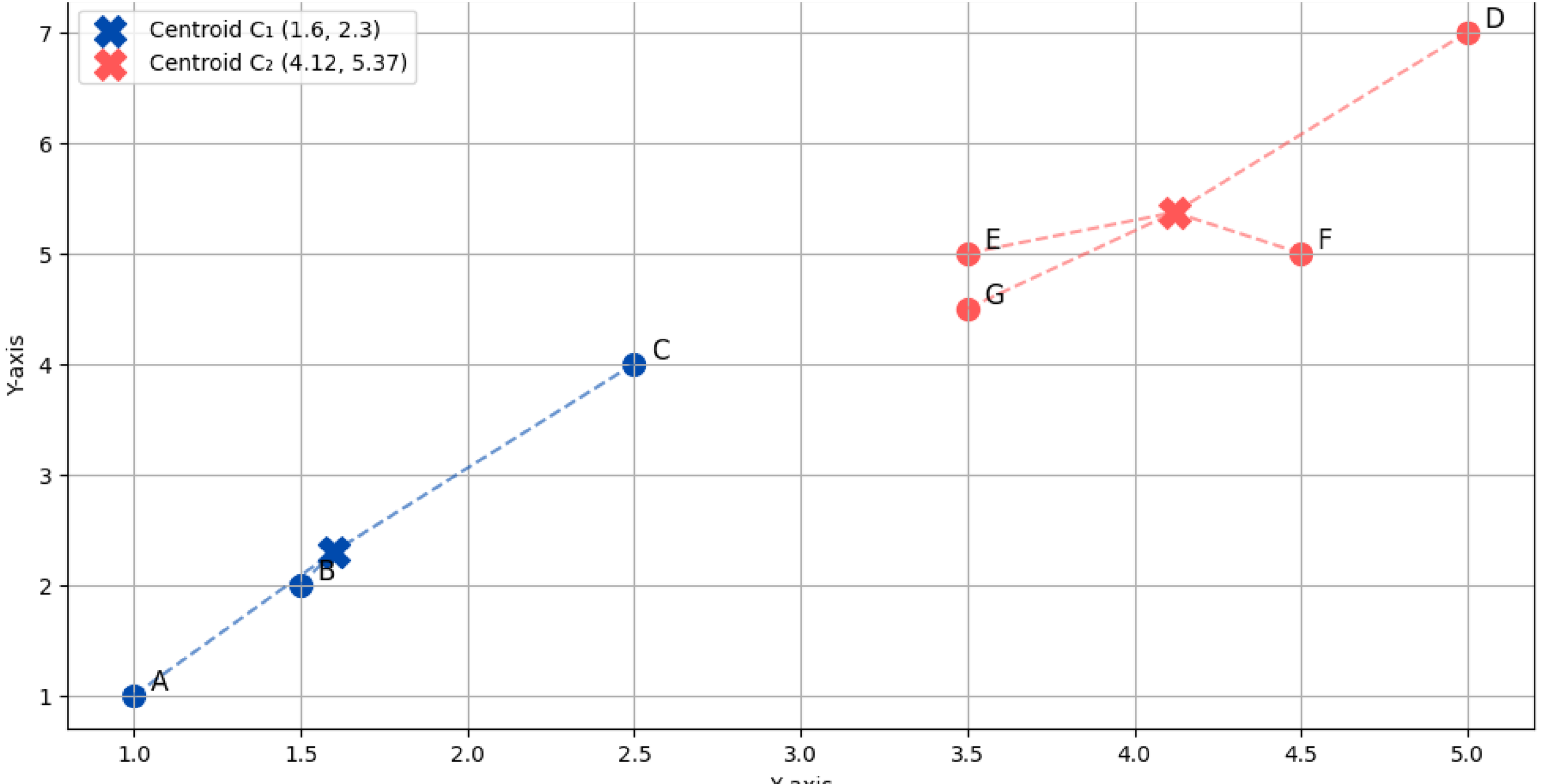
The movements between iterations are small, meaning the centroids have nearly converged.

This indicates that the K-Means algorithm has reached convergence, and Iteration 3 is the final iteration.

Centroid Movement in K-Means



K-means Clustering Visualization (K=2)



Thank you for your time